chapter-8
counters

⇒ Multiple flip-flops are connected together to form counters.

Counters are classified into two broad categories according to the way they are clocked : asynchronous and synchronous.

⇒ In asynchronous counters, commonly called ripple counters, the first flip-flop is clocked by the external clock pulse and then each successive flip-flop is clocked by the output of the preceding flip-flop.

⇒ In synchronous counters, the clock input is connected to all of the flip-flops so that they are clocked simultaneously.

Asynchronous counter operation

Asynchronous: The term asynchronous refers to events that do not have a fixed time relationship with each other and generally do not occur at the same time.

Asynchronous counter: An asynchronous counter is one in which the flip-flops within the counter do not change states at exactly the same time because they do not have a common clock pulse.

⇒ **counters:** Counter is a device that stores (and sometime displays) the number of times a particular event or process has occurred, often in a relationship to a clock.

A counter circuit is usually constructed of a number of flip-flops connected in cascade.

* How many flip flops are required to design mod-n counter?

n = number of flip flops.

N = MOD number.

$$2^n \geq N$$

MOD 5,

$2^2 \geq 5$ ✗

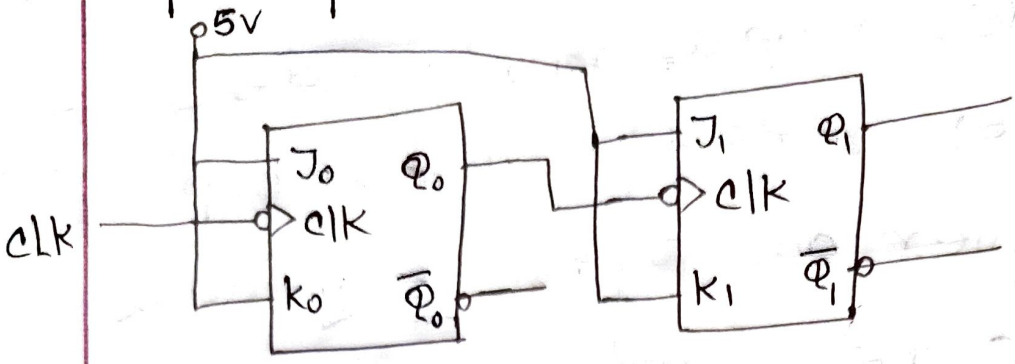
$2^3 \geq 5$ ✓

$8 \geq 5$ ∴ 3 flip-flops are needed for MOD 5 Counter.

Synchronous Counter	Asynchronous Counter
1] In synchronous counter, all flip flops are triggered with same clock simultaneously.	1] In asynchronous counter, different flip flops are triggered with different clock, not simultaneously.
2] Synchronous counter is faster than asynchronous counter in operation.	2] Asynchronous counter is slower than synchronous counter in operation.
3] Synchronous counter does not produce any decoding errors.	3] Asynchronous counter produces decoding error.
4] Synchronous counter is also called parallel counter.	4] Asynchronous counter is also called serial counter.

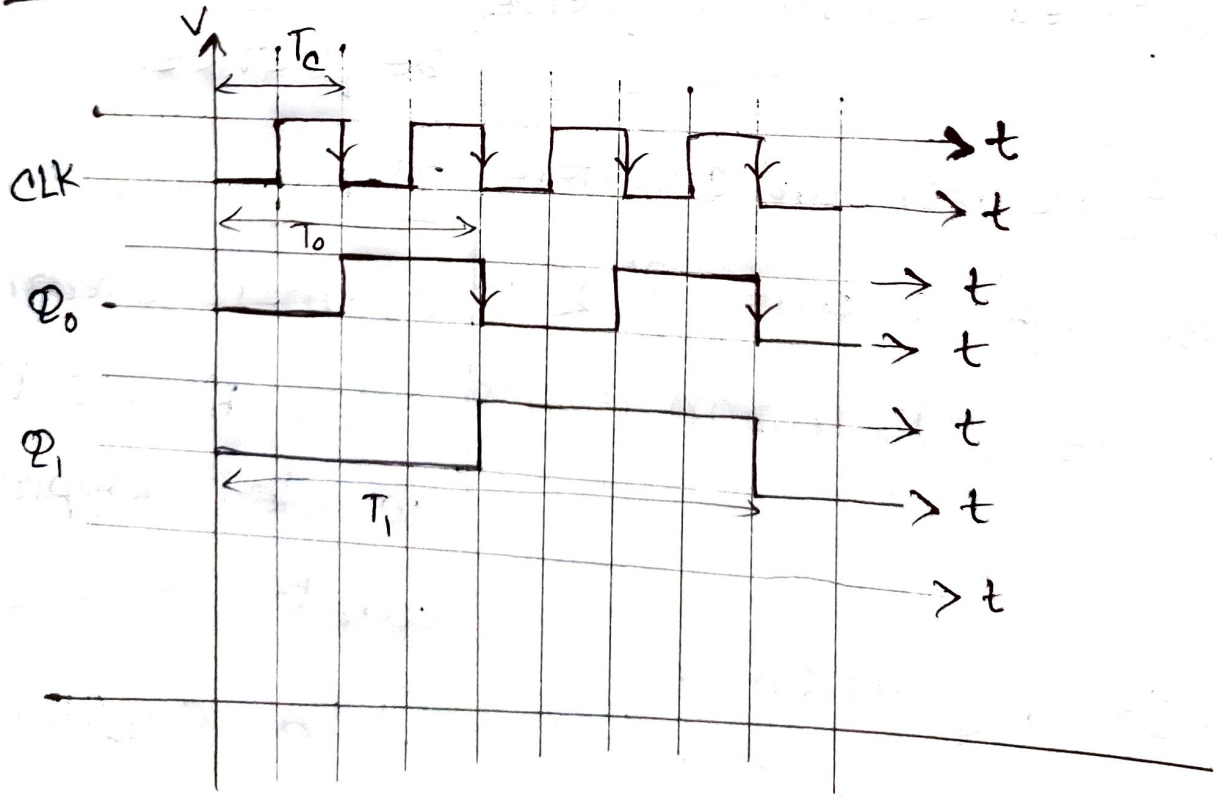
Synchronous counter	Asynchronous counter
5] Synchronous counter designing as well implementation are complex due to increasing the number of states.	5] Asynchronous counter designing as well as implementation is very easy.
6] Synchronous counter will operate in any desired count sequence.	6] Asynchronous counter will operate only in fixed count sequence.
7] Synchronous counter examples are: Ring counter, Johnson counter	7] Asynchronous counter examples are: Ripple UP counter, Ripple DOWN counter.
8] In synchronous counter, propagation delay is less.	8] In asynchronous counter, there is high propagation delay.

⇒ Flip-flop as divide by 2 circuit :-



The asynchronous counters are also known as ripple counters.

Timing diagram



$\therefore T_0 = 2T_c$.

$$\frac{1}{f_0} = \frac{2}{f_c}$$

$\therefore f_0 = \frac{f_c}{2}$

$T_1 = 2T_0$

$\frac{1}{f_1} = \frac{2}{f_0}$

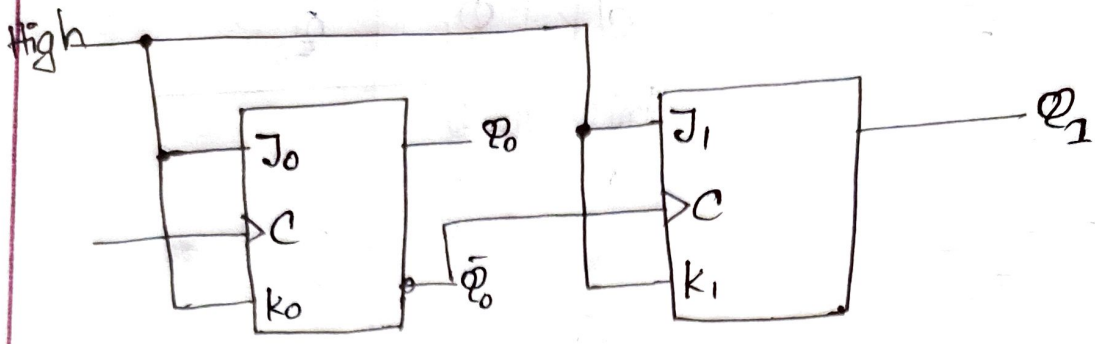
$f_1 = \frac{f_0}{2} = \frac{f_c}{4}$

CLK	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4(recycle)	0	0

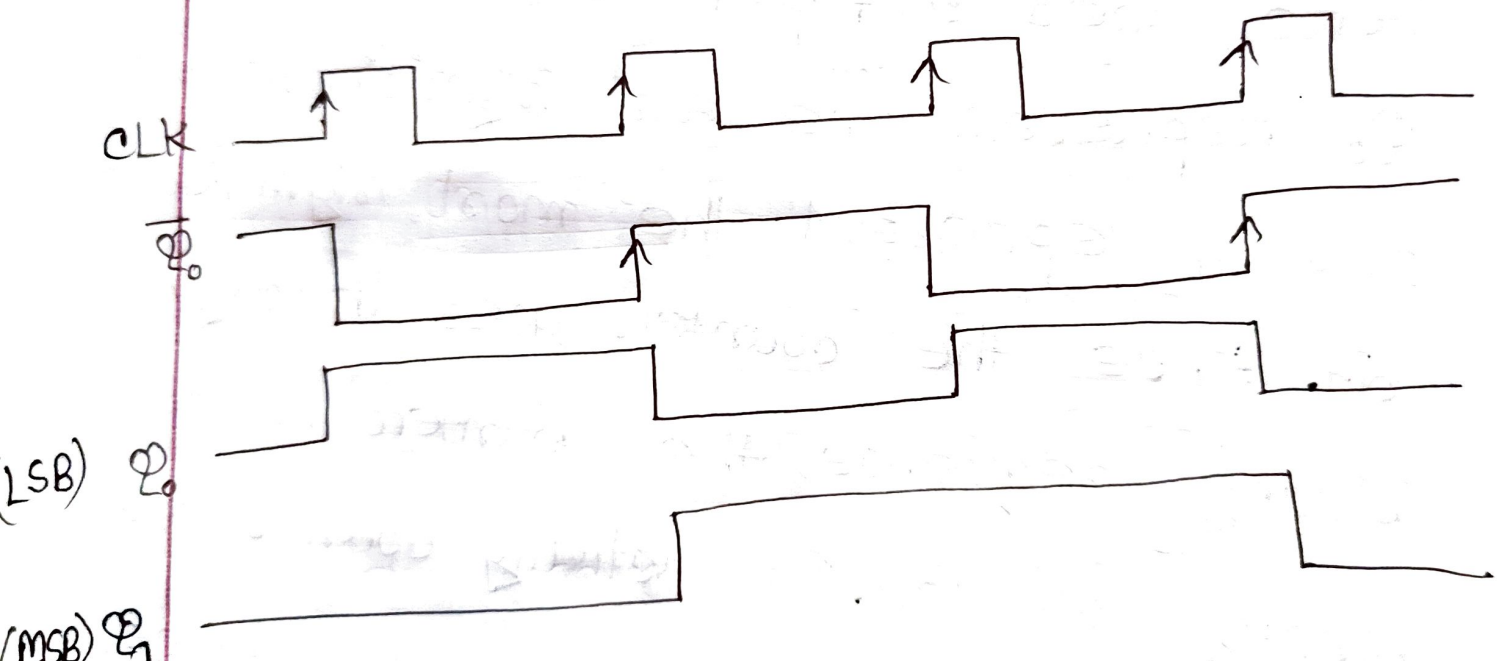
⇒ Here used 2 flip-flops ($2^2 = 4$). Q_0 represent the least significant bit and Q_1 represent the most significant bit. Since the counter goes through a binary sequence, the counter is a binary counter. It actually counts the number of clock pulses up to three, and on the fourth pulse it recycles to its original state ($Q_0 = 0, Q_1 = 0$).

⇒ The term recycle is commonly applied to counter operation, it refers to the transition of the counter from its final state back to its original state.

A 2-Bit Asynchronous Binary Counter



Timing diagram



* Two-bit counter exhibits four different states, as you would expect with two flip-flops ($2^2=4$). Q_0 represents the least significant bit (LSB) and Q_1

represents the most significant bit (MSB), the sequence of counter states represents a sequence of binary numbers. In digital logic, Q_0 is always the LSB unless otherwise specified.

Clock pulse	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4 (recycles)	0	0

Since it goes through a binary sequence, the counter is called a binary counter. It actually counts the number of clock pulses up to three and on the fourth pulse it recycles to its original state.

$2^{bit} = 2^2 = 4$ state [0-3]

The term recycle is commonly applied to counter operation, it refers to the transition of the counter from its final state to its original state.

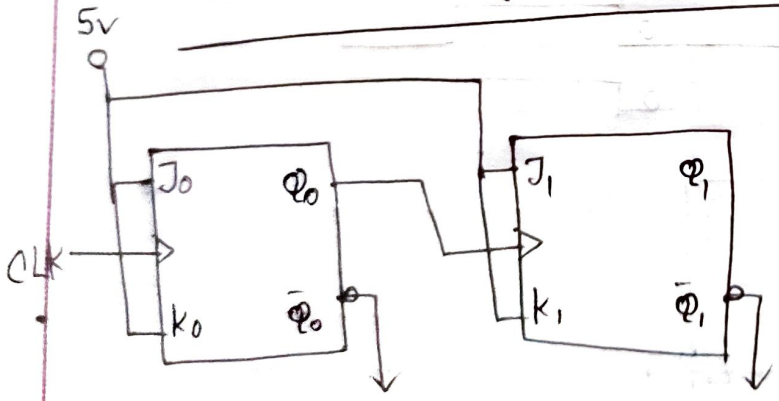
A 3-bit Asynchronous Binary counter

3-bit counter has eight states, due to its three flip-flops. This counter counts 0 through 7 and then recycles to the zero state. This counter can be easily expanded for higher count, by connecting additional toggle flip-flops.

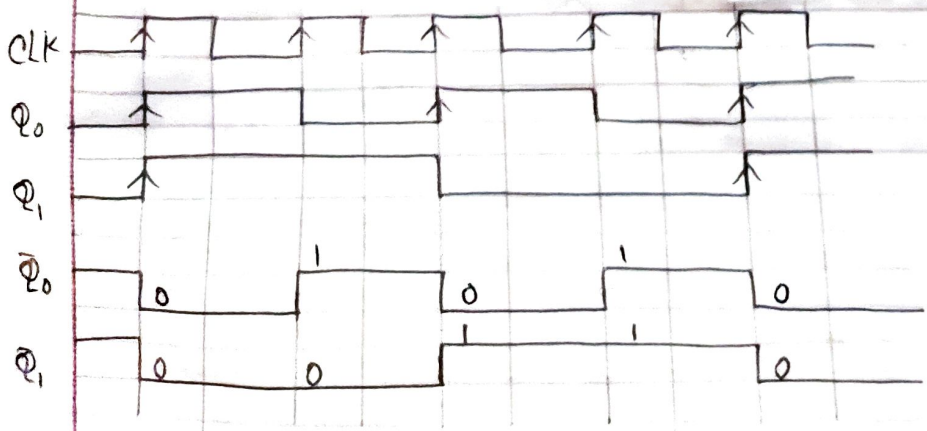
clock pulse	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8 (recycles)	0	0	0

Problem

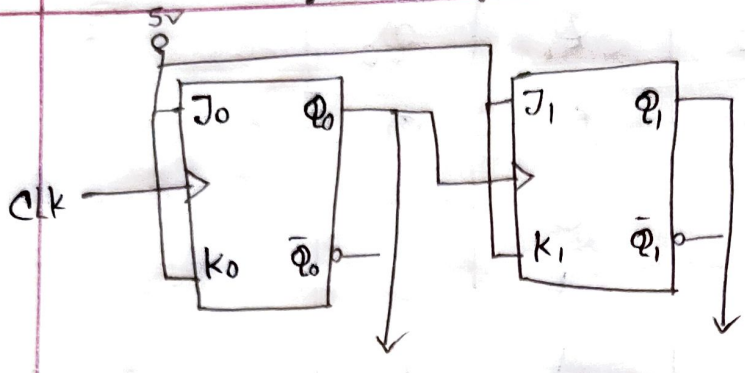
2-bit asynchronous up counter



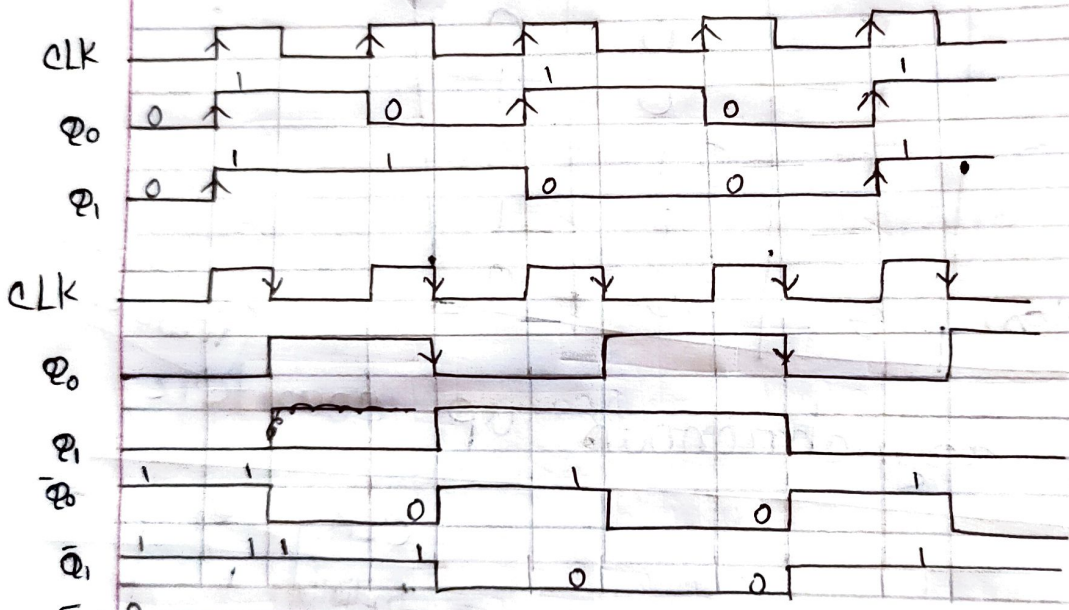
CLK	Q_1	Q_0
Initially	1	1
1	0	0
2	0	1
3	1	0
4	1	1
5	0	0

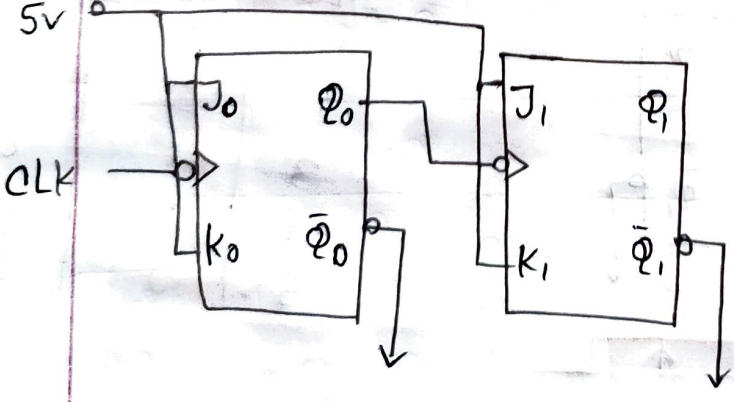


2-bit asynchronous down counter

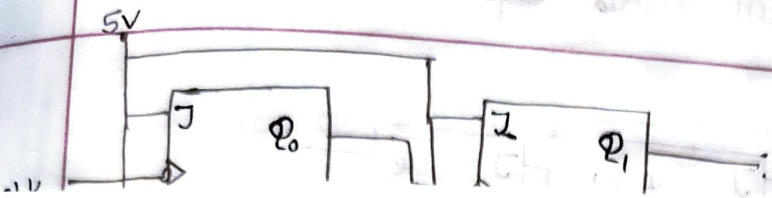


CLK	Q_1	Q_0
Initially	0	0
1	1	1
2	1	0
3	0	1
4	0	0
5	1	1



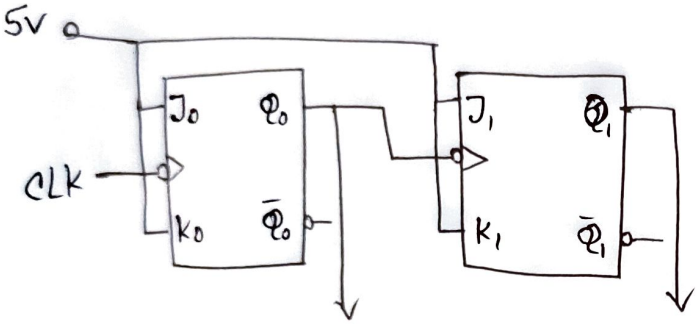


2-bit asynchronous down counter

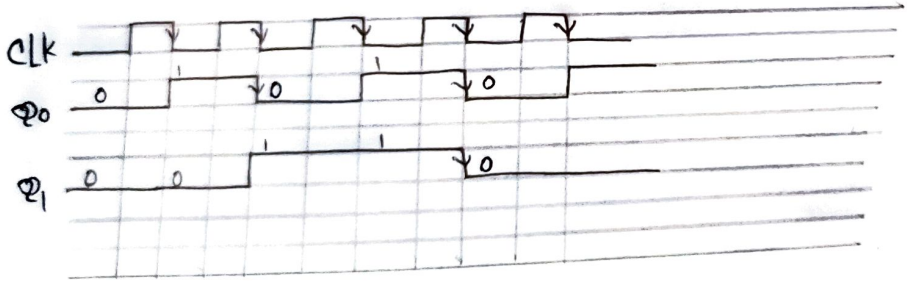


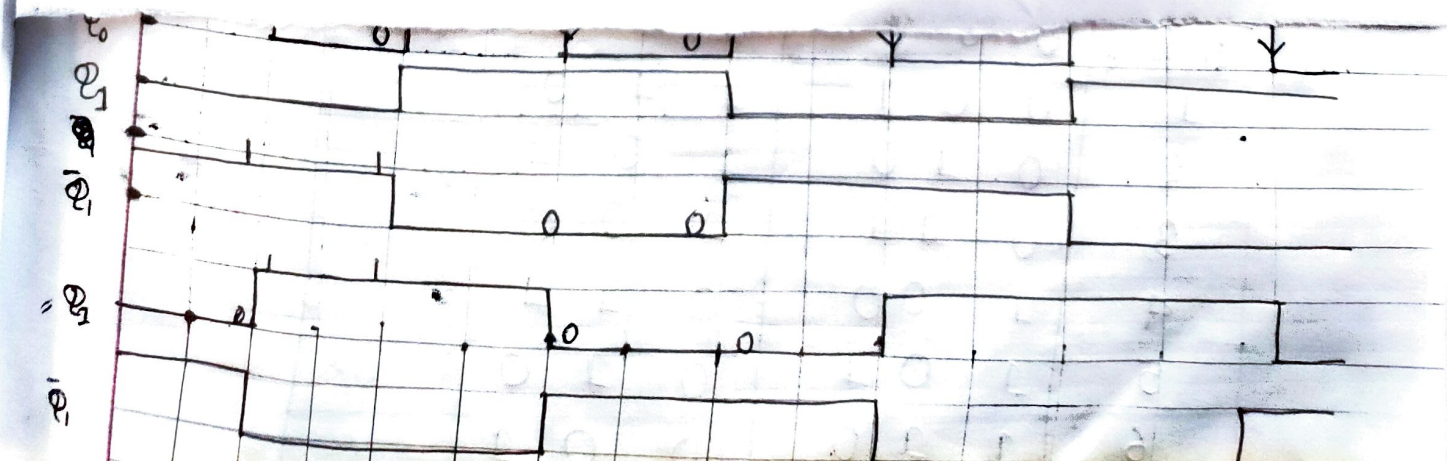
	up		down	
CLK	Q_1	Q_0	$\bar{Q}_1$	$\bar{Q}_0$

2-bit asynchronous up counter

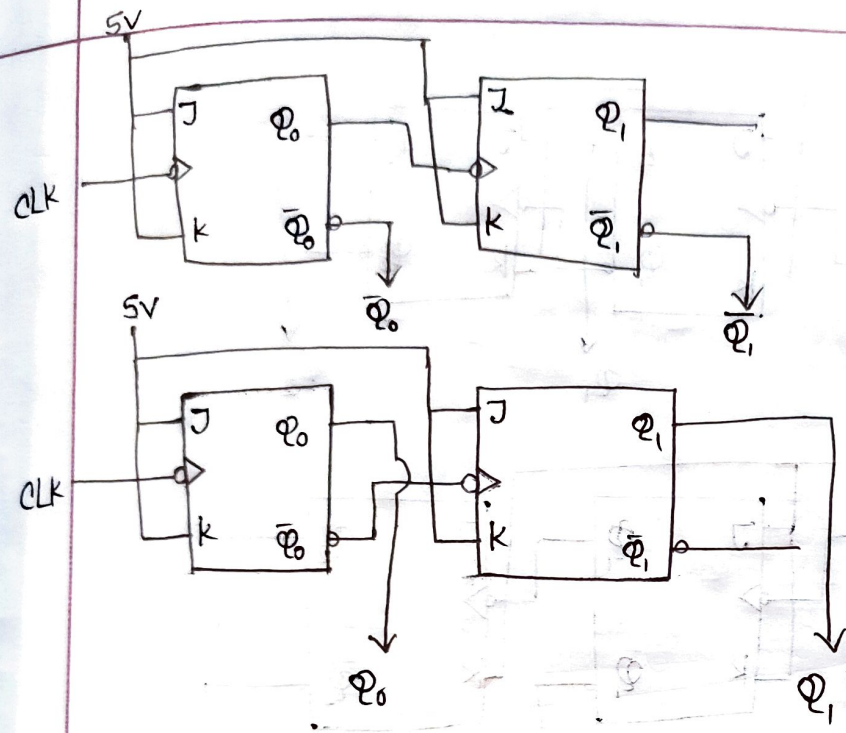


CLK	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4	0	0

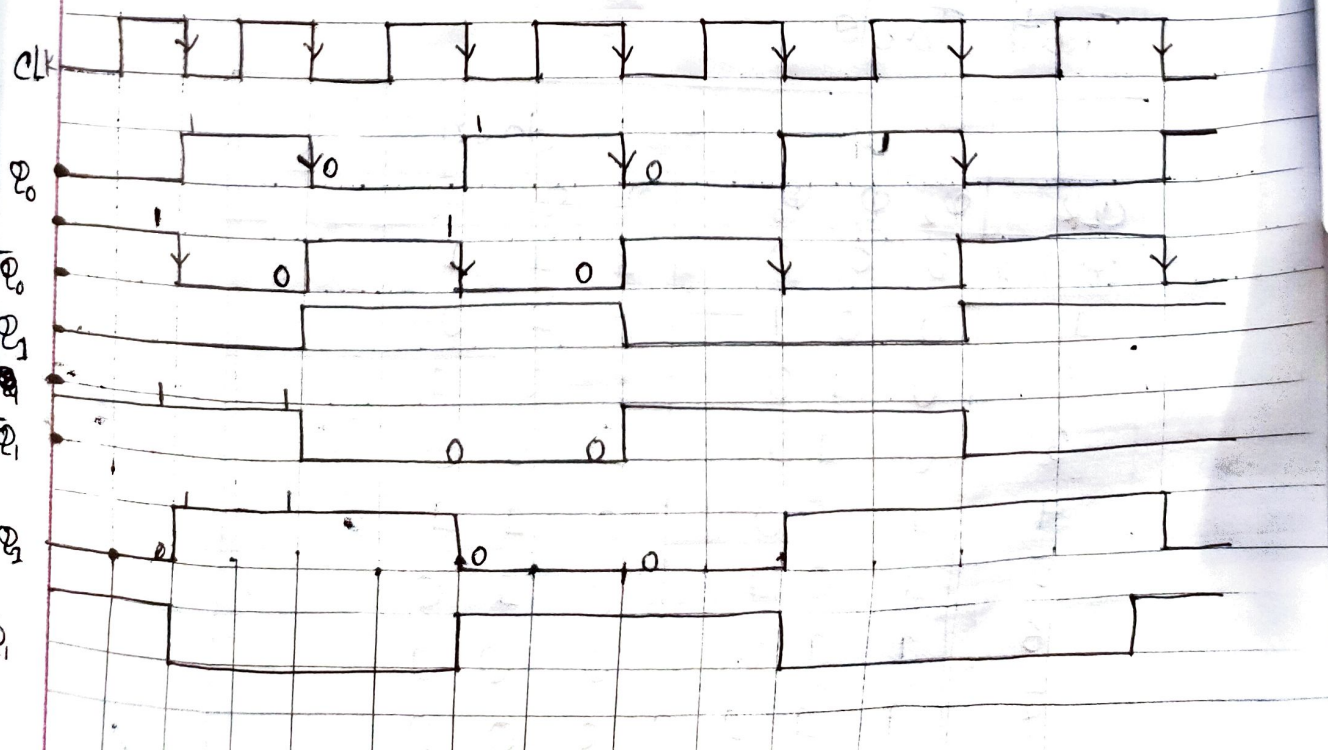




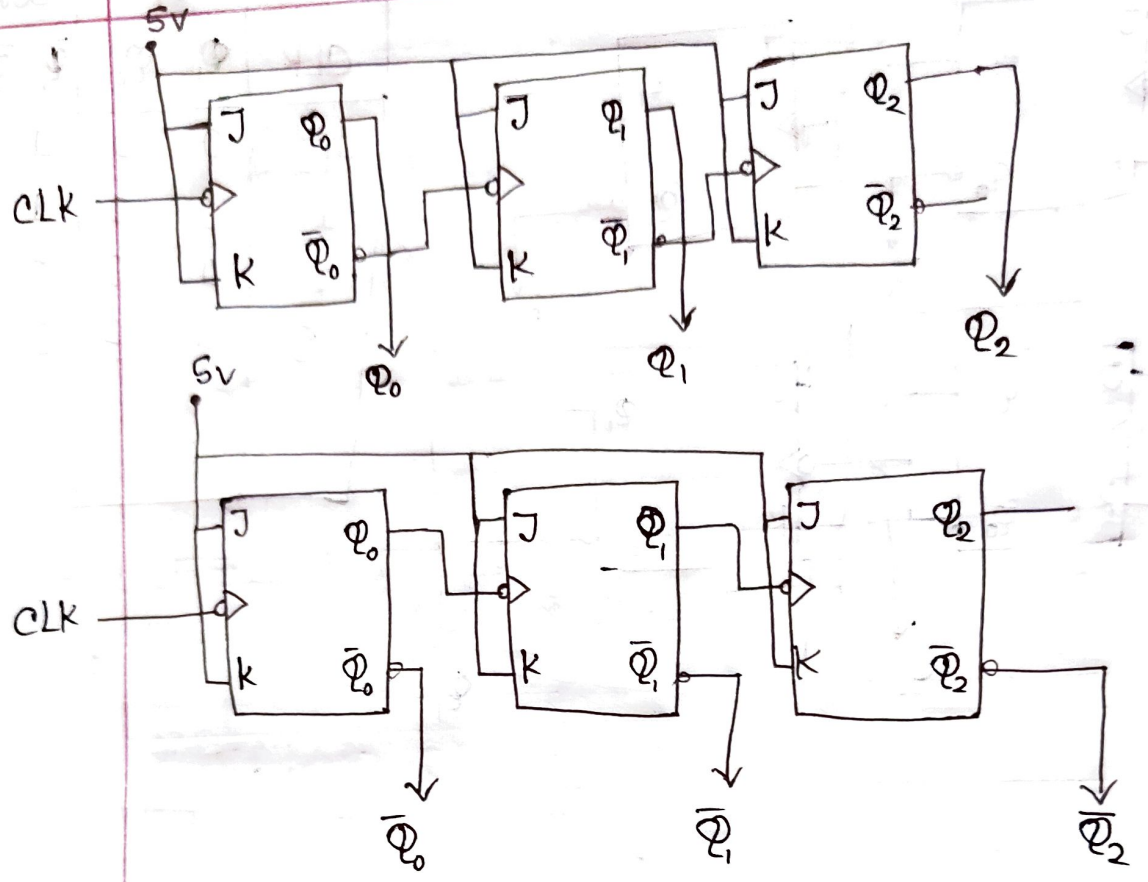
2-bit asynchronous down counter



	up		down	
CLK	Q_1	Q_0	$\bar{Q}_1$	$\bar{Q}_0$
Initially	0	0	1	1
1	0	1	1	0
2	1	0	0	1
3	1	1	0	0

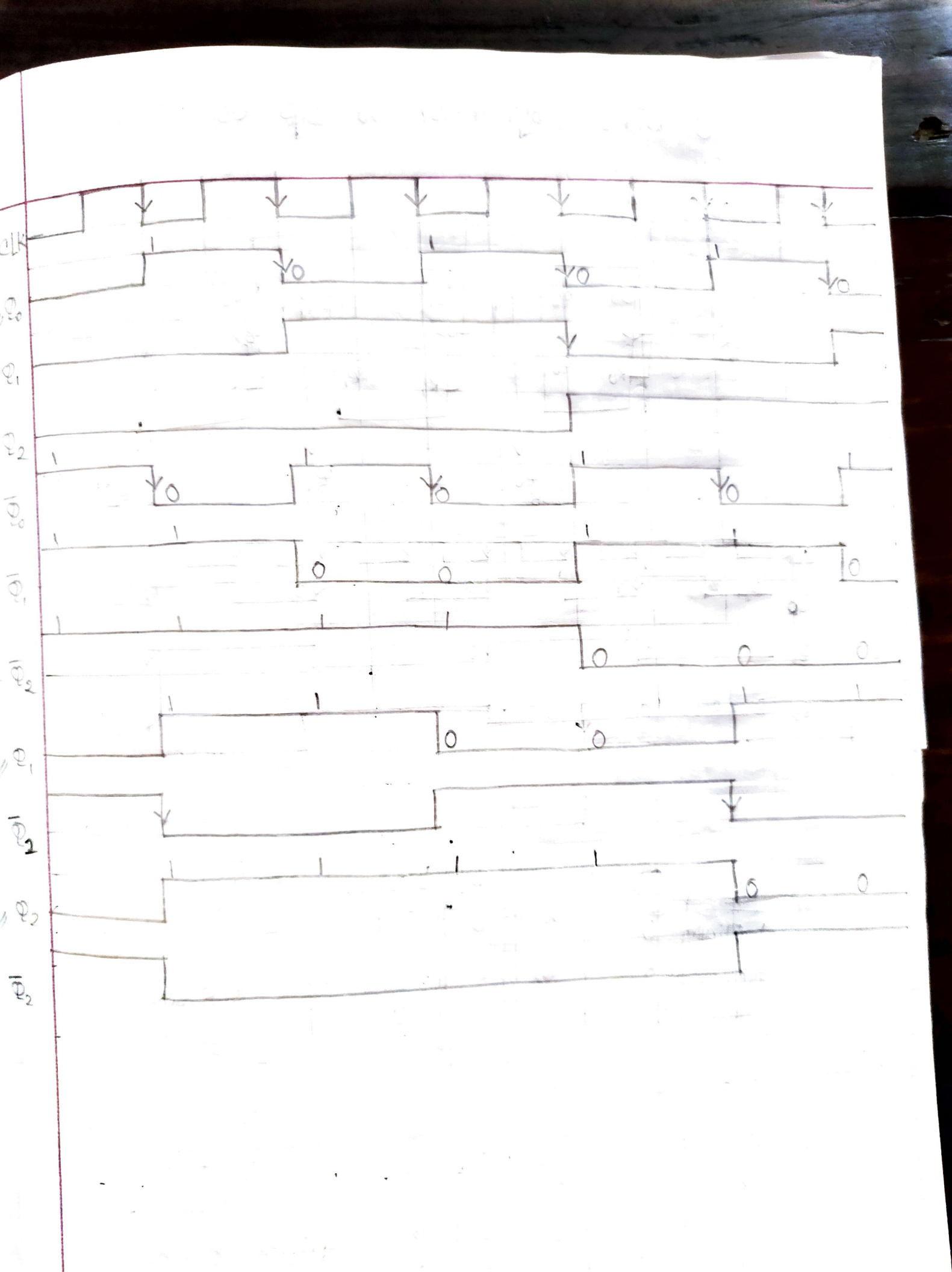


3-bit asynchronous ~~up~~ down counter

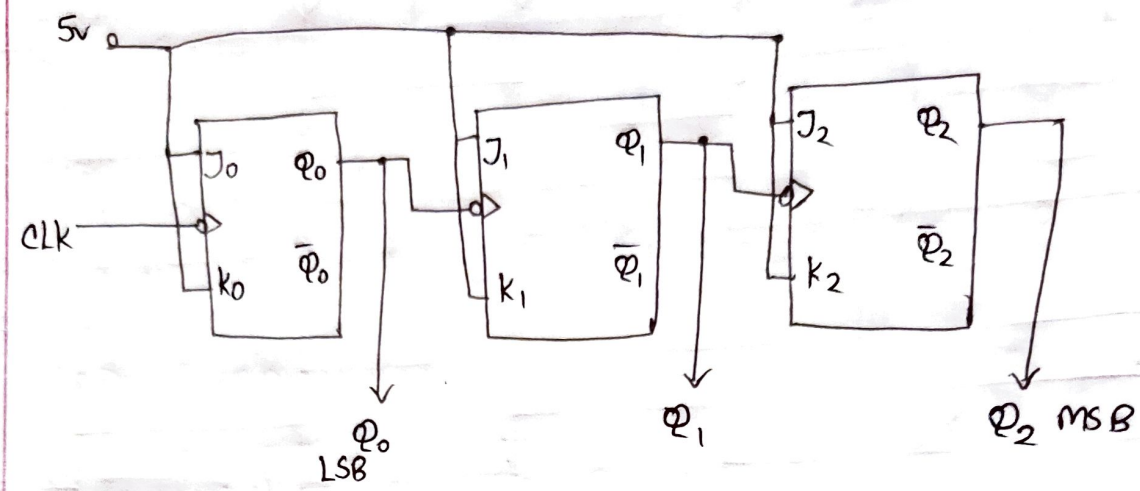


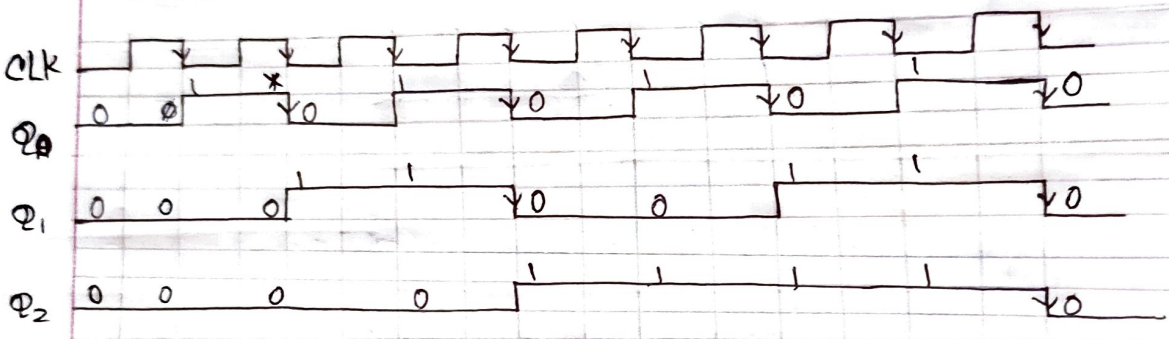
Truth Table

	up			down		
CLK	Q_2	Q_1	Q_0	$\bar{Q}_2$	$\bar{Q}_1$	$\bar{Q}_0$
Initially	0	0	0	1	1	1
1	0	0	1	1	1	0
2	0	1	0	1	0	1
3	0	1	1	1	0	0
4	1	0	0	0	1	1
5	1	0	1	0	1	0
6	1	1	0	0	0	1
7	1	1	1	0	0	0


CLK
Q_0
Q_1
Q_2
$\bar{Q}_0$
$\bar{Q}_1$
$\bar{Q}_2$
Q_1
$\bar{Q}_2$
Q_2
$\bar{Q}_2$

3-bit asynchronous up counter.

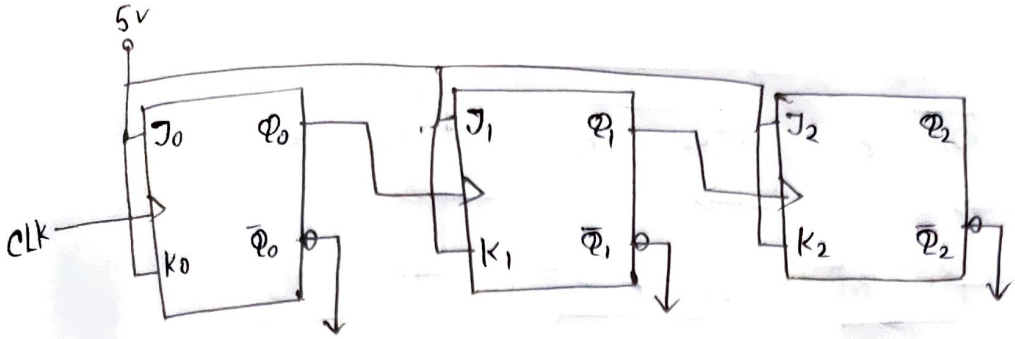


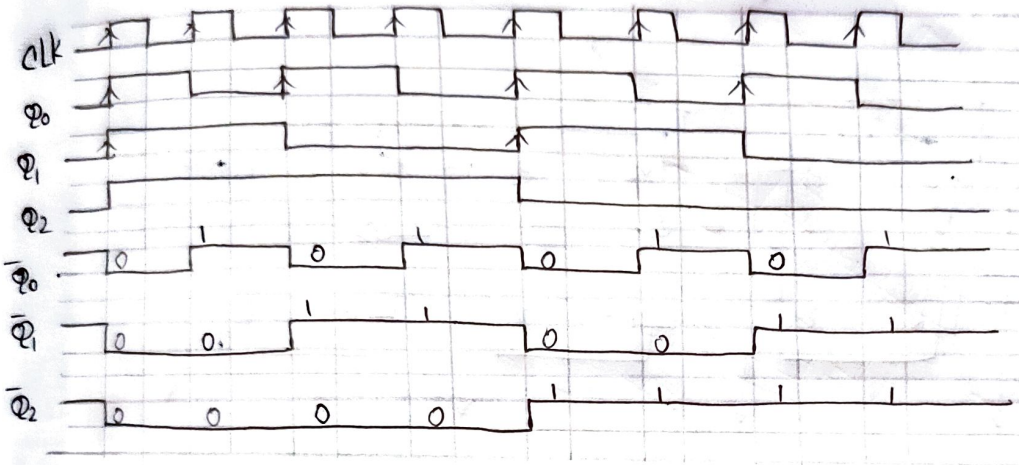


CLK	Q_2	Q_1	Q_0	Decimal equivalent
Initially	0	0	0	0
1st (↓)	0	0	1	1
2nd	0	1	0	2
3rd	0	1	1	3
4th	1	0	0	4
5th	1	0	1	5
6th	1	1	0	6
7th	1	1	1	7
8th	0	0	0	0

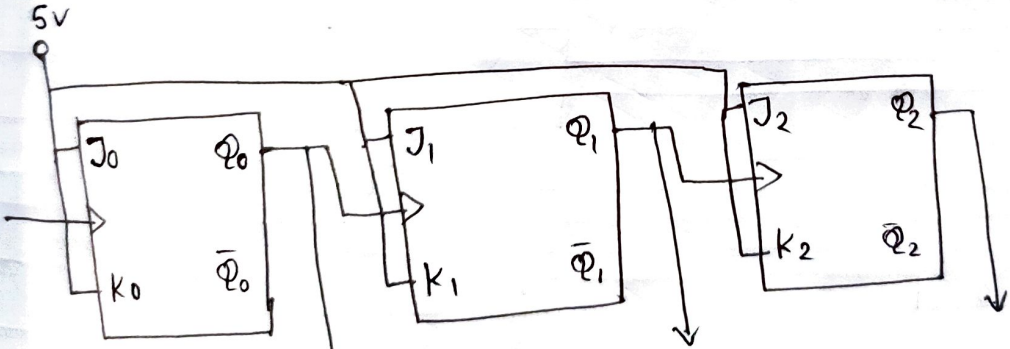
8 states
$= 2^3$
3 bit
3 flip-flop needed.
Max count $= 2^n - 1$
$= 8 - 1$

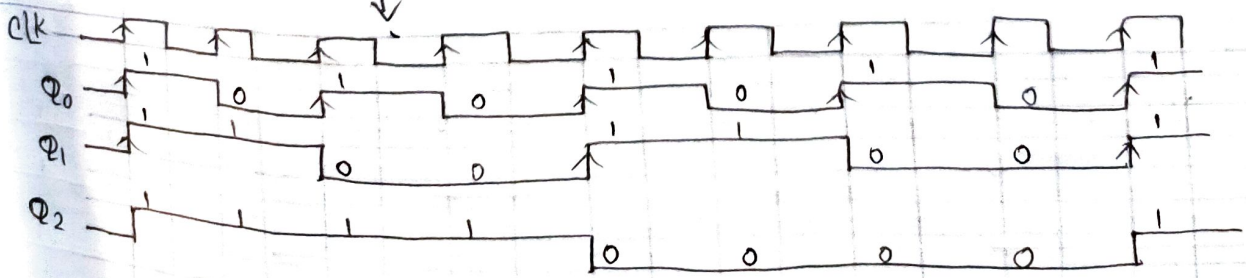
3 bit asynchronous up counter



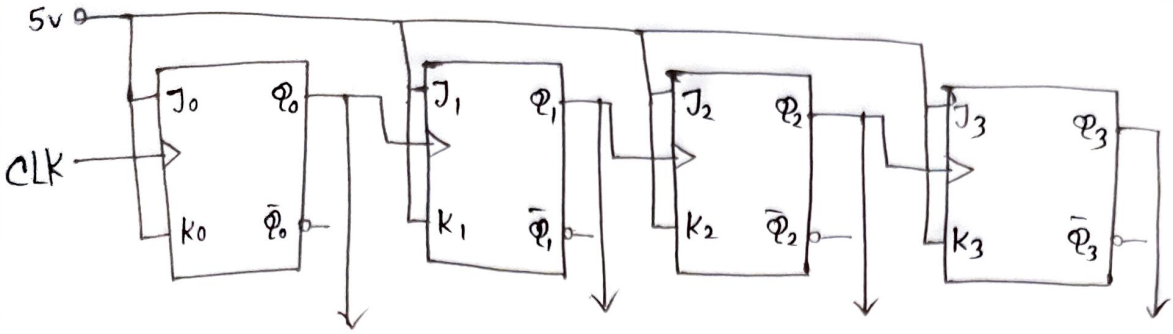


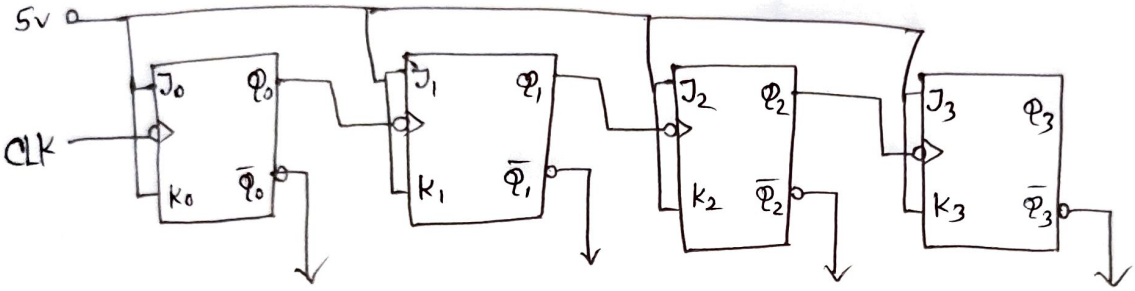
3 bit asynchronous down counter

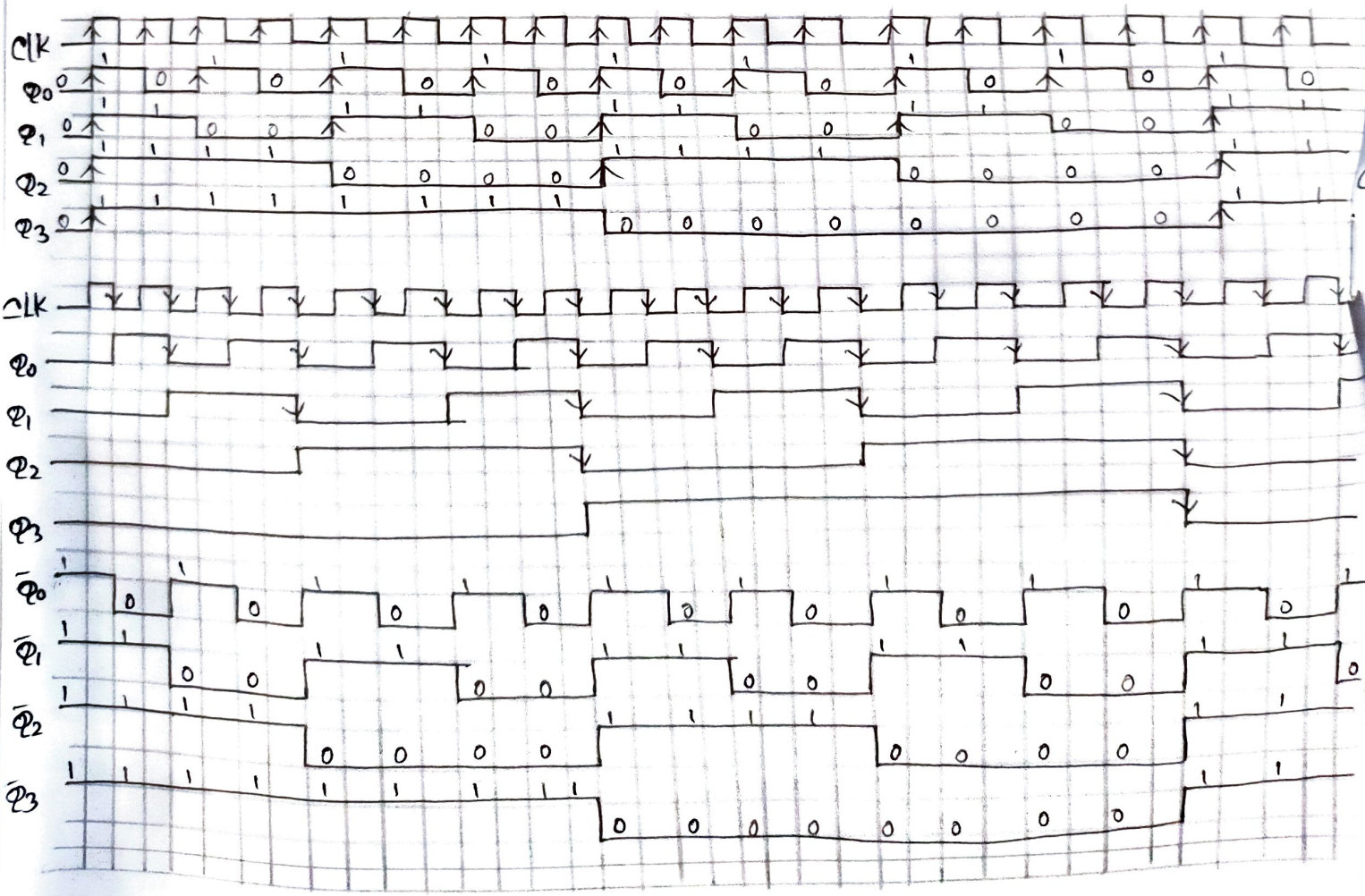




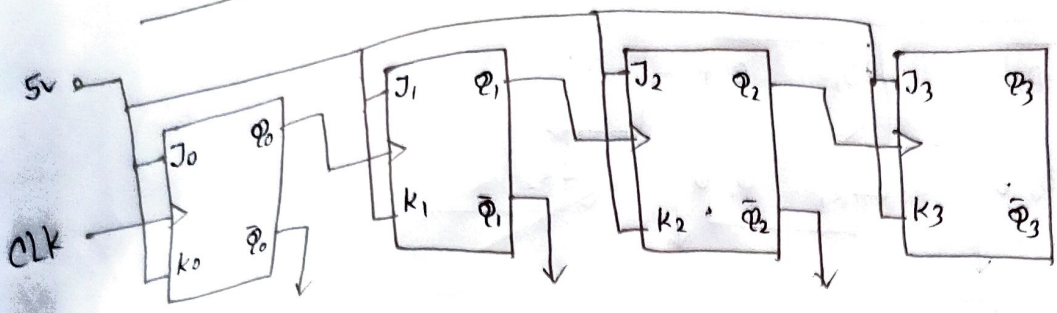
4-bit asynchronous down counter

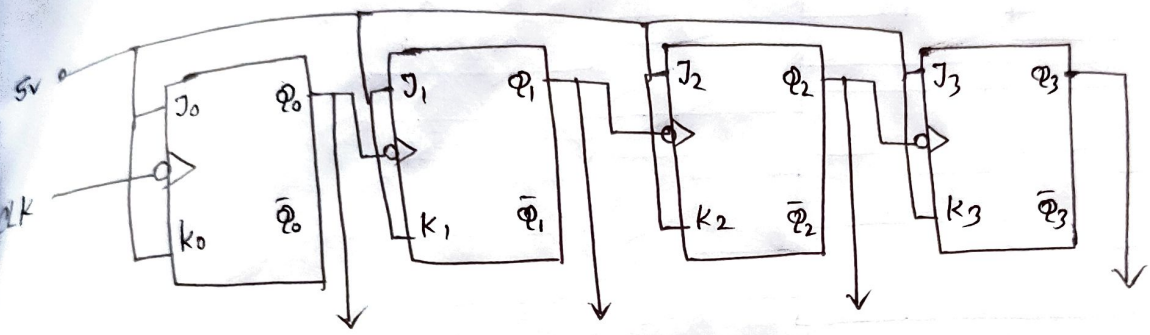


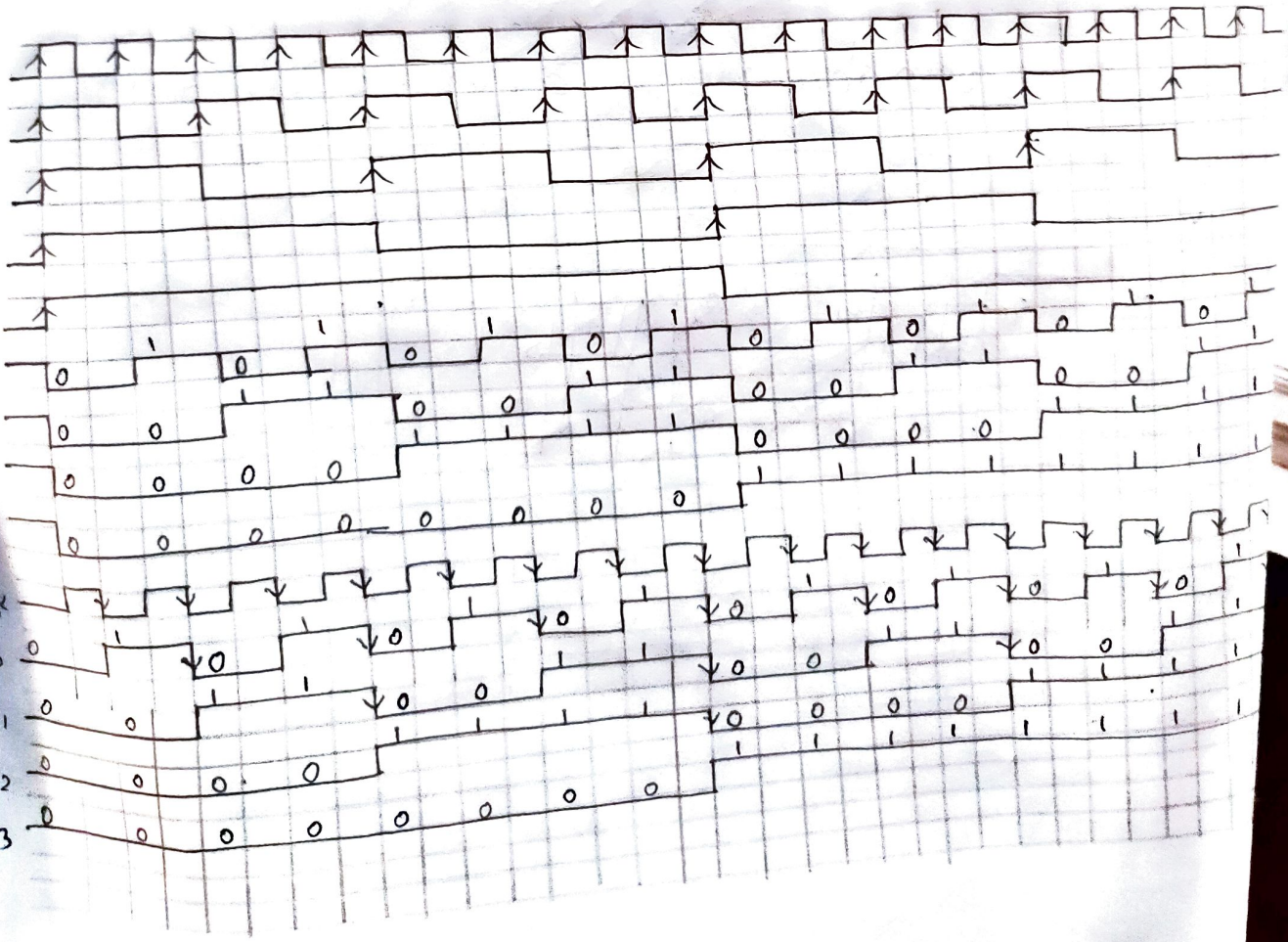




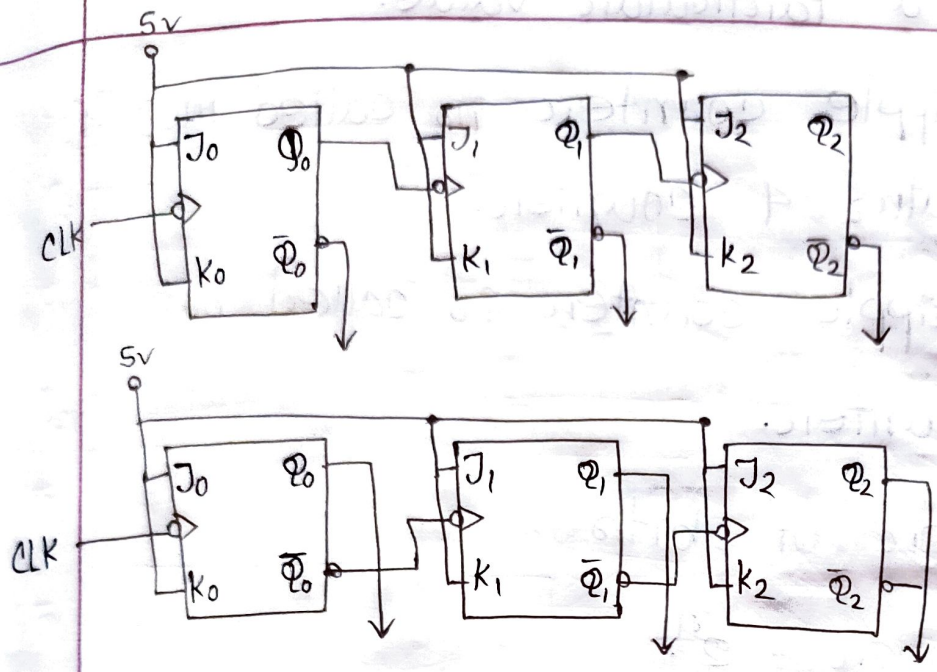
4-bit asynchronous up counter

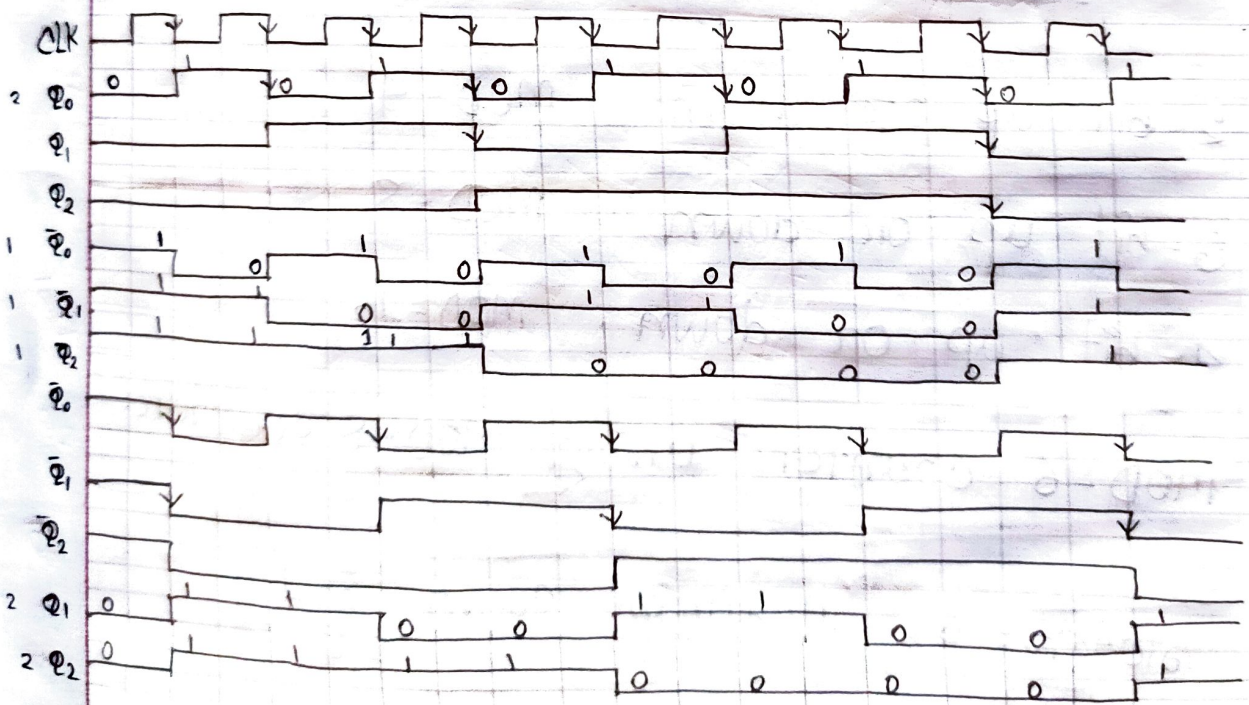






3-bit asynchronous down counter.





CLK	Q_2	Q_1	Q_0	$\bar{Q}_2$	$\bar{Q}_1$	$\bar{Q}_0$
Initially	0	0	0	1	1	1
1	0	0	1	1	1	0
2	0	1	0	1	0	1
3	0	1	1	1	0	0
4	1	0	0	0	1	1
5	1	0	1	0	1	0
6	1	1	0	0	0	1
7						

Modulus of the counter & counting up to a particular value.

⇒ 2-bit ripple counter is called MOD-4 or modulus 4 counter.

⇒ 3-bit ripple counter is called is MOD-8 counter.

n = number of bits

MOD number = 2^n.

2-bit up or down MOD-4

3-bit up or down MOD-8

4-bit up or down MOD-16

* MOD-6 counter using MOD-8 counter.

=

states = 6, Maximum count = 6-1
= 5

0	0 0 0	MOD-6
1	0 0 1	
2	0 1 0	
3	0 1 1	
4	1 0 0	
5	1 0 1	
	1 1 0	X
	1 1 1	

CLK

CLK

Q_0

Q_1

Q_2

Connect $\overline{PRE}$ so that $\overline{PRE} = 1$ & never the counter is set, $Q \neq 1$.

If PRE = 0, $Q = 1$

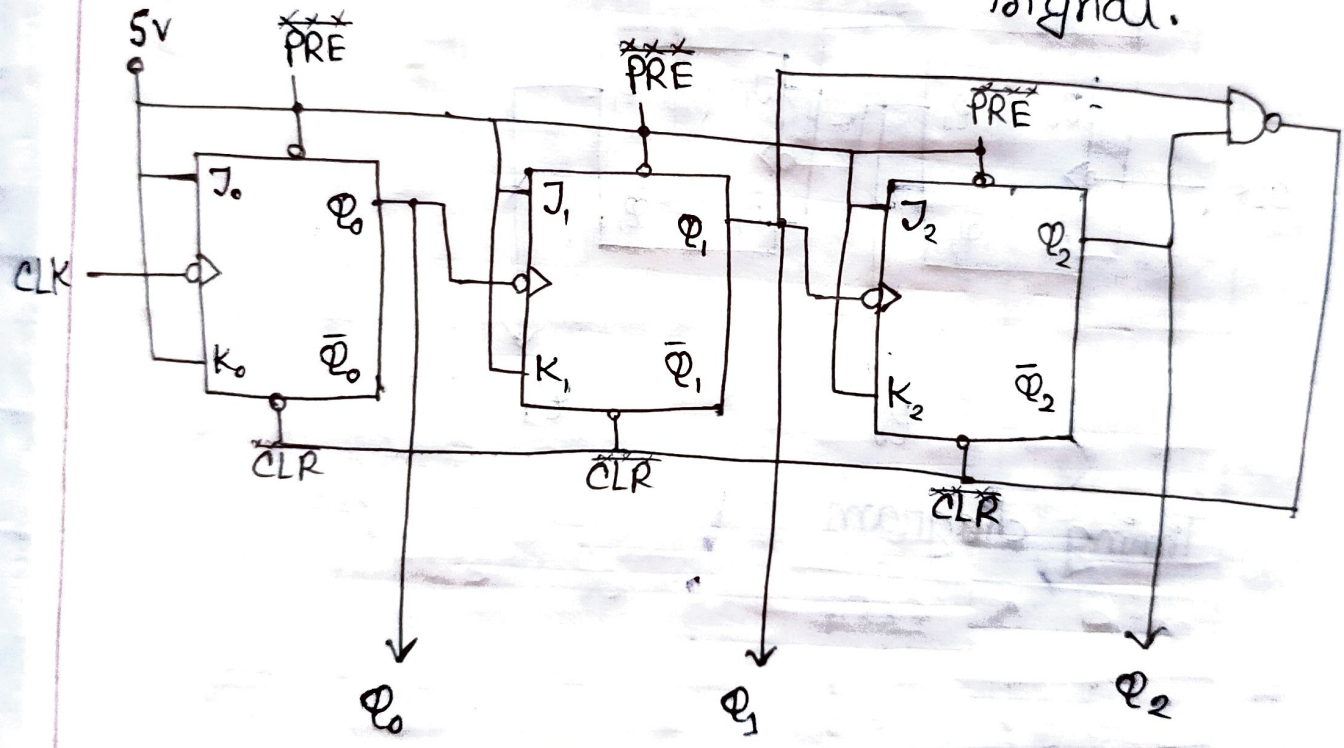
CLR = 0, $Q = 0$

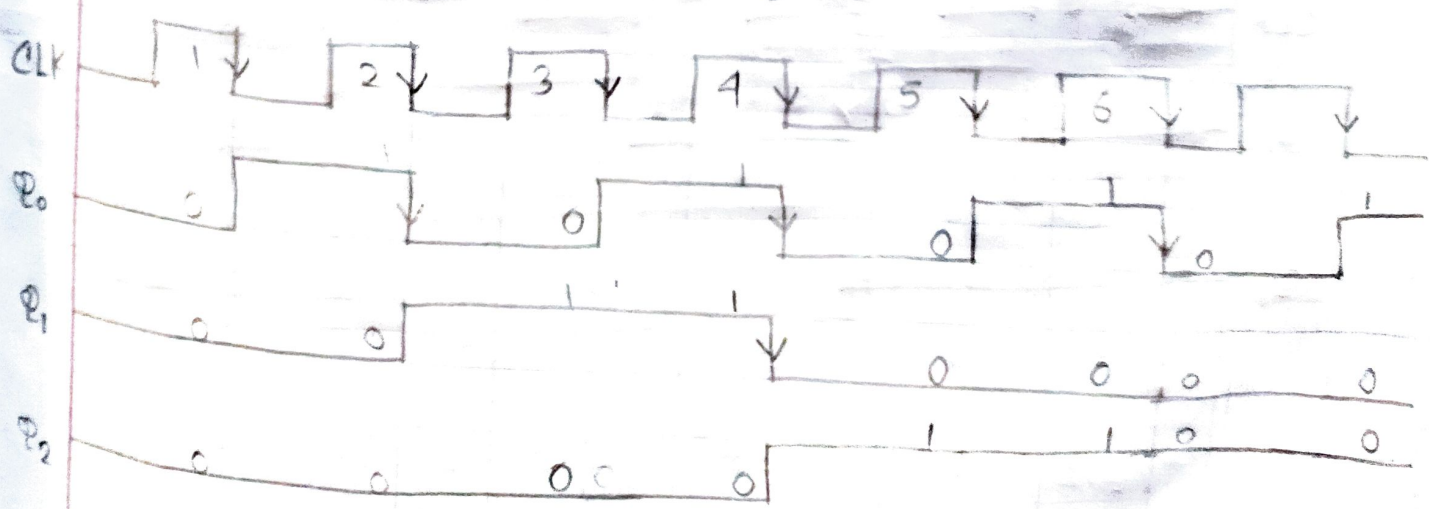
$\overline{PRE} = 1$, $Q = 1$

$\overline{CLR} = 1$, $Q = 0$

$\overline{PRE}$ ○ — Q / $\overline{CLR}$

bubble shows here means that they are active low signal.





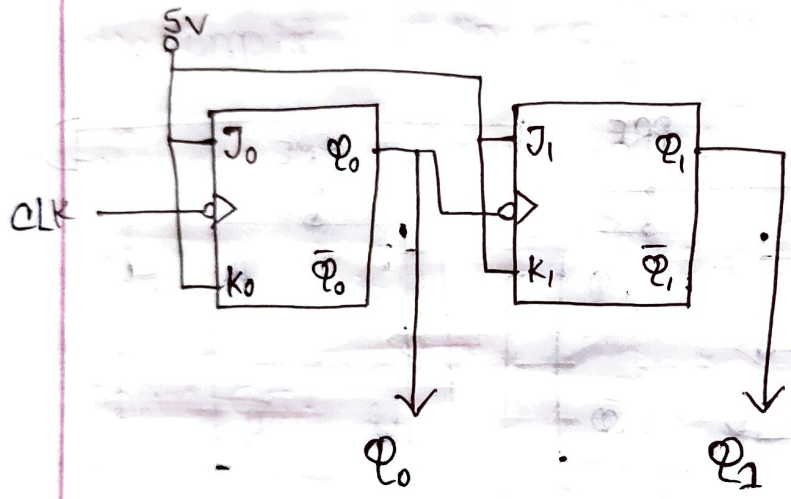
MOD-4 counter

	Q_1	Q_0
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

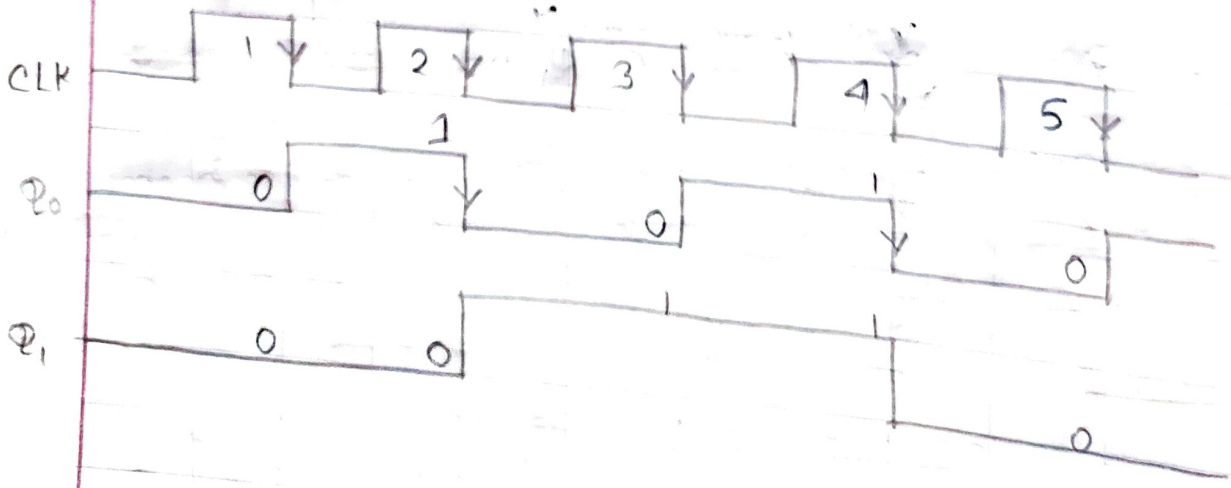
state = 4

Maximum count = 4 - 1
= 3

$2^2 = 4$



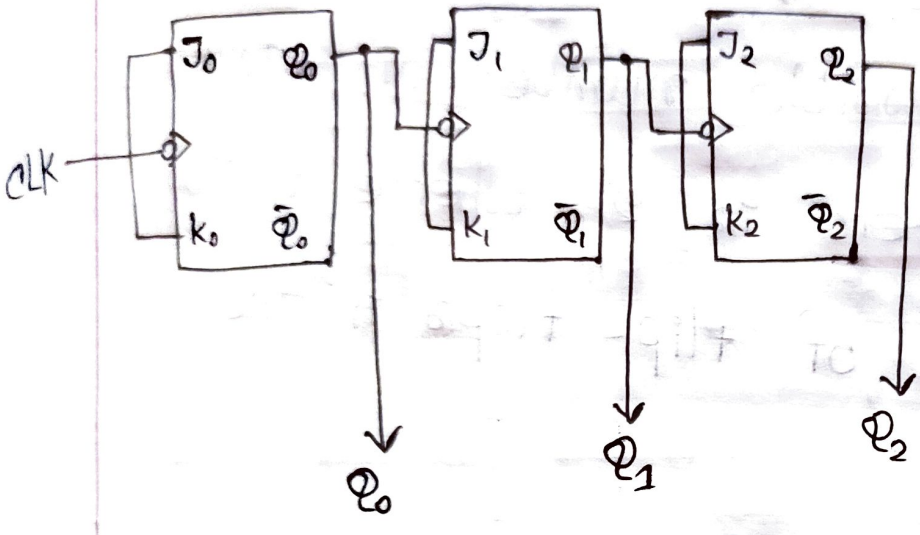
Timing diagram



MOD - 8 counter

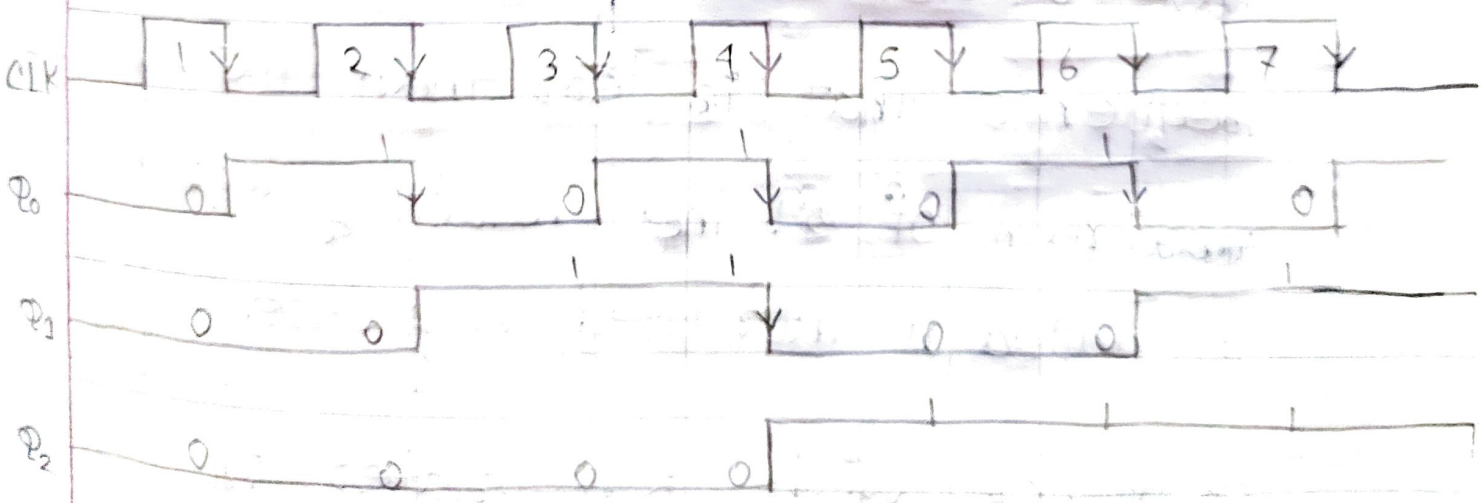
State = 8, Maximum count = 8 - 1
= 7

$2^3 = 8$, 3 flip-flops



	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
	0	0	0

Timing diagram



Asynchronous decade counters

The modulus of a counter is the number of unique states that the counter will sequence through. The maximum possible number of states of a counter is 2^n, where n is the number of flip-flops in the counter.

Counters can also be designed to have a number of states in their sequence that is less than the maximum of 2^n. The resulting sequence is called a truncated sequence.

Counters with ten states in their sequence are called decade counters. A decade counter with a

count sequence of zero through nine in a BCD decade counter because its ten state sequence produces the BCD code.

MOD-10 counter

	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0

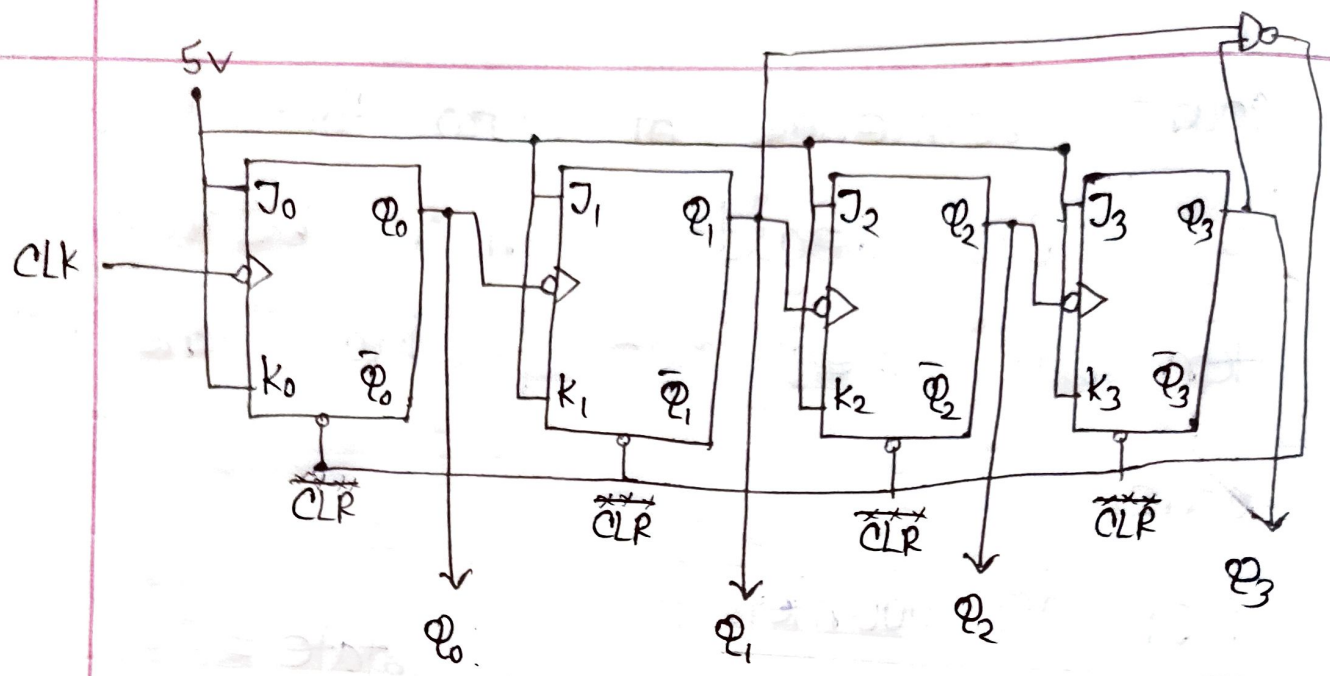
state = 10

Maximum count = 10-1
= 9

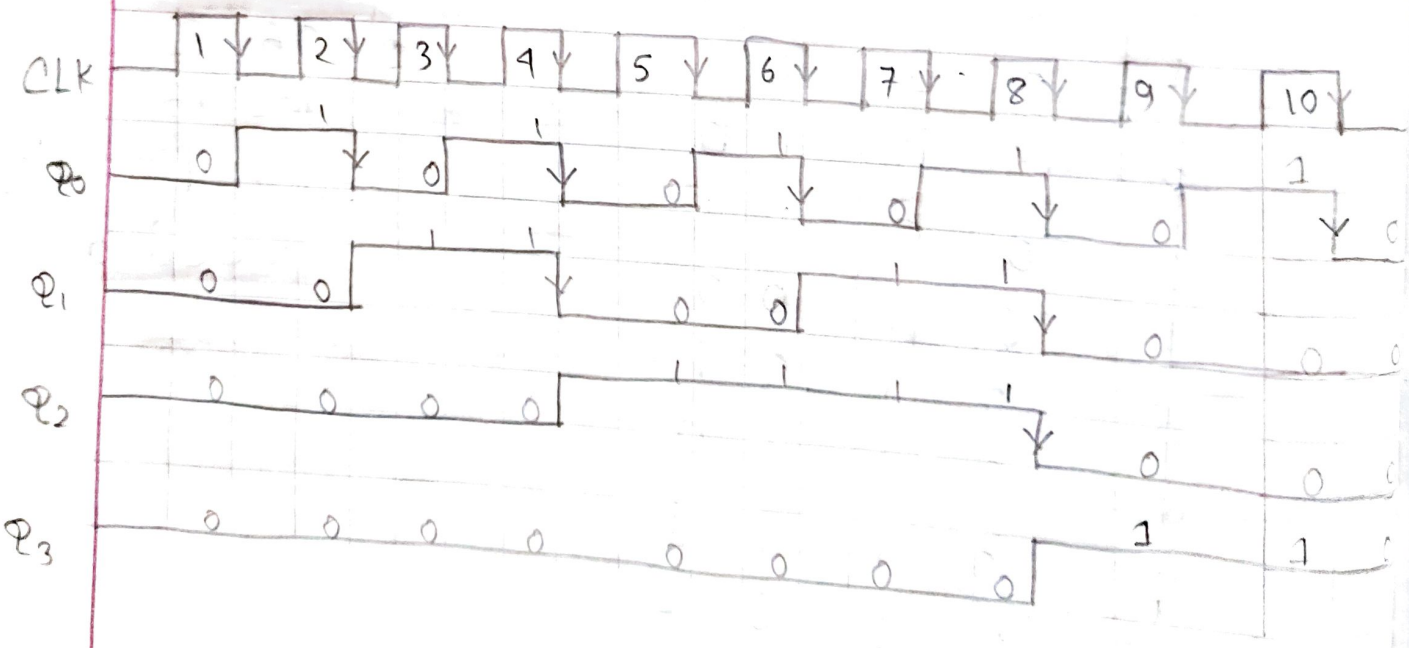
$2^4 = 16$

$10 < 16$

So, 4 flip-flops needed.



Timing diagram

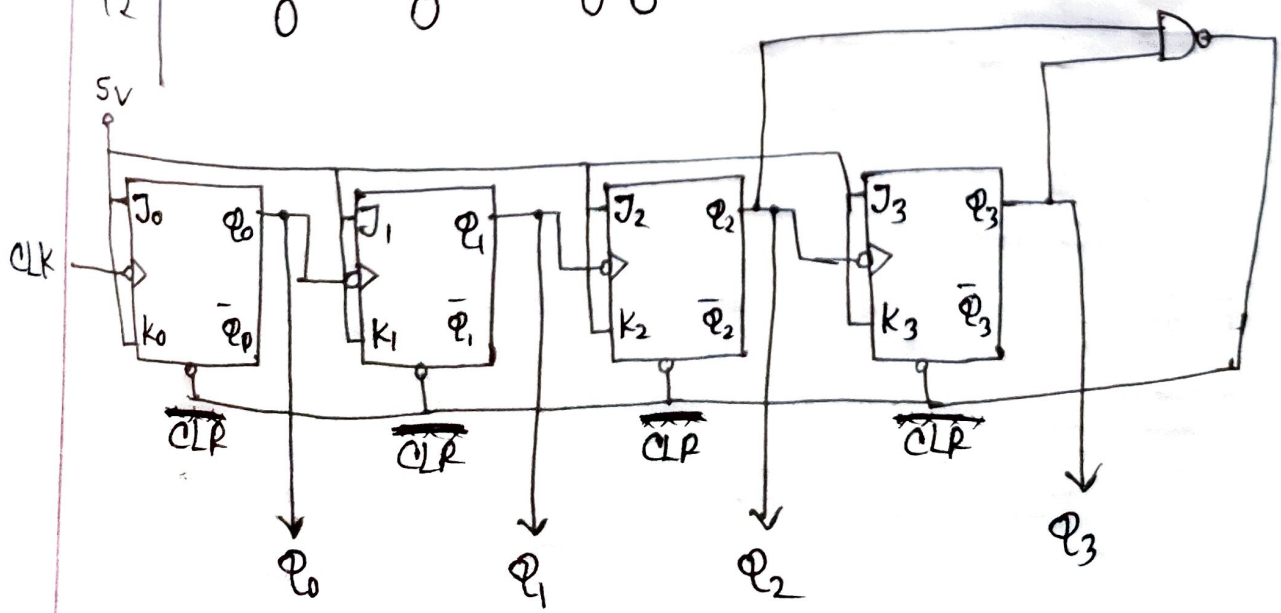


MOD-12 counter

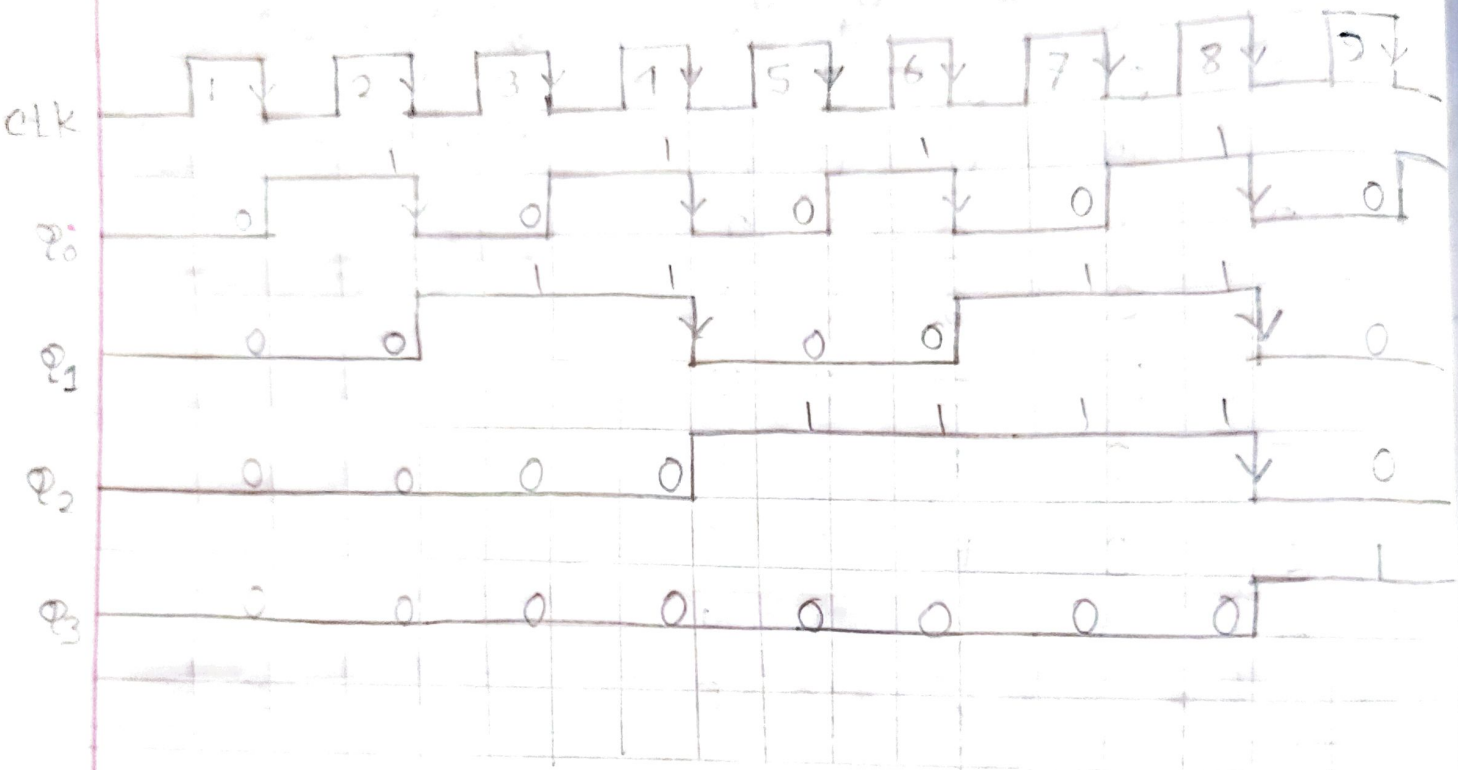
	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	0	0	0	0

State = 12,

Maximum count = 12-1
= 11.



Timing diagram



Important points

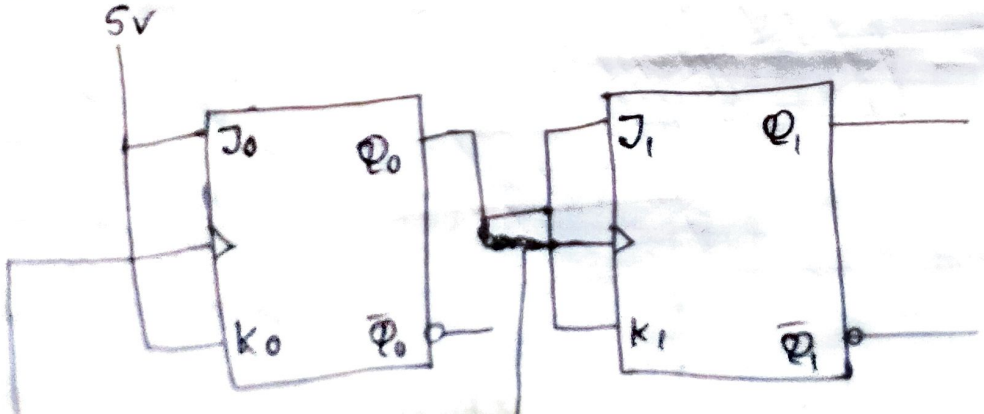
i) Negative edge triggered → Q in CLK → up counter

ii) Negative edge triggered → $\bar{Q}$ in CLK → down counter

iii) Positive edge triggered → $\bar{Q}$ in CLK → up counter

iv) Positive edge triggered → Q in CLK → down counter

2-bit synchronous up Binary counter



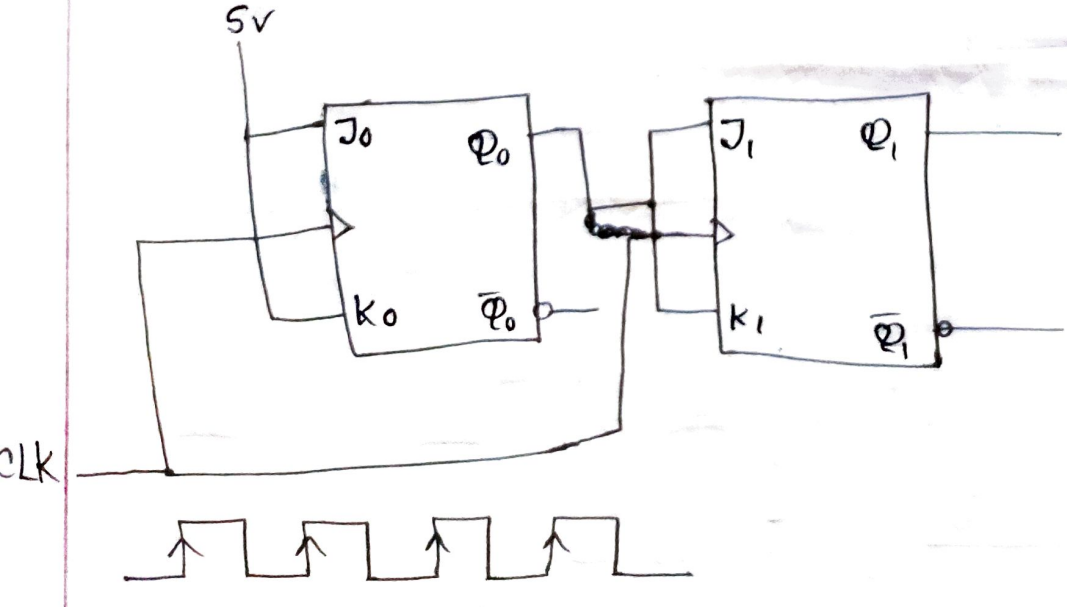
Section review

1) The term asynchronous mean in relation to counters is that each flip-flop after the first one is enabled by the output of the preceding flip-flop

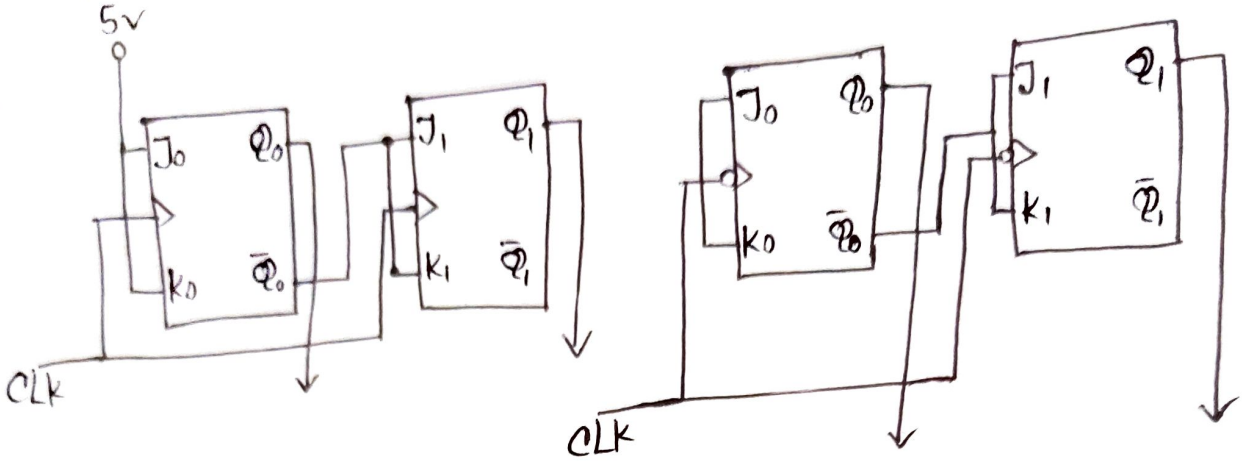
2) Modulus - 14 counter have 14 states. Minimum 4 flip-flops are required.

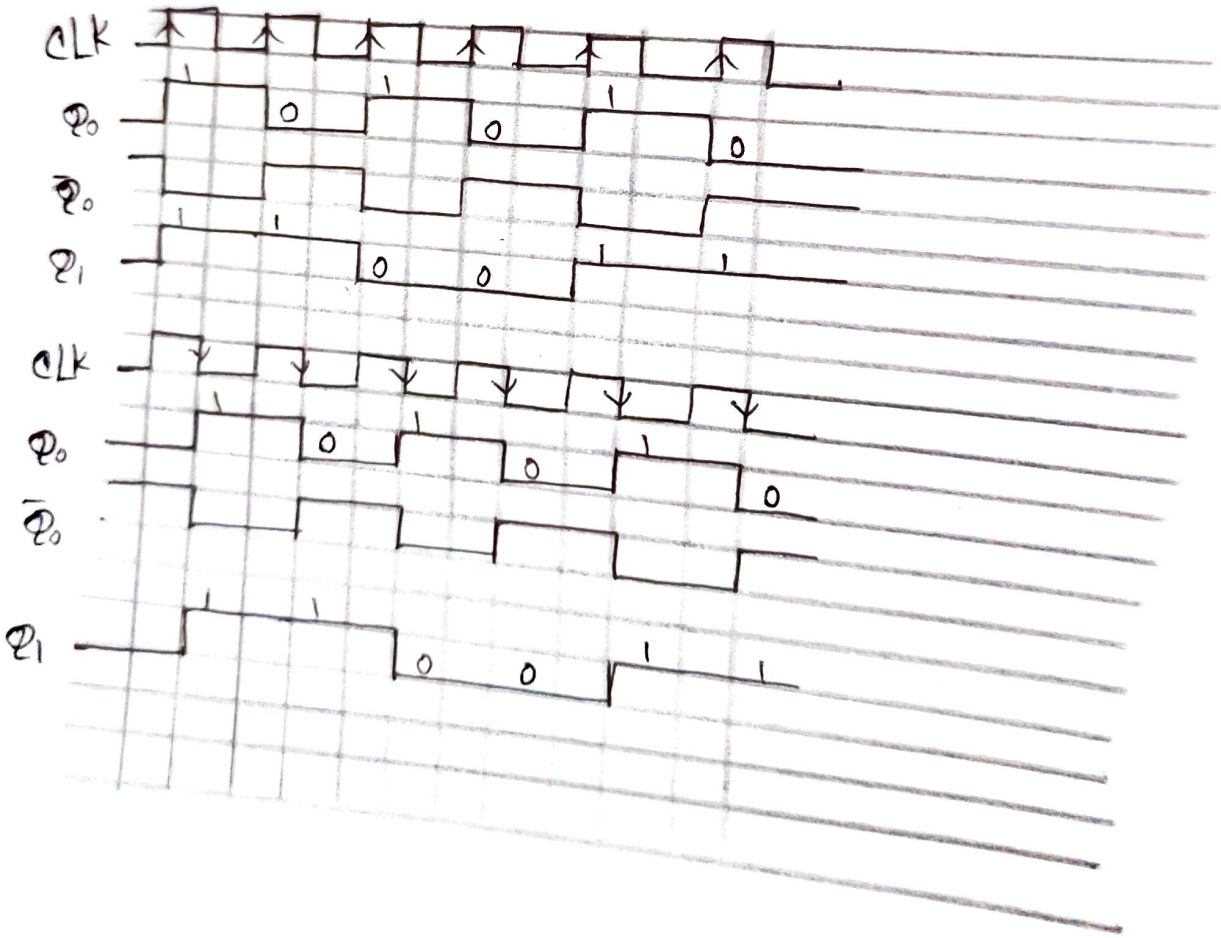
Synchronous counter operation

2-bit synchronous UP Binary counter



2-bit synchronous down counter

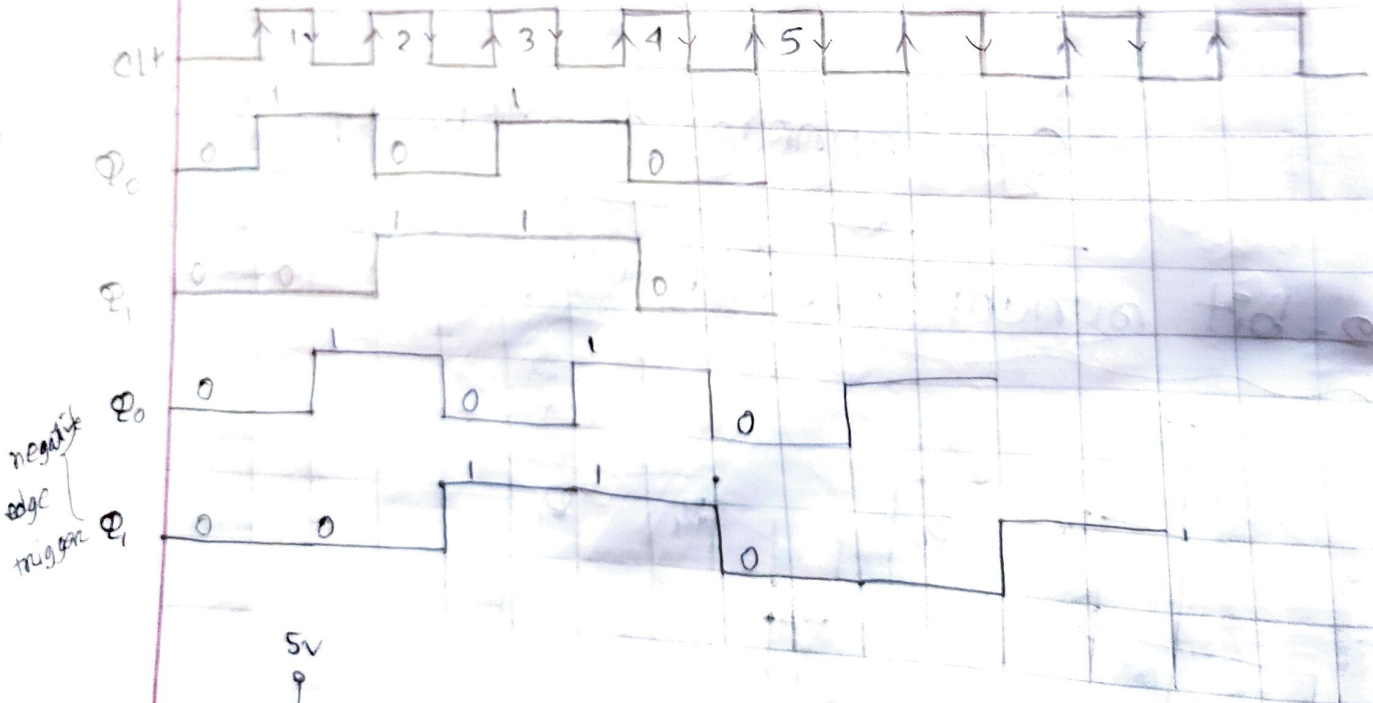


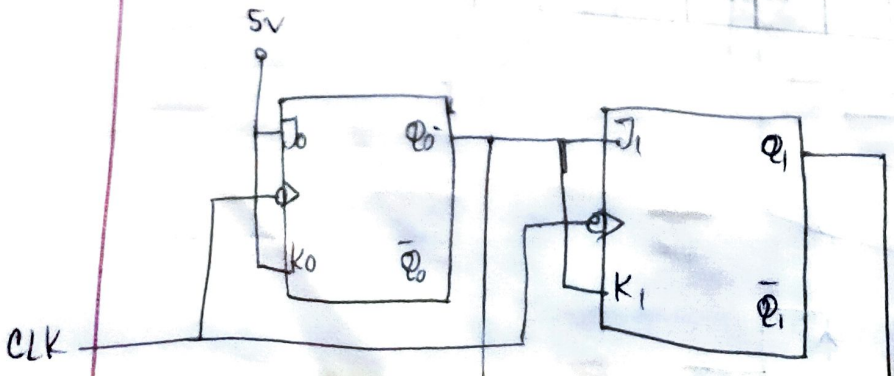


up counter

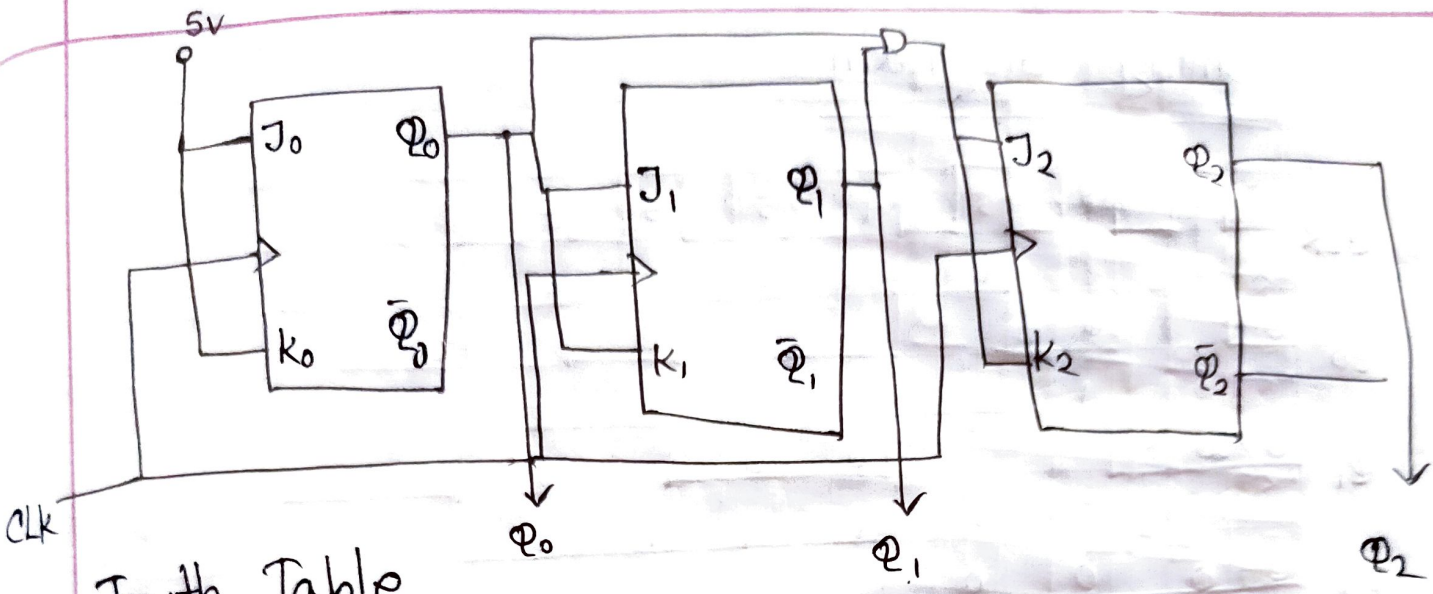
CLK	Q_1	Q_0
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

Timing diagram





A 3-bit synchronous UP Binary counter

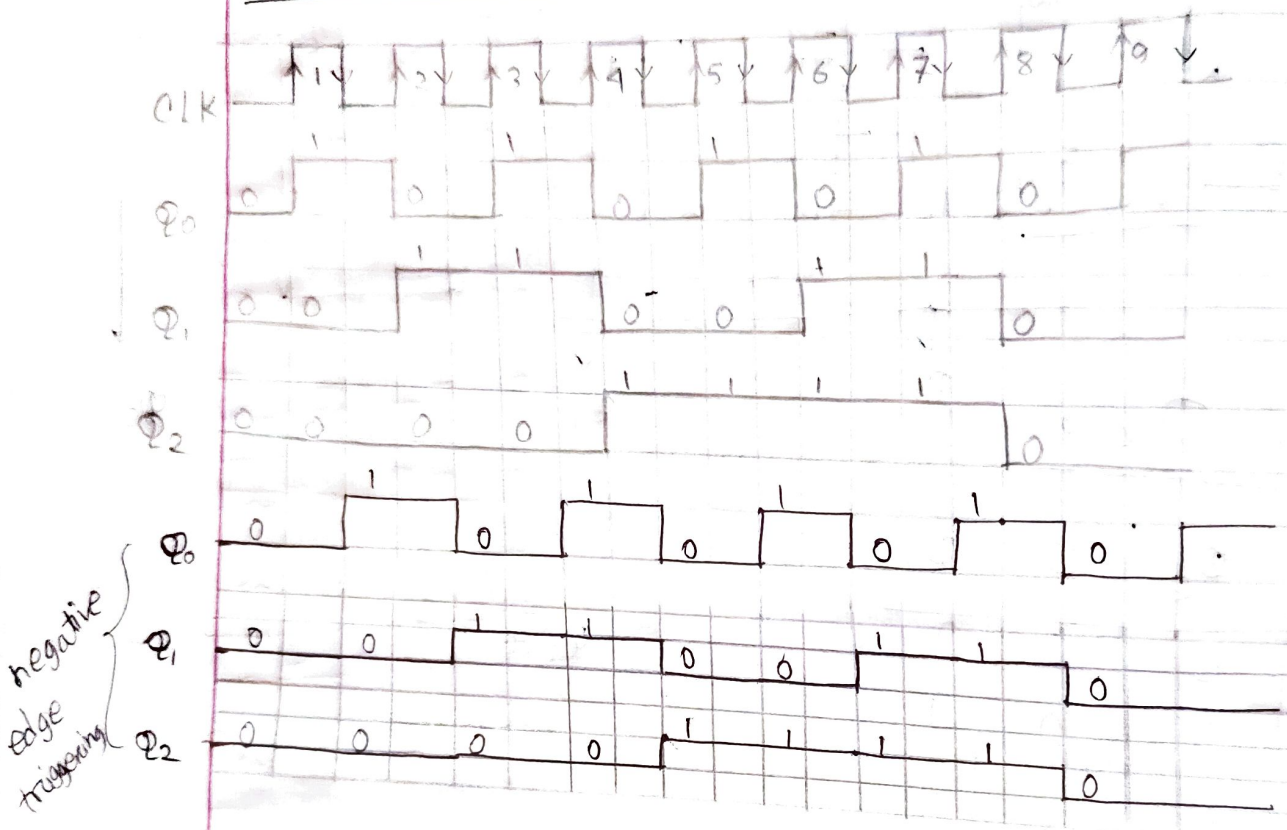


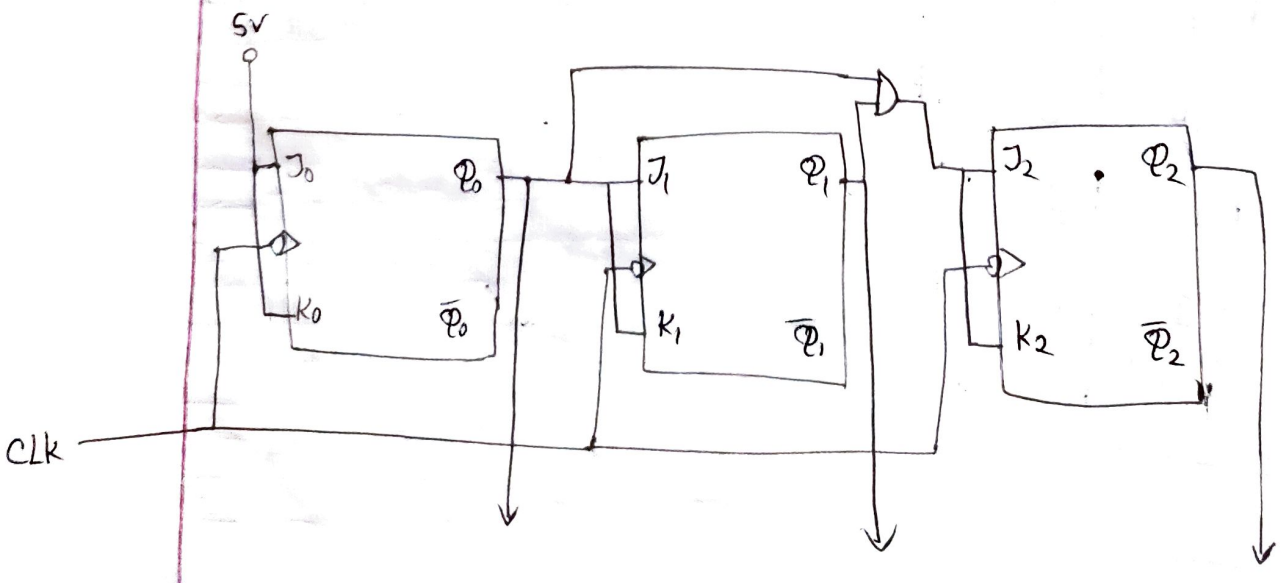
Truth Table

Clk	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1 ↙Toggle	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0 ↙Toggle	0	0

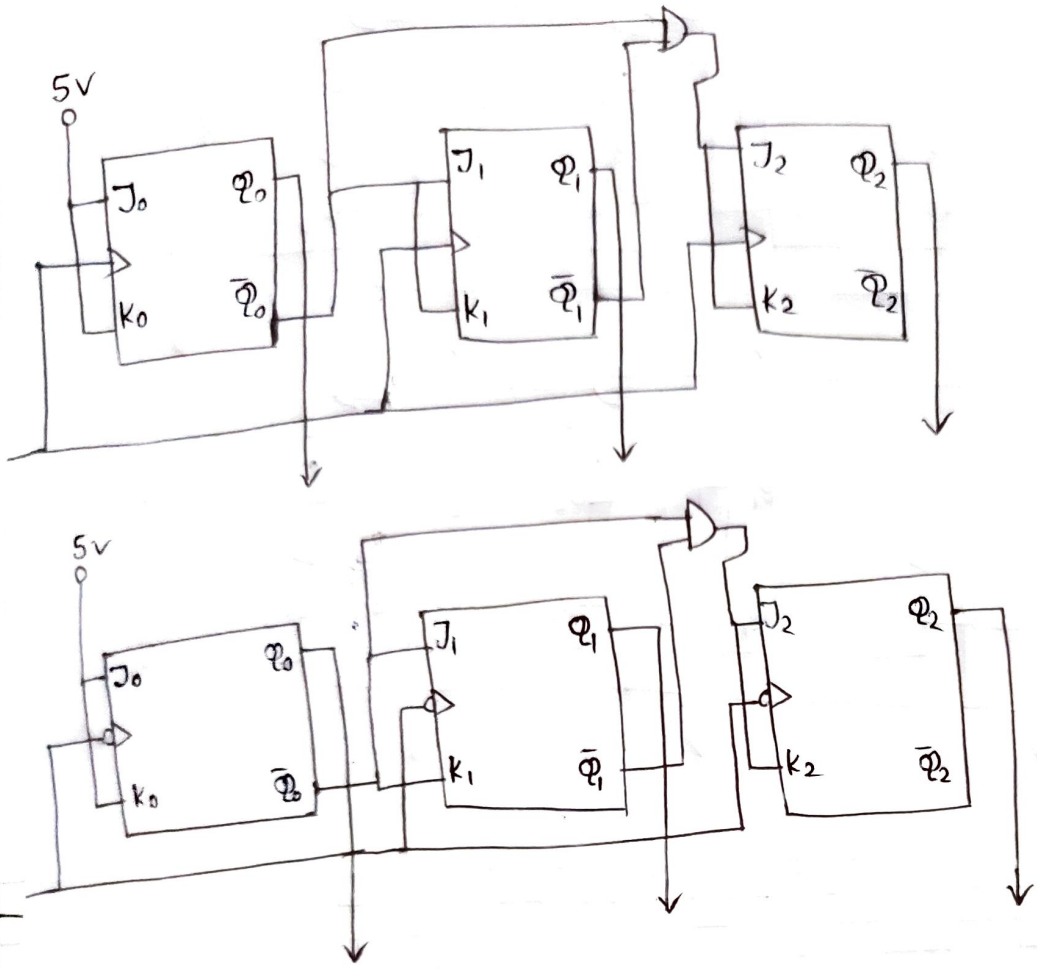
$Q_1Q_0 = 1$, then toggle

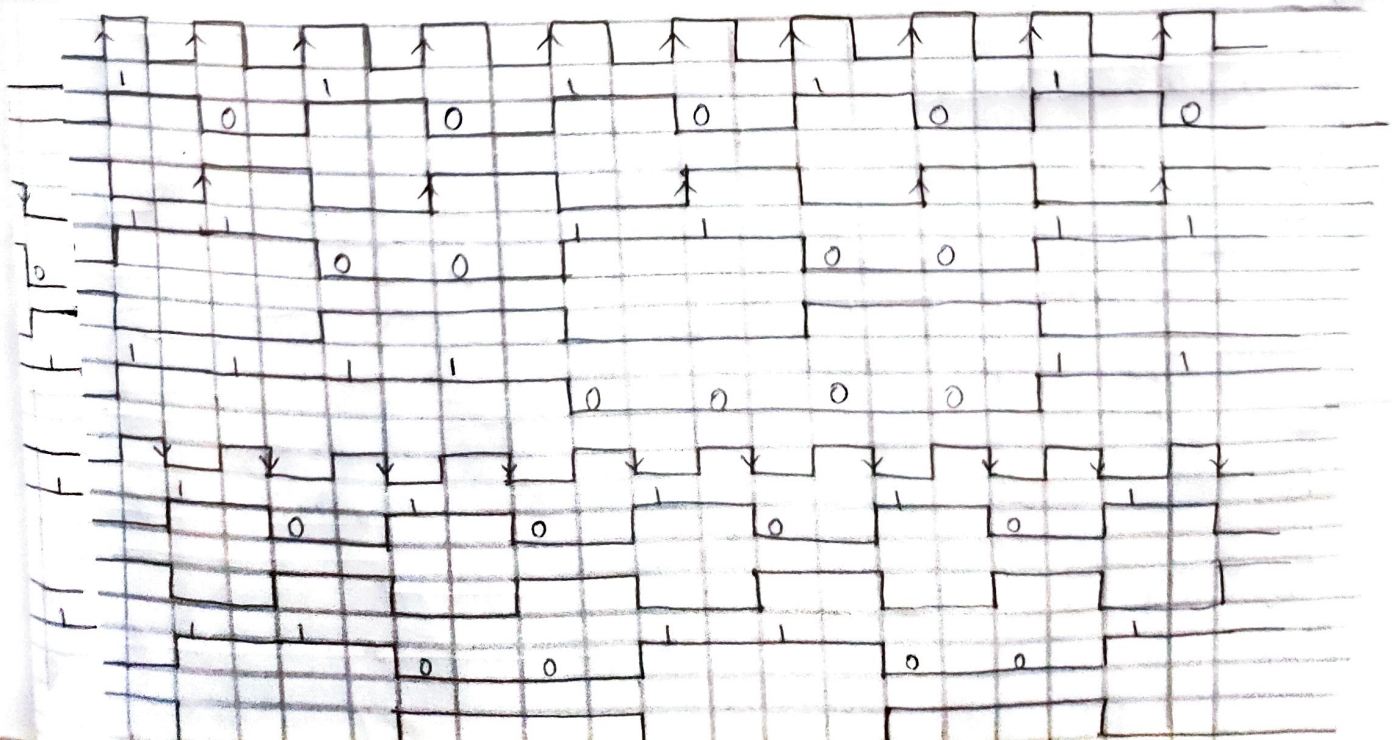
Timing diagram



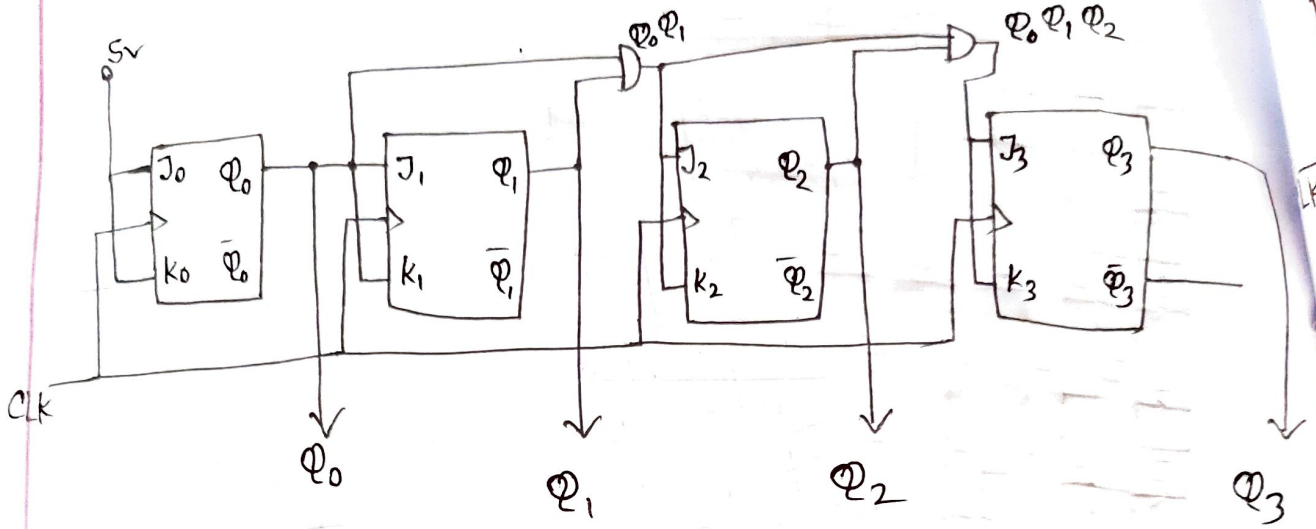


3-bit synchronous down counter





4-bit synchronous ↑ (up) Binary Counter



Truth Table

CLK	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
16	0	0	0	0

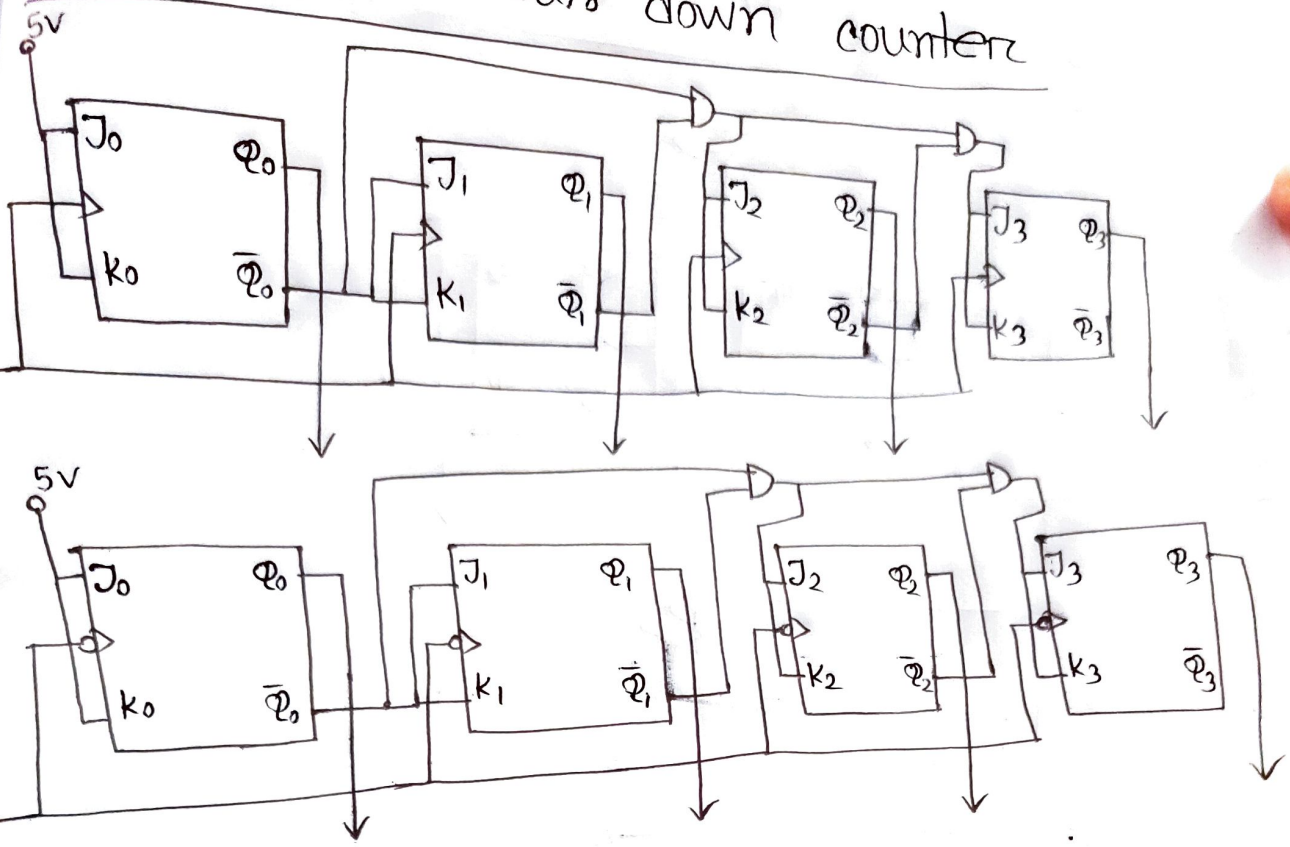
Q_2 Toggle, when

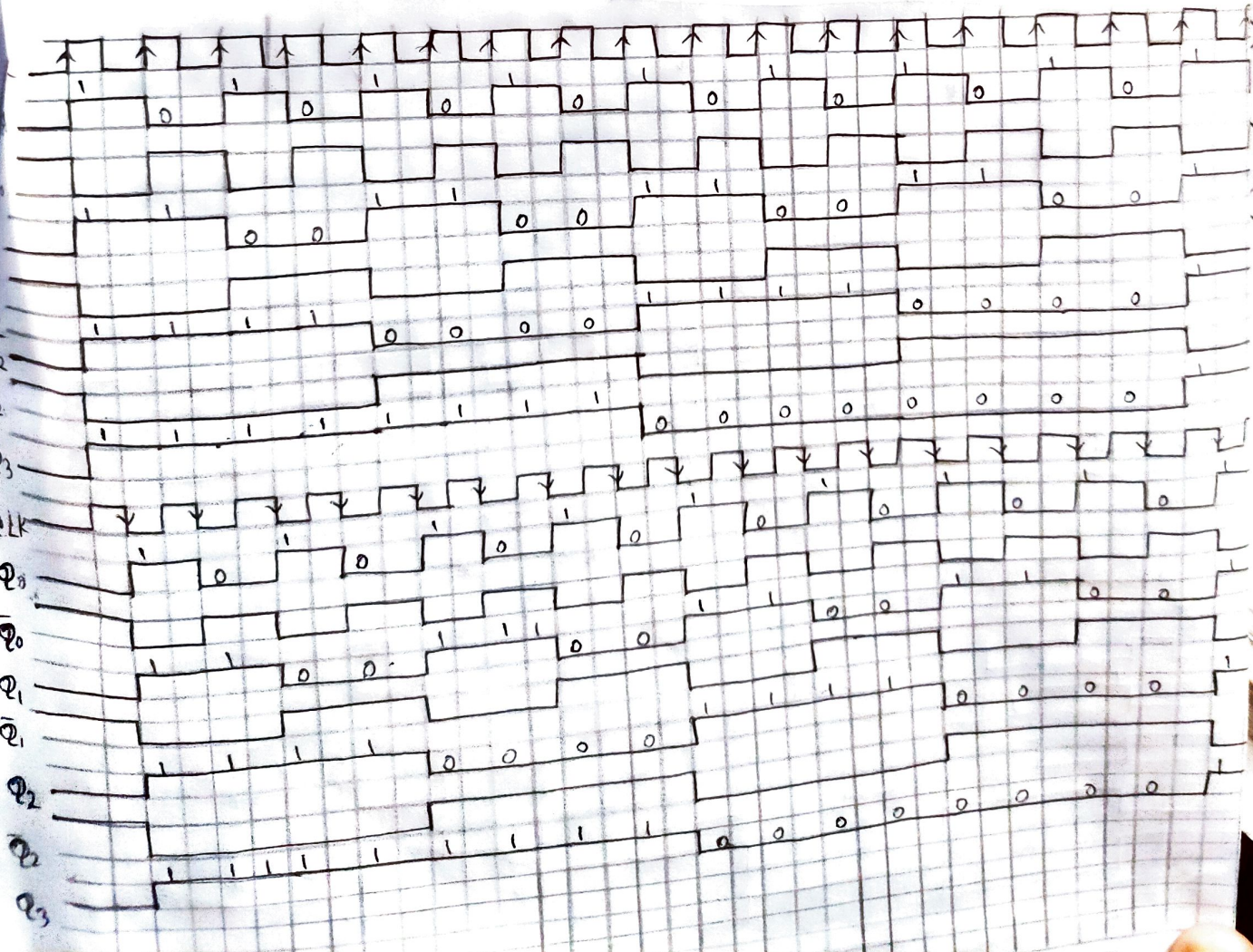
$Q_1Q_0 = 1$

Q_3 Toggle, when

$Q_2Q_1Q_0 = 1$

4-bit synchronous down counter





1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18

MOD-6 synchronous Binary counter.

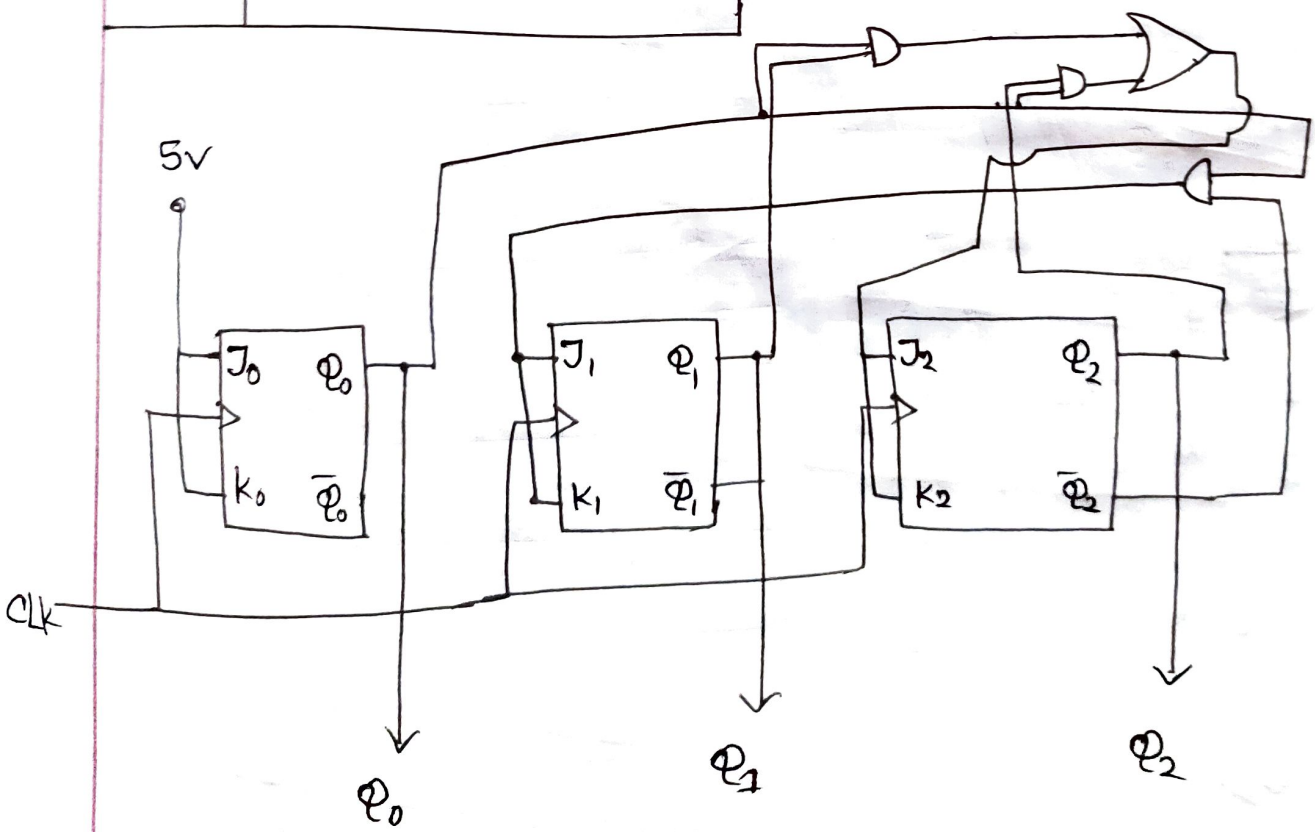
Truth Table

CLK	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	0	0	0

$J_0 = K_0 = 1$

$J_1 = K_1 = \bar{Q}_2 Q_0$

$J_2 = K_2 = Q_1 Q_0 + Q_2 Q_0$



How to design synchronous Counters

Step 1: Decide the number of flip flops.

step 2: Excitation table of flip flop.

step 3: State diagram and circuit excitation table.

Step 4: Obtain simplified equations using K' map.

Step 5: Draw the logic diagram.

Ques Design 2-bit synchronous up counter

Step-1: Number of flip-flops = 2
J-K flip flop.

Step-2: Excitation table of J-K flip flop.

Q^t	Q^{t+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1

How to design synchronous counters

Step 1: Decide the number of flip flops.

Step 2: Excitation table of flip flop.

Step 3: State diagram and circuit excitation table.

Step 4: Obtain simplified equations using K' map.

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Ques Design 2-bit synchronous up counter

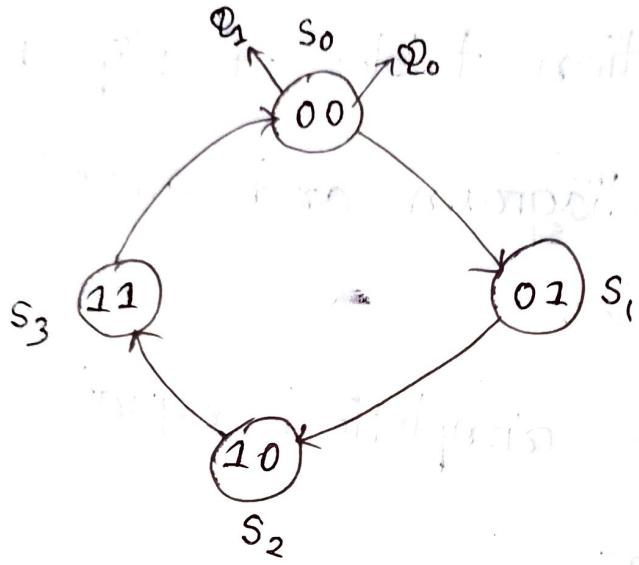
step-1: Number of flip-flops = 2

J-K flip flop.

step-2: Excitation table of J-K flip flop.

Q^t	Q^{t+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Step-3: State diagram

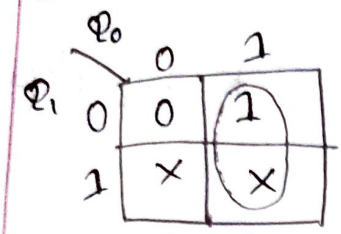


Circuit excitation table:

Present state		Next state					
Q_1	Q_0	Q_1^*	Q_0^*	J_1	K_1	J_0	K_0
0	0	0	1	0	x	1	x
0	1	1	0	1	x	x	1
1	0	1	1	x	0	1	x
1	1	0	0	x	1	x	1

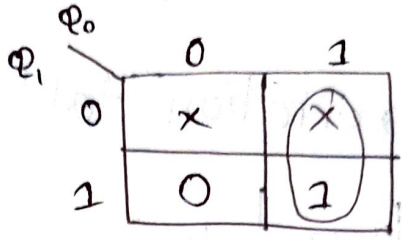
Step 4: Obtain simplified equations using k'map

for J_1:



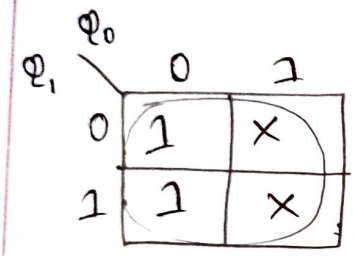
$J_1 = Q_0$

for K_1:



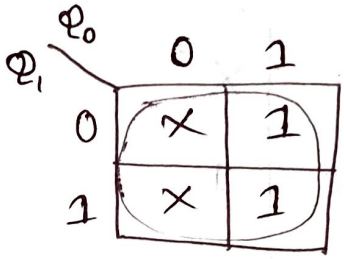
$K_1 = Q_0$

for J_0:



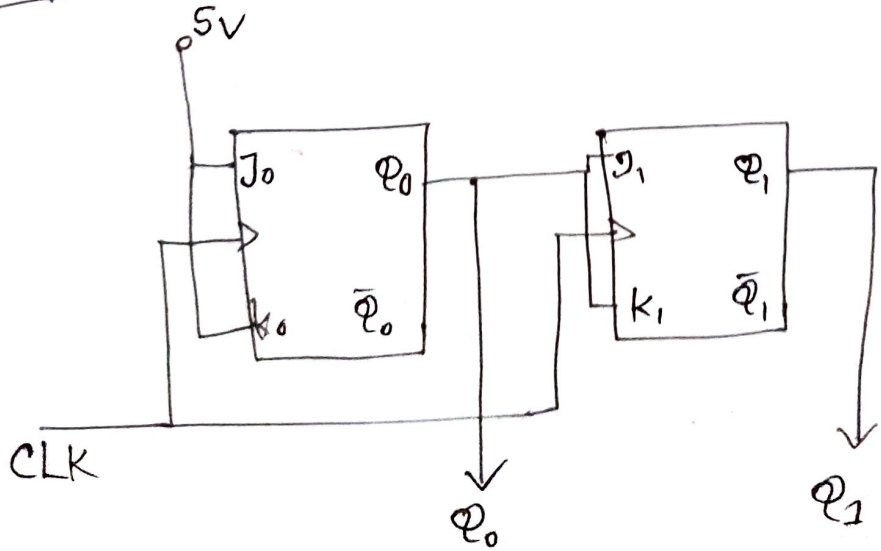
$J_0 = 1$

for K_0:

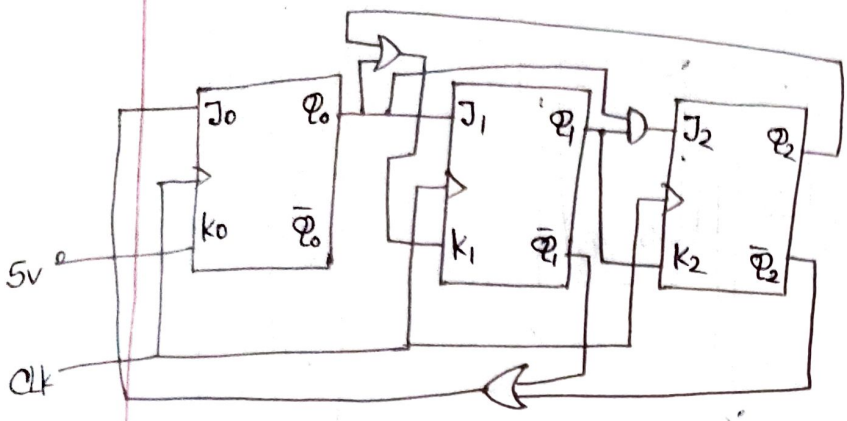


$K_0 = 1$

Step 5: Draw the logic diagram



MOD-7 synchronous up counter

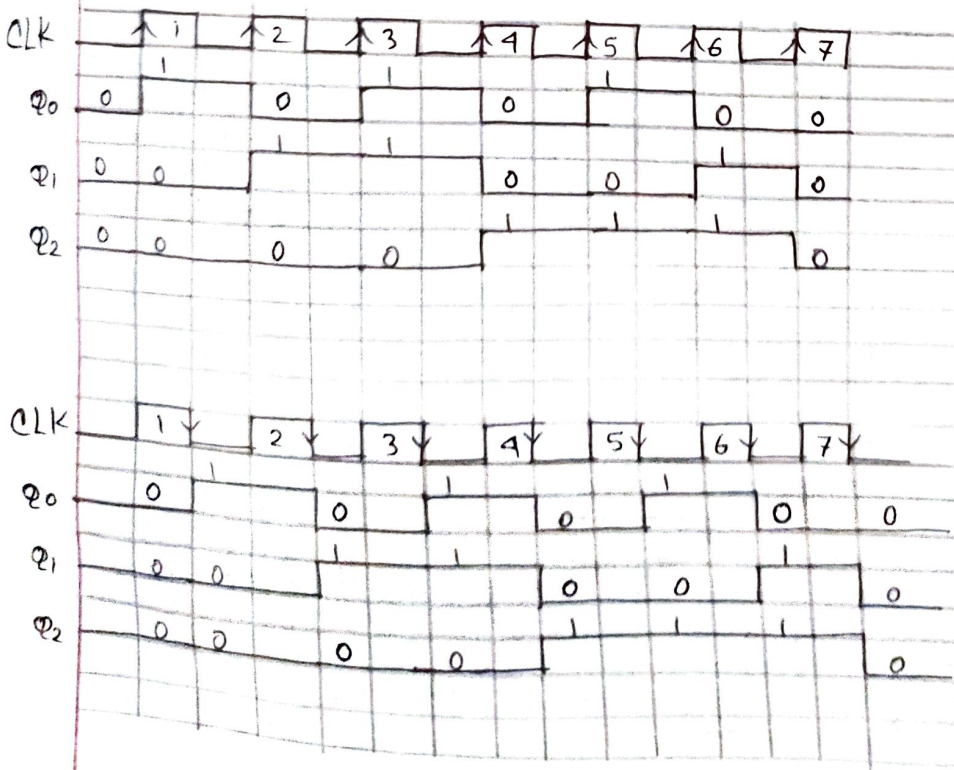


CLK	Q_2	Q_1	Q_0
	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	0	0	0

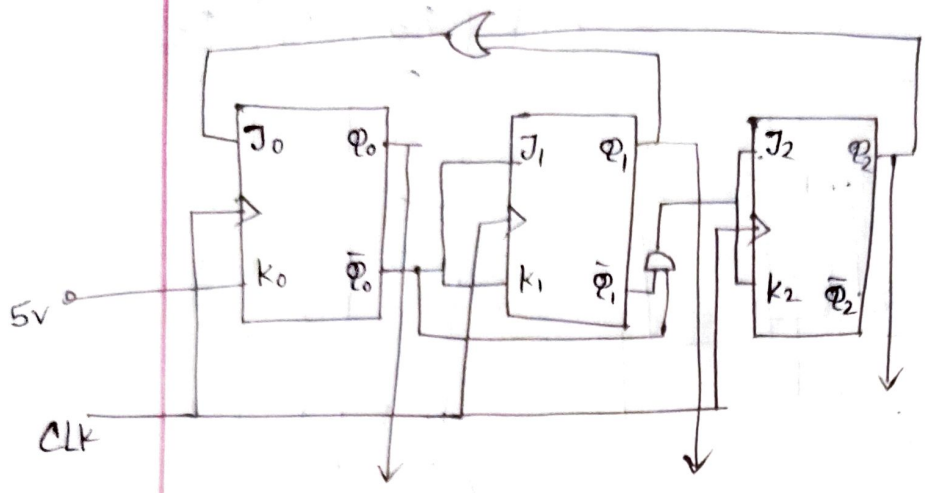
$J_0 = \bar{Q}_2 + \bar{Q}_1$, $K_0 = 1$

$J_1 = Q_0$, $K_1 = Q_2 + Q_0$

$J_2 = Q_1 Q_0$, $K_2 = Q_1$



Mod-7 synchronous down counter



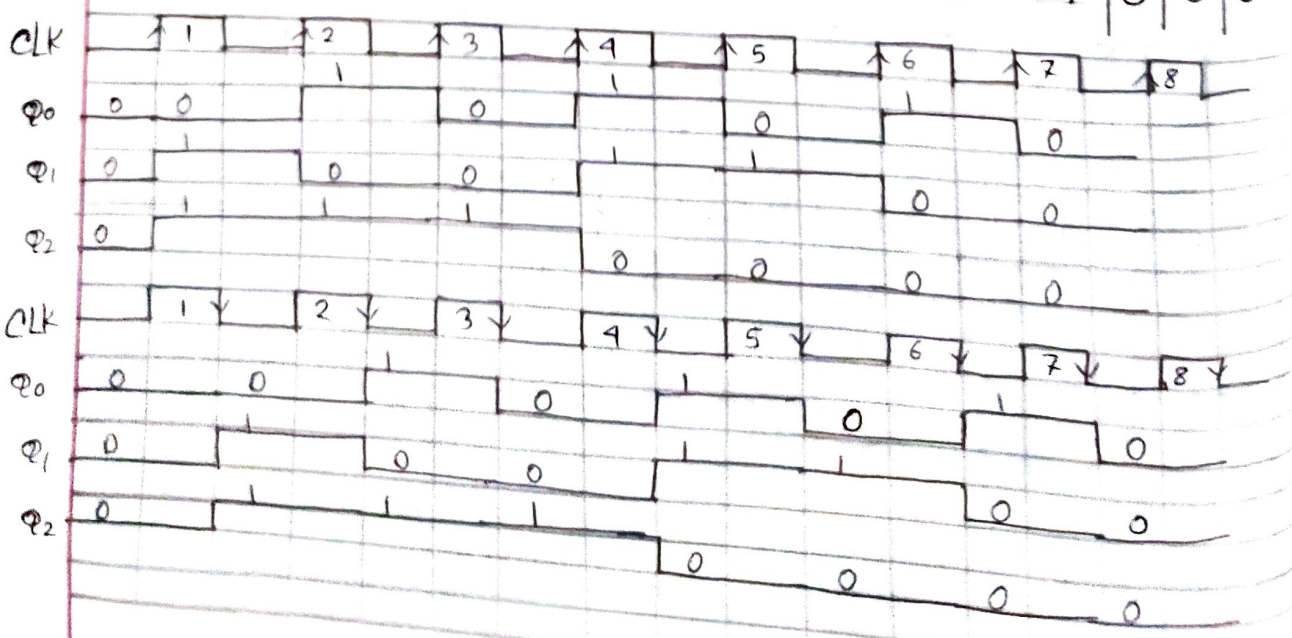
CLK	Q_2	Q_1	Q_0
	0	0	0
	0	0	1
	0	1	0
	0	1	1
	1	0	0
	1	0	1
	1	1	0

$J_0 = Q_2 + Q_1$, $K_0 = 1$

$J_1 = \bar{Q}_0$, $K_1 = \bar{Q}_0$

$J_2 = \bar{Q}_1 \bar{Q}_0$, $K_2 = \bar{Q}_1 \bar{Q}_0$

CLK	Q_2	Q_1	Q_0
	0	0	0
1	1	1	0
2	1	0	1
3	1	0	0
4	0	1	1
5	0	1	0
6	0	0	1
7	0	0	0



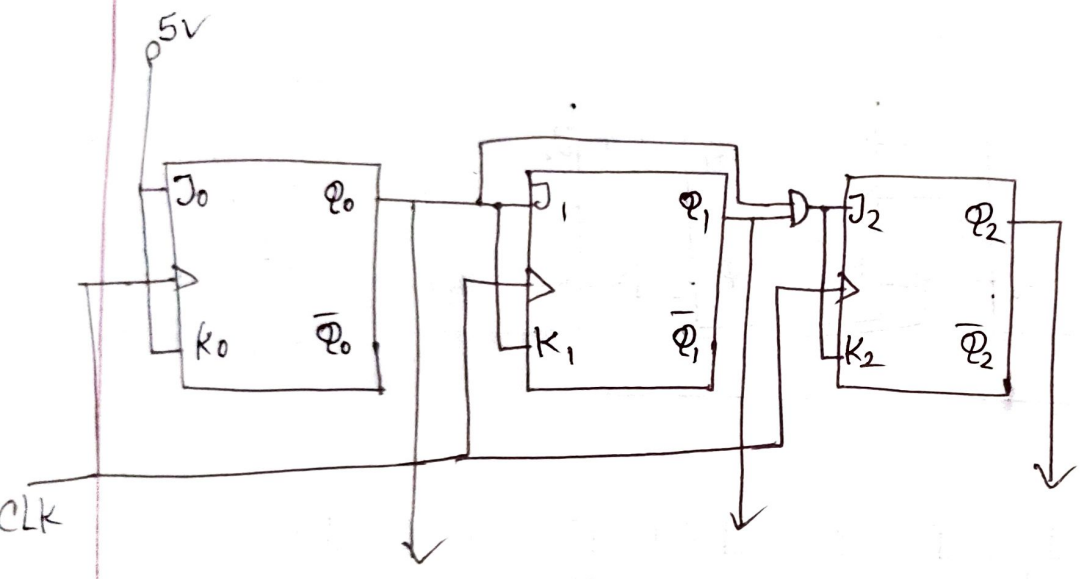
MOD-8 synchronous up counter

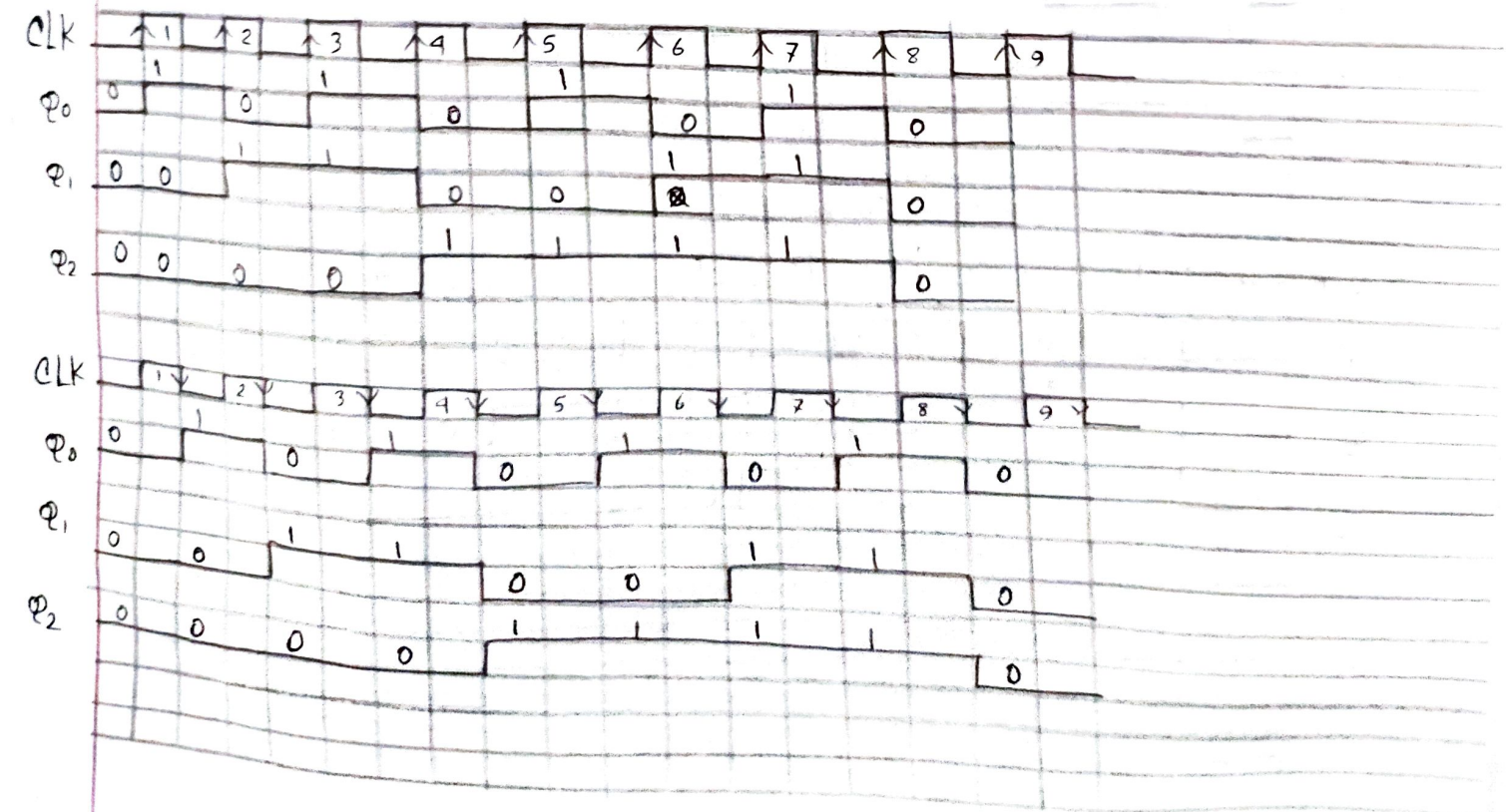
$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

$J_2 = K_2 = Q_1 Q_0$

CLK	Q_2	Q_1	Q_0
	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0



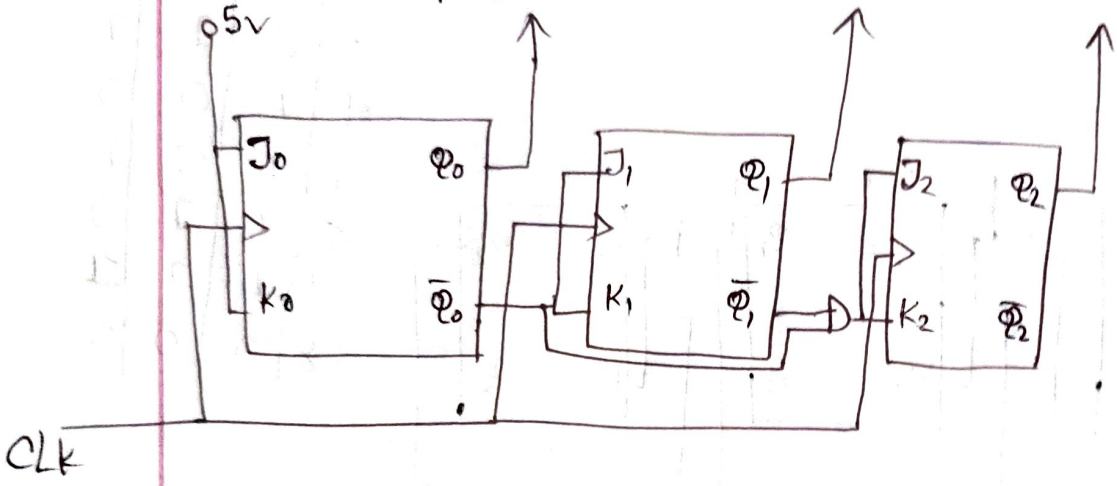


MOD - 8 synchronous down counter.

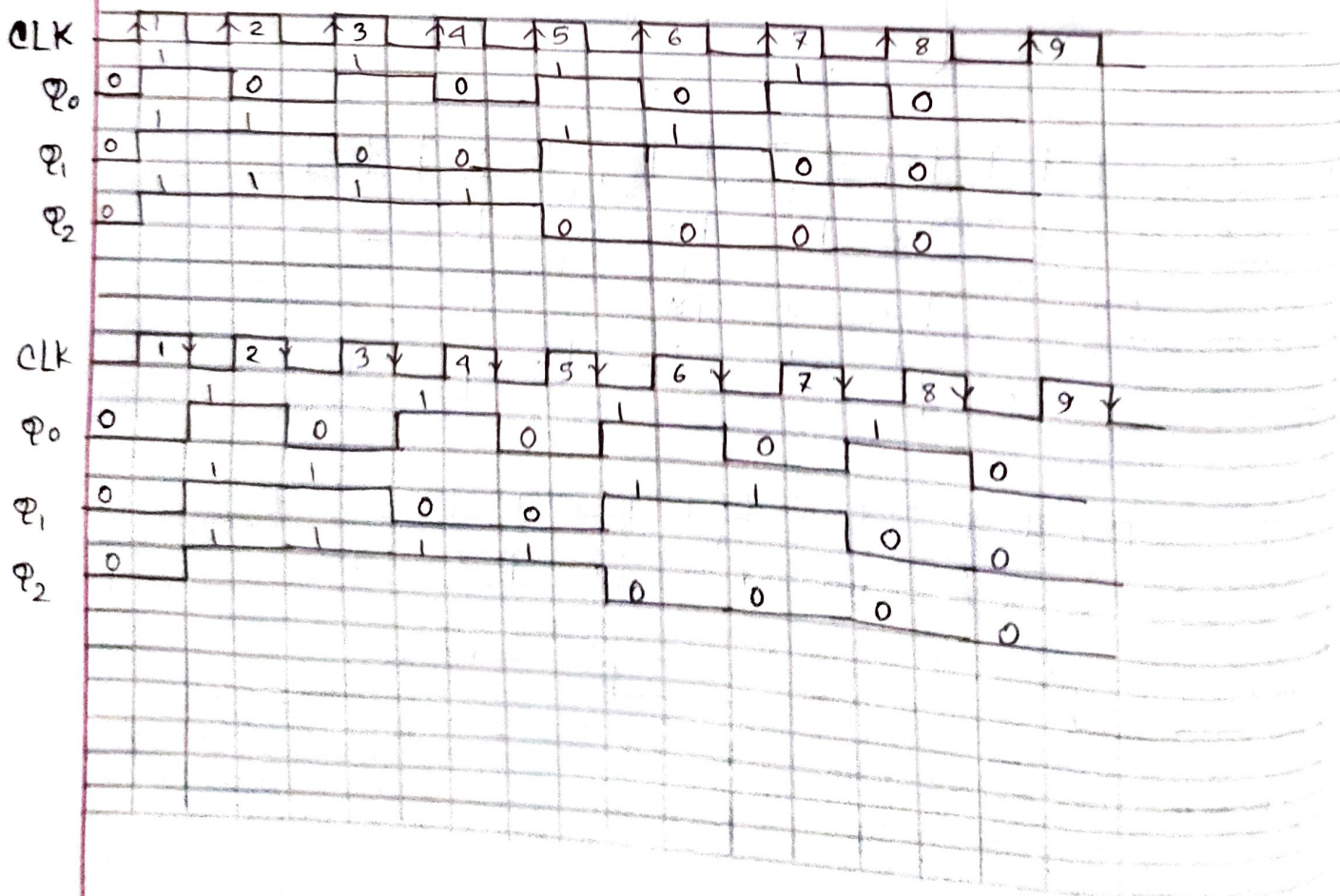
$J_0 = K_0 = 1$

$J_1 = K_1 = \bar{Q}_0$

$J_2 = K_2 = \bar{Q}_1 \bar{Q}_0$



CLK	Q_2	Q_1	Q_0
	0	0	0
1	1	1	1
2	1	1	0
3	1	0	1
4	1	0	0
5	0	1	1
6	0	1	0
7	0	0	1
8	0	0	0



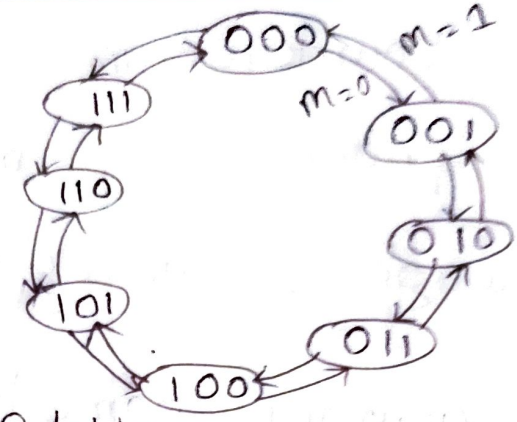
Up/down synchronous counters.

An up/down counter is one that is capable of progressing in either direction through a certain sequence.

An up/down counter, sometimes called a bidirectional counter, can have any specified sequence of states. A 3-bit binary counter that advances upward through its sequence (0, 1, 2, 3, 4, 5, 6, 7) and then can be reversed so that it goes through the sequence in the opposite direction (7, 6, 5, 4, 3, 2, 1, 0) is an illustration of up/down sequential operation.

3-bit UP/Down Synchronous counter

Step 1: State diagram



m = 0 , up
m = 1 , down

Step 2: Next state table

	Present			next		
m	Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	1	1
0	0	1	1	1	0	0
0	1	0	0	1	0	1
0	1	0	1	1	1	0
0	1	1	0	1	1	1
0	1	1	1	0	0	0
1	0	0	0	1	1	1
1	0	0	1	0	0	0
1	0	1	0	0	0	1
1	0	1	1	0	1	0
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

up: rows where m = 0

down: rows where m = 1

Step 3: Flip-flop transition table

Q^t	Q^{t+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Step 4: Karnaugh Map,

J_0,

MQ_2 \ Q_1Q_0	00	01	11	10
00	1	*	*	1
01	1	*	*	1
11	1	*	*	1
10	1	*	*	1

$J_0 = 1$

K_0,

MQ_2 \ Q_1Q_0	00	01	11	10
00	*	1	1	*
01	*	1	1	*
11	*	1	1	*
10	*	1	1	*

$K_0 = 1$

J_1,

MQ_2 \ Q_1Q_0	00	01	11	10
00	0	1	*	*
01	0	1	*	*
11	1	0	*	*
10	1	0	*	*

$J_1 = M\bar{Q}_0 + \bar{M}Q_0$

K_1,

MQ_2 \ Q_1Q_0	00	01	11	10
00	*	*	1	0
01	*	*	1	0
11	*	*	0	1
10	*	*	0	1

$K_1 = M\bar{Q}_0 + \bar{M}Q_0$

J_2,

MQ_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	*	*	*	*
11	*	*	*	*
10	1	0	0	0

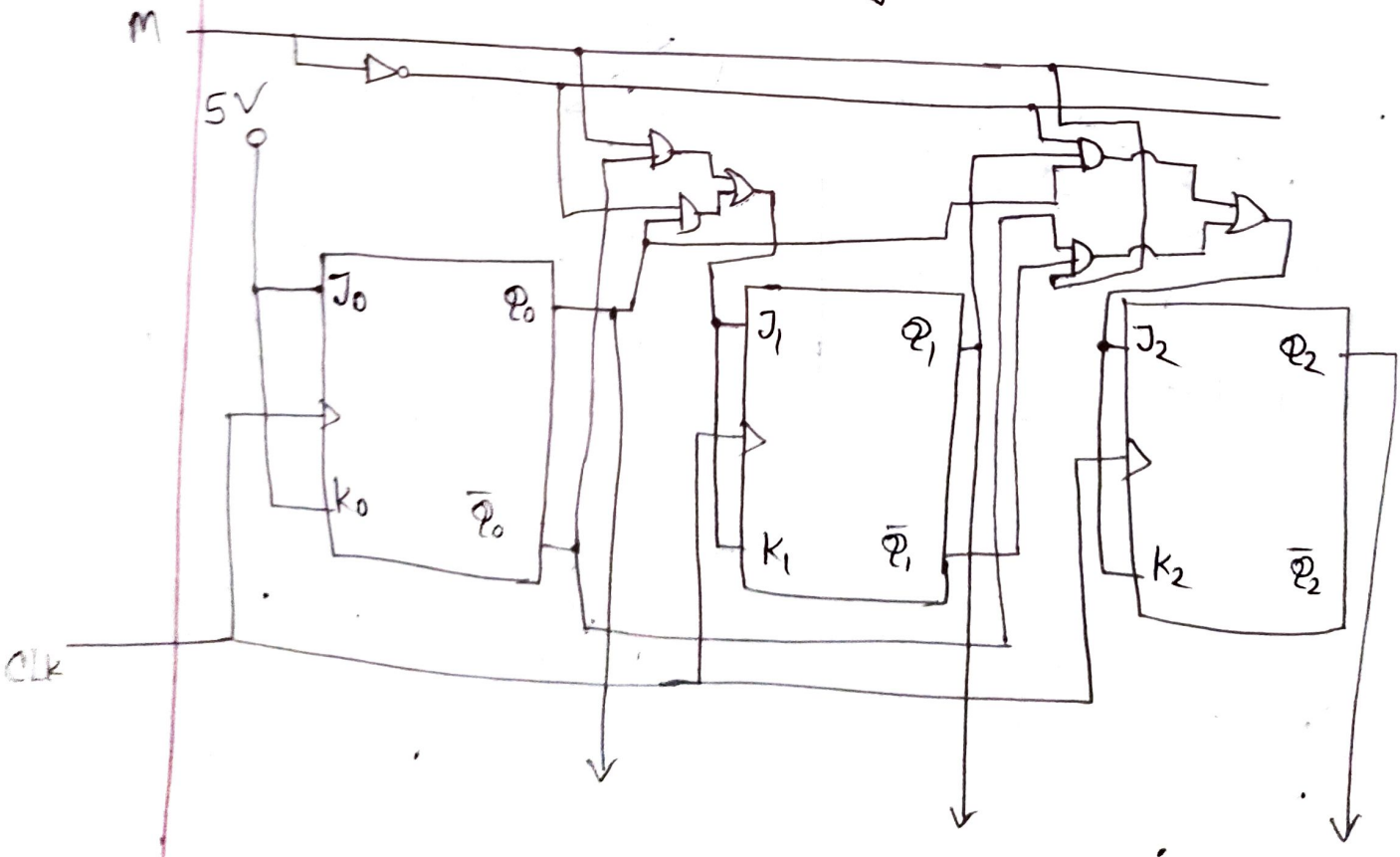
$J_2 = M\bar{Q}_1\bar{Q}_0 + \bar{M}Q_1Q_0$

K_2

MQ_2 \ Q_1Q_0	00	01	11	10
00	*	*	*	*
01	0	0	1	0
11	1	0	0	0
10	*	*	*	*

$K_2 = M\bar{Q}_1\bar{Q}_0 + \bar{M}Q_1Q_0$

Step-5: Circuit diagram,



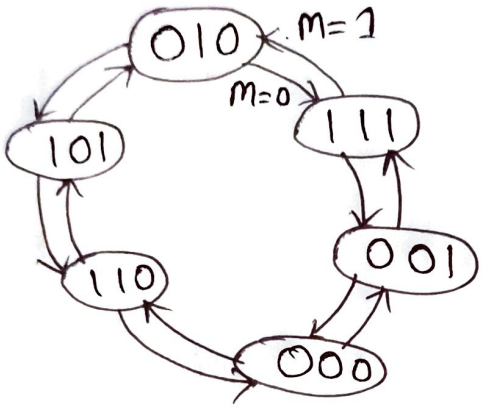
Assignment

2 ⇄ 7 ⇄ 1 ⇄ 0 ⇄ 6 ⇄ 5

Step 1: State diagram

M = 0, up

M = 1, down



Step 2: Next state table

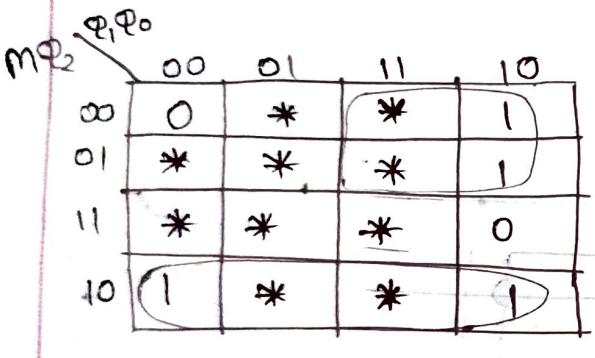
	Present state			Next State		
M	Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	1	0	1	1	1
0	1	1	1	0	0	1
0	0	0	1	0	0	0
0	0	0	0	1	1	0
0	1	1	0	1	0	1
0	1	0	1	0	1	0
0						
0						
1	0	1	0	1	0	1
1	1	1	1	0	1	0
1	0	0	1	1	1	1
1	0	0	0	0	0	1
1	1	1	0	0	0	0
1	1	0	1	1	1	0
1						
1						

step 3: flip-flop transition table

Q^t	Q^{t+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

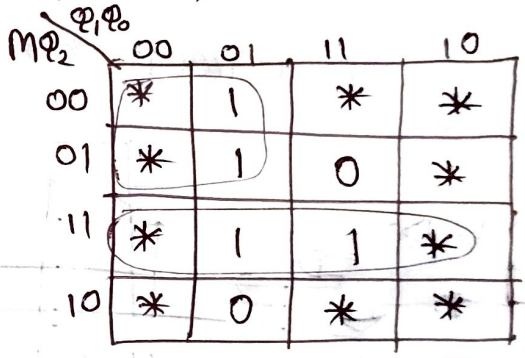
step 4: karnaugh Map.

For, J_0



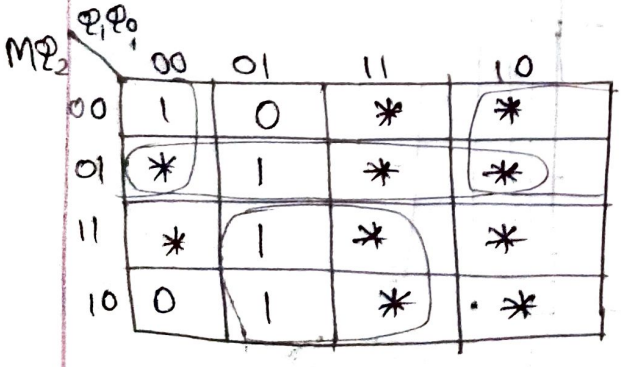
$$J_0 = M\bar{Q}_2 + \bar{M}Q_1$$

For K_0,



$$K_0 = MQ_2 + \bar{M}\bar{Q}_1$$

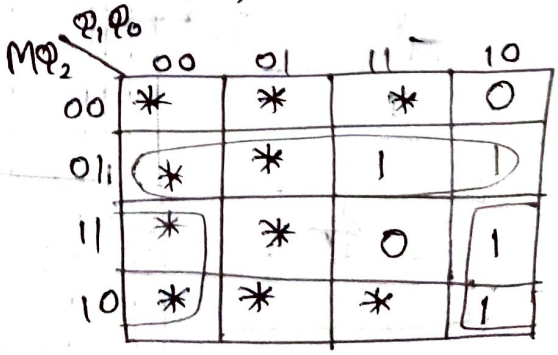
For, J_1



$$J_1 = MQ_0 + \bar{M}Q_2 + \bar{M}\bar{Q}_0$$

$$= MQ_0 + \bar{M}(Q_2 + \bar{Q}_0)$$

For K_1,



$$K_1 = M\bar{Q}_0 + \bar{M}Q_2$$

For J_2,

MQ_1 \ Q_1Q_0	00	01	11	10
00	1	0	*	1
01	*	*	*	*
11	*	*	*	*
10	0	1	*	1

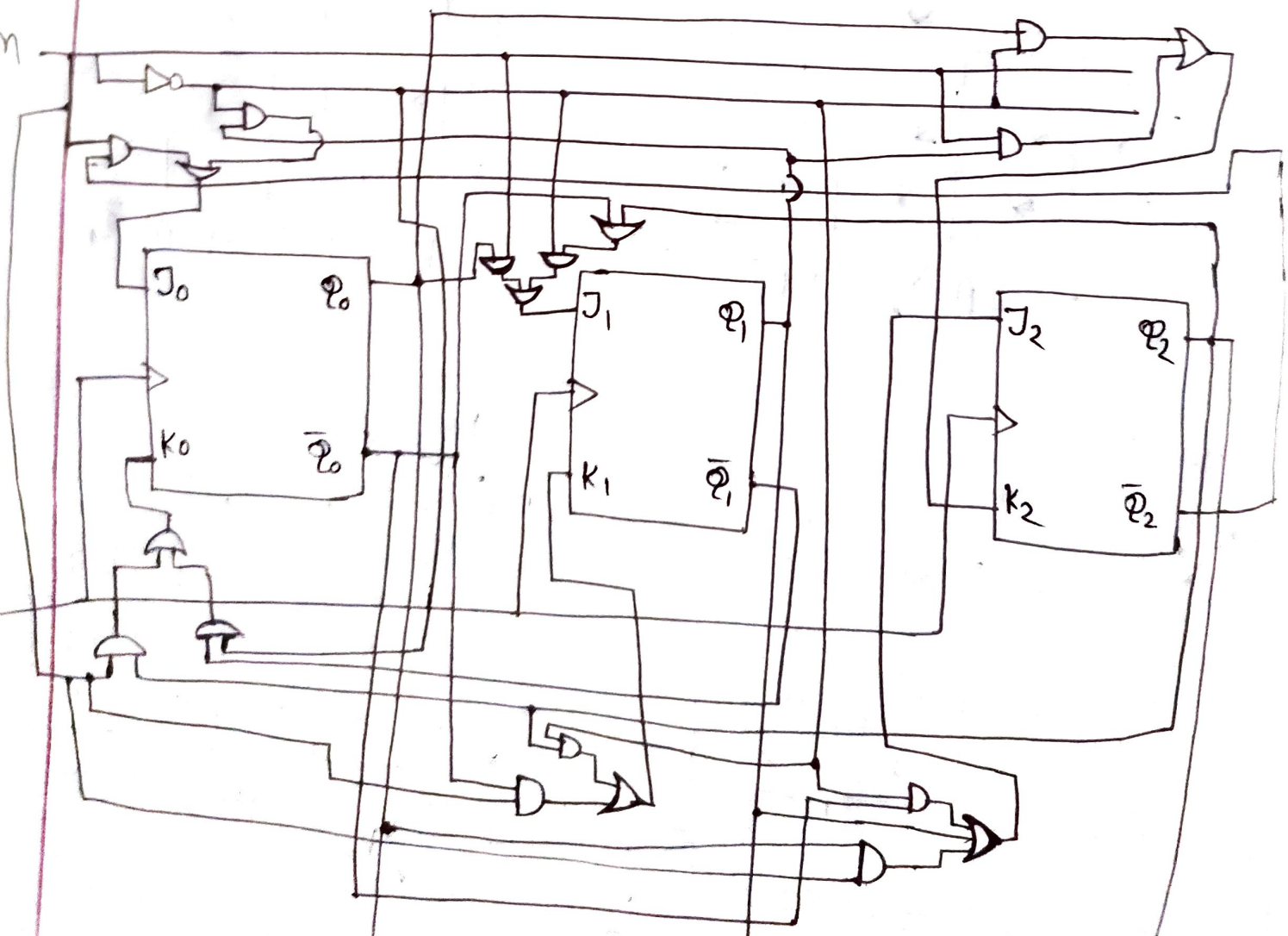
$$J_2 = Q_1 + MQ_0 + \bar{M}\bar{Q}_0$$

For K_2,

MQ_2 \ Q_1Q_0	00	01	11	10
00	*	*	*	*
01	*	1	1	0
11	*	0	1	1
10	*	*	*	*

$$K_2 = MQ_1 + \bar{M}Q_0$$

step-5: circuit diagram



For J_2,

MQ_2 \ Q_1Q_0	00	01	11	10
00	1	0	*	1
01	*	*	*	*
11	*	*	*	*
10	0	1	*	1

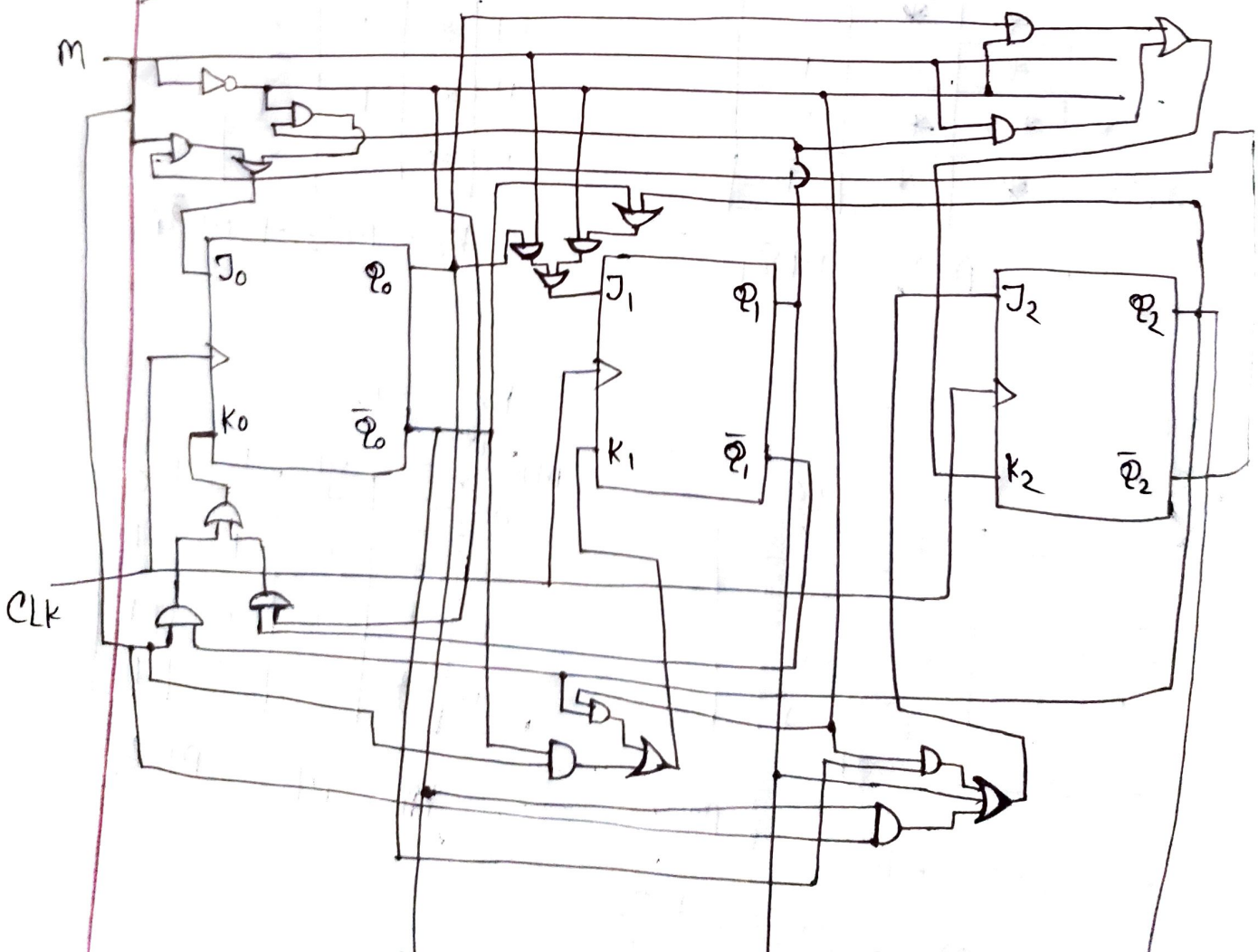
$$J_2 = Q_1 + MQ_0 + \bar{M}\bar{Q}_0$$

For K_2,

MQ_2 \ Q_1Q_0	00	01	11	10
00	*	*	*	*
01	*	1	1	0
11	*	0	1	1
10	*	*	*	*

$$K_2 = MQ_1 + \bar{M}Q_0$$

step-5: circuit diagram



2-1-2023
Monday
class: 13

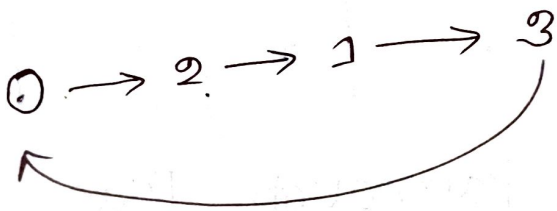
Design of a counter to produce the following binary sequences using J-K flip-flop.

Q_1	Q_0
0	0
1	0
0	1
1	1
0	0
:	:

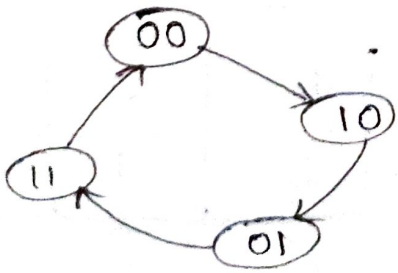
= State = 4

number of flip-flop = 2

$[2^2 = 4]$.



Step-1: State diagram



Step-2: Next state table

Present state		Next state	
Q_1	Q_0	Q_1	Q_0
0	0	1	0
1	0	0	1
0	1	1	1

2-1-2023
Monday
class: 13

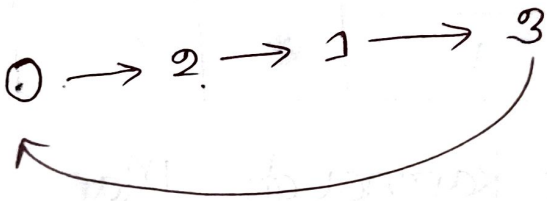
Design of a counter to produce the following binary sequences using J-K flip-flop.

Q_1	Q_0
0	0
1	0
0	1
1	1
0	0
:	:

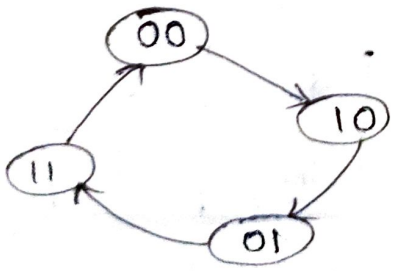
= State = 4

number of flip-flop = 2

$[2^2 = 4]$.



Step-1: State diagram



Step-2: Next state table

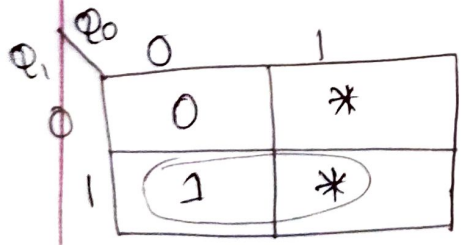
Present state		Next state	
Q_1	Q_0	Q_1	Q_0
0	0	1	0
1	0	0	1
0	1	1	1
1	1	0	0

Step-3: Flip-flop transition table using J-k flip-flop.

output		Input	
Q^t	Q^{t+1}	J	k
0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

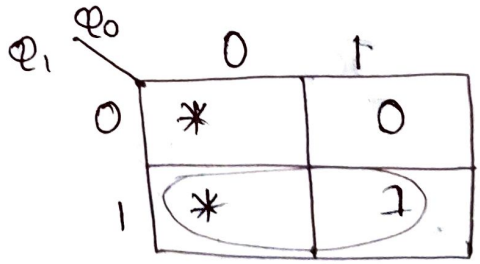
Step-4: karnaugh Map

For J_0,



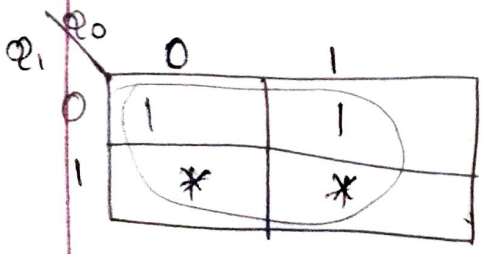
$J_0 = Q_1$

For k_0,



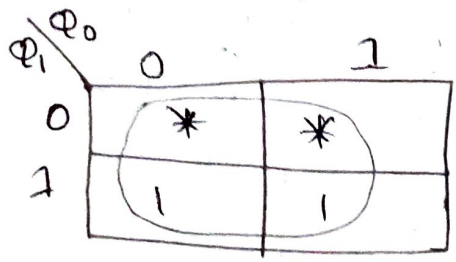
$k_0 = Q_1$

For J_1,

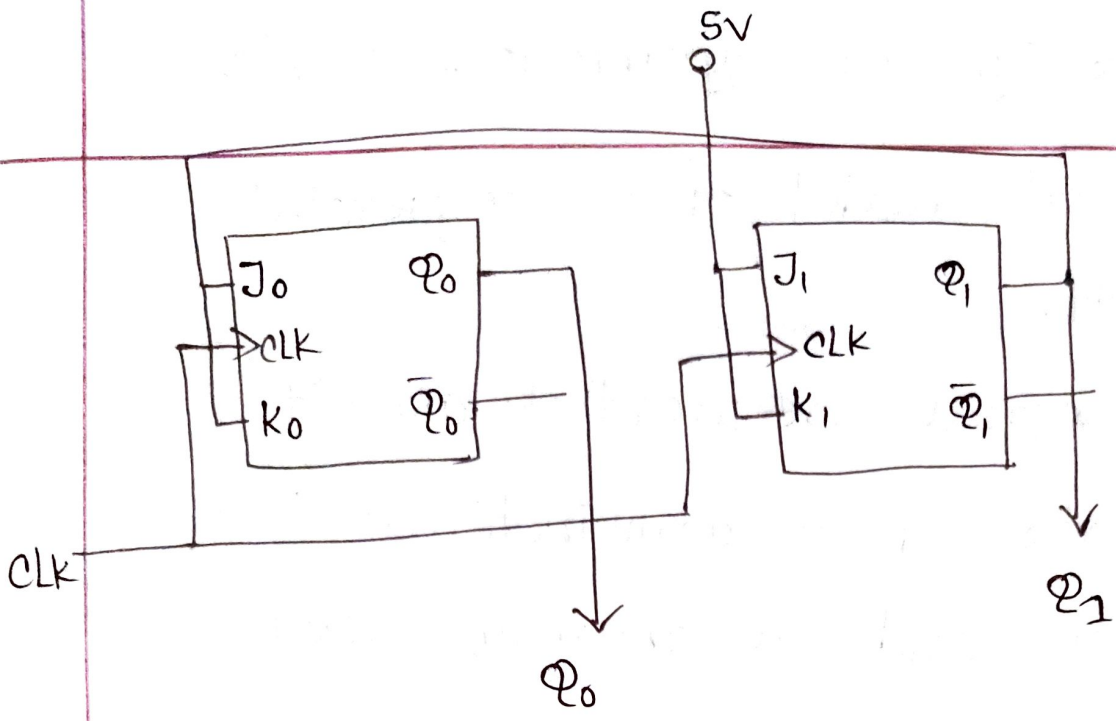


$J_1 = 1$

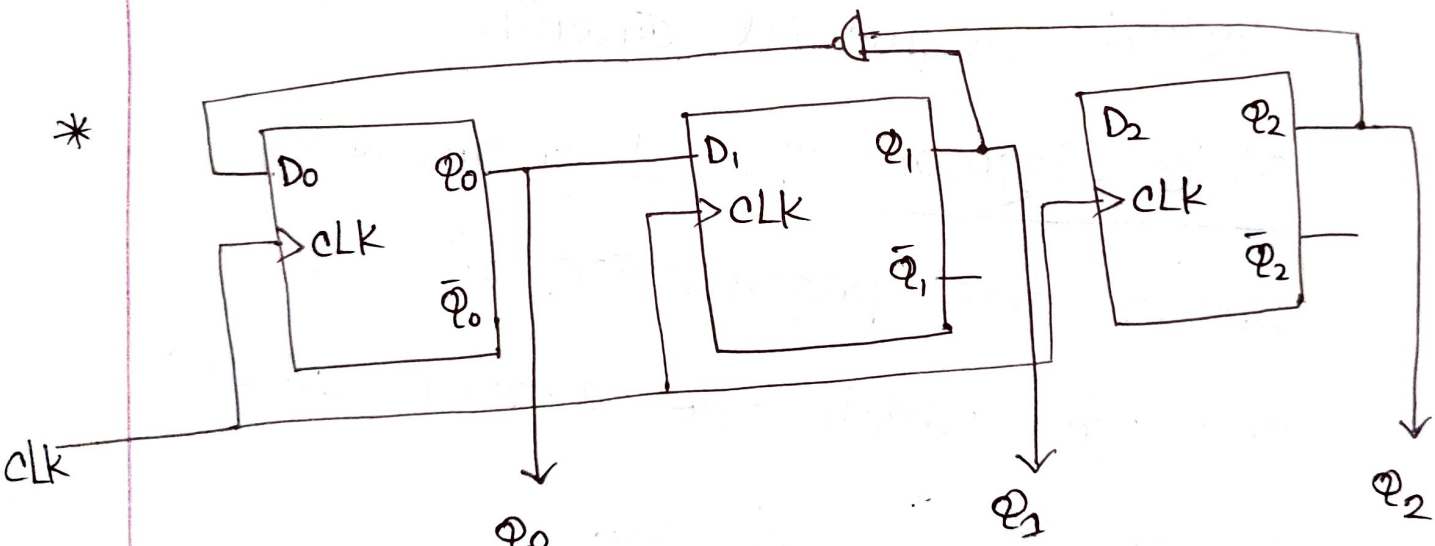
For k_1,



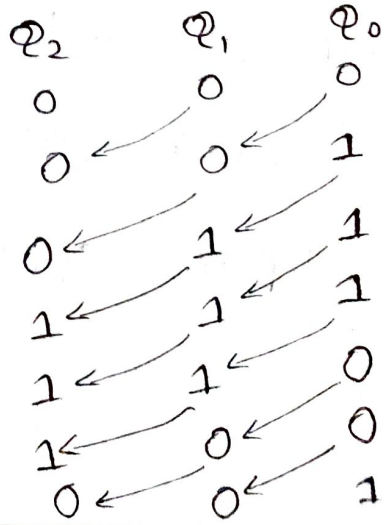
$k_1 = 1$



* D flip-flop used in cache memory.



CLK	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	1
3	1	1	1
4	1	1	0
5	1	0	0
6	0	0	1



Design of synchronous counters

* General model of a sequential circuit.

A general sequential circuit consists of a combinational logic section and a memory section

=> Counters are a special case of clocked sequential circuits.

State diagram: The diagram which shows the progression of states through which the counter advances when it is clocked.

=> Next state is the state that the counter goes to from its present state upon application of a clock pulse.

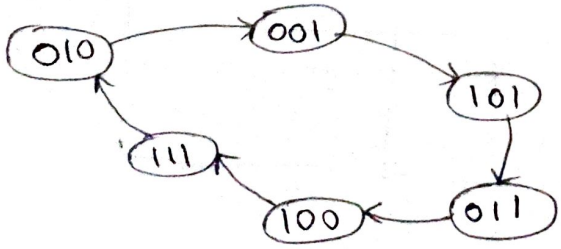
Transition table: The table where all possible output transitions are, listed by showing the Q output of the flip-flop going from present states to next states. Q^t is the present state of the flip-flop (before a clock pulse) and Q^{t+1} is the next state (after a clock pulse).

* Design a irregular synchronous counter.

1 → 5 → 3 → 4 → 7 → 2

=

Step-1: state diagram



Step-2: Next state table

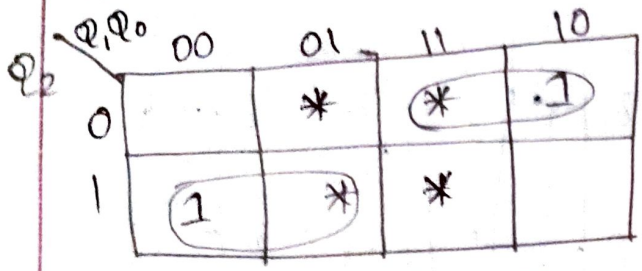
Present State			Next State		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	1	1	0	1
1	0	1	0	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	1	1	0	1	0
0	1	0	0	0	1

Step-3: Flip-flop transition table using J-K flip flop.

Output		Input	
Q^t	Q^{t+1}	J	K
0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

Step-4: karnaugh Map

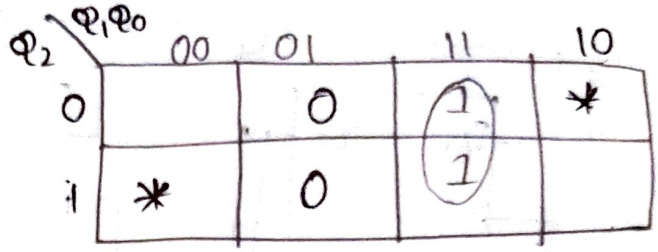
For J_0,



$J_0 = Q_2\bar{Q}_1 + \bar{Q}_2 Q_1$

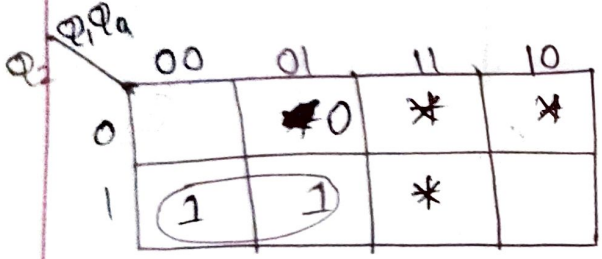
$= Q_2 \oplus Q_1$

For K_0,



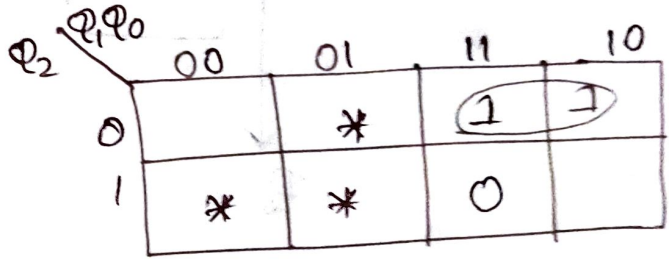
$K_0 = Q_1 Q_0$

For J_1,



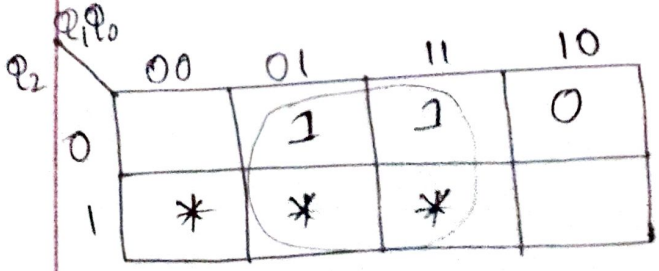
$J_1 = Q_2\bar{Q}_1$

For K_1,



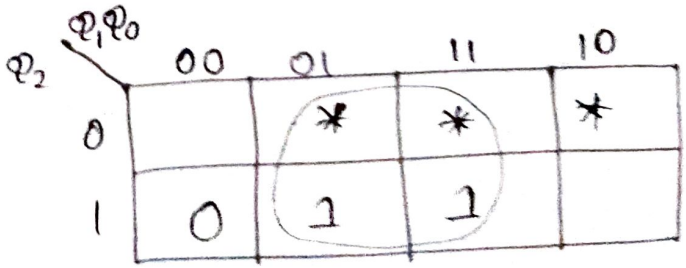
$K_1 = Q_1\bar{Q}_2$

For J_2,



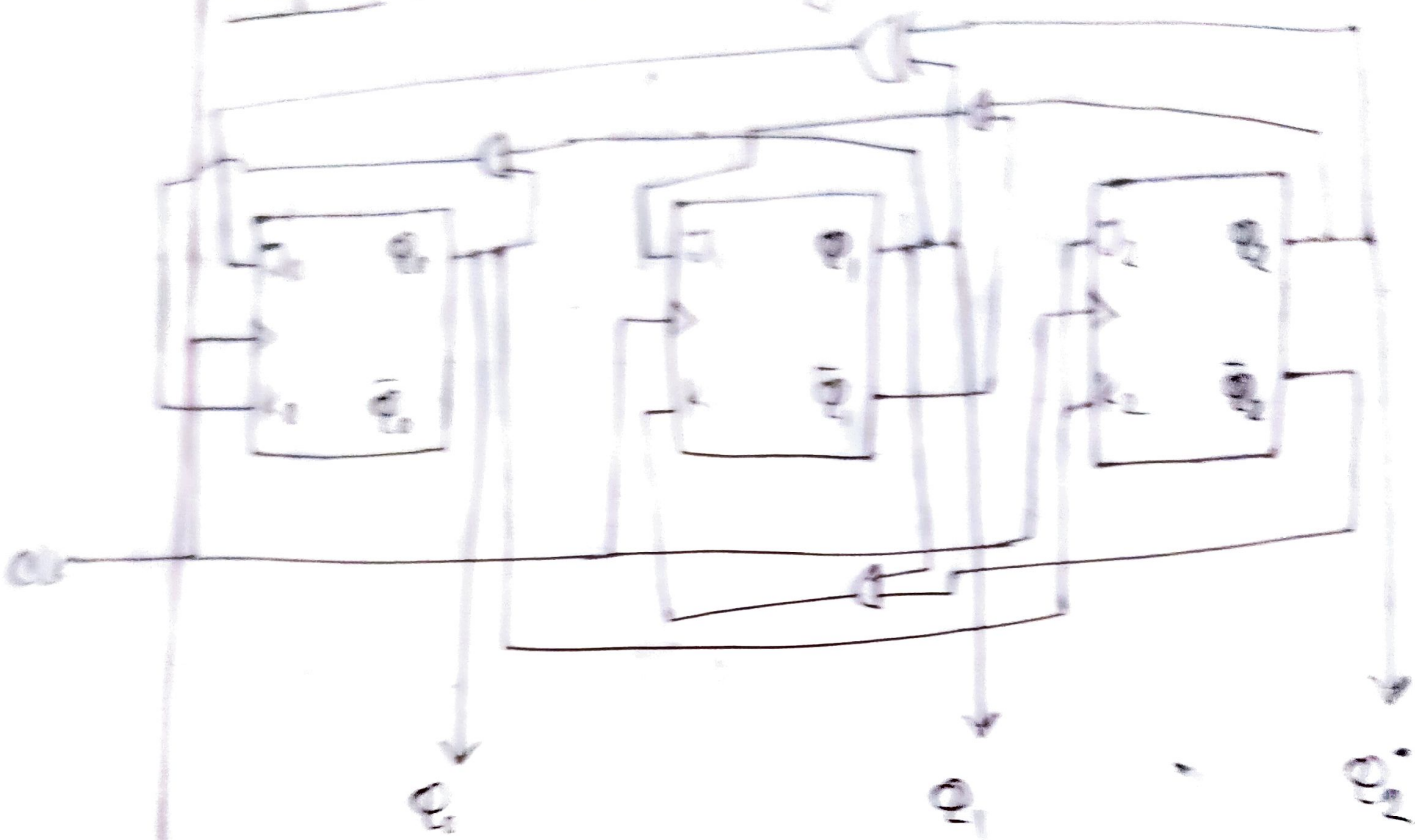
$J_2 = Q_0$

For K_2,

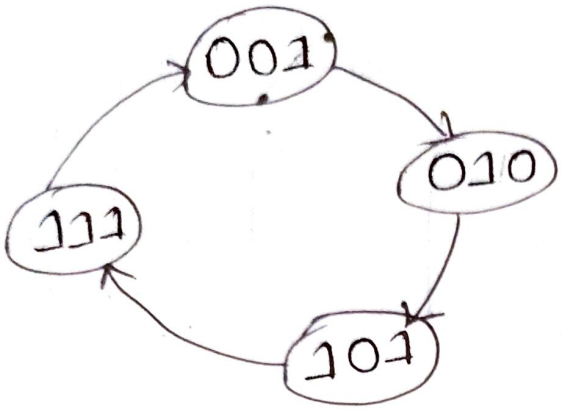


$K_2 = Q_0$

Step - 5 : Circuit diagram



Ex-8.5 : Design a counter with the irregular binary count sequence.



=

Step-2: Next state table

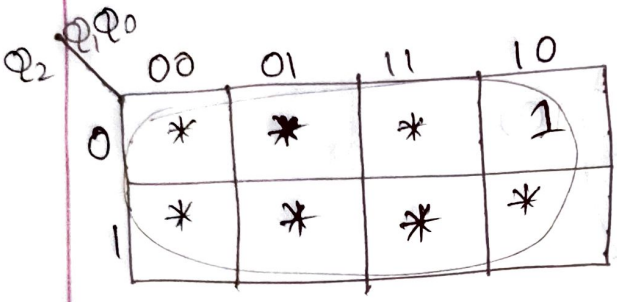
Present State			Next State		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	1	0	1	0
0	1	0	1	0	1
1	0	1	1	1	1
1	1	1	0	0	1

Step-3 : Flip-flop transition table

output		Input	
Q^t	Q^{t+1}	J	k
0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

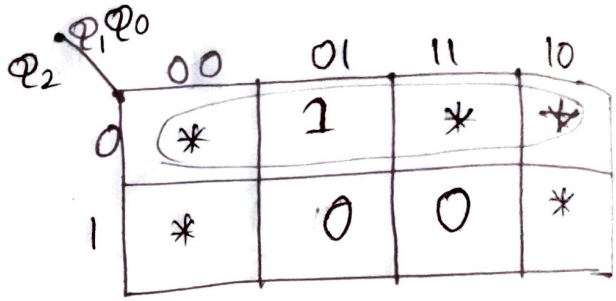
Step-4: karnaugh Map

For J_0,



$J_0 = 1$

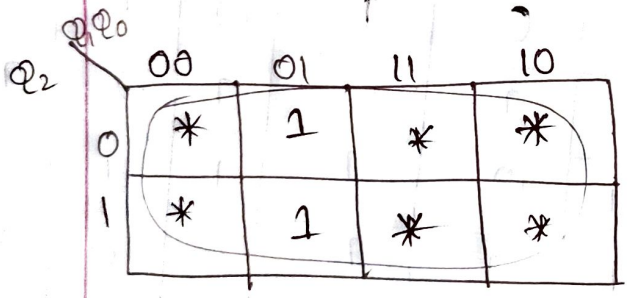
For K_0,



$K_0 = \bar{Q}_2$

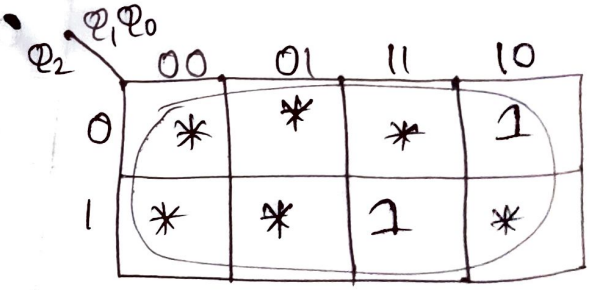
CLK

For J_1,



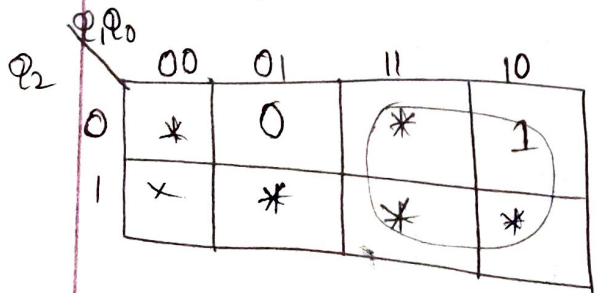
$J_1 = 1$

For K_1,



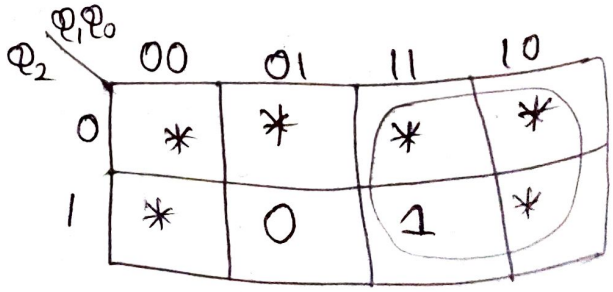
$K_1 = 1$

For J_2,

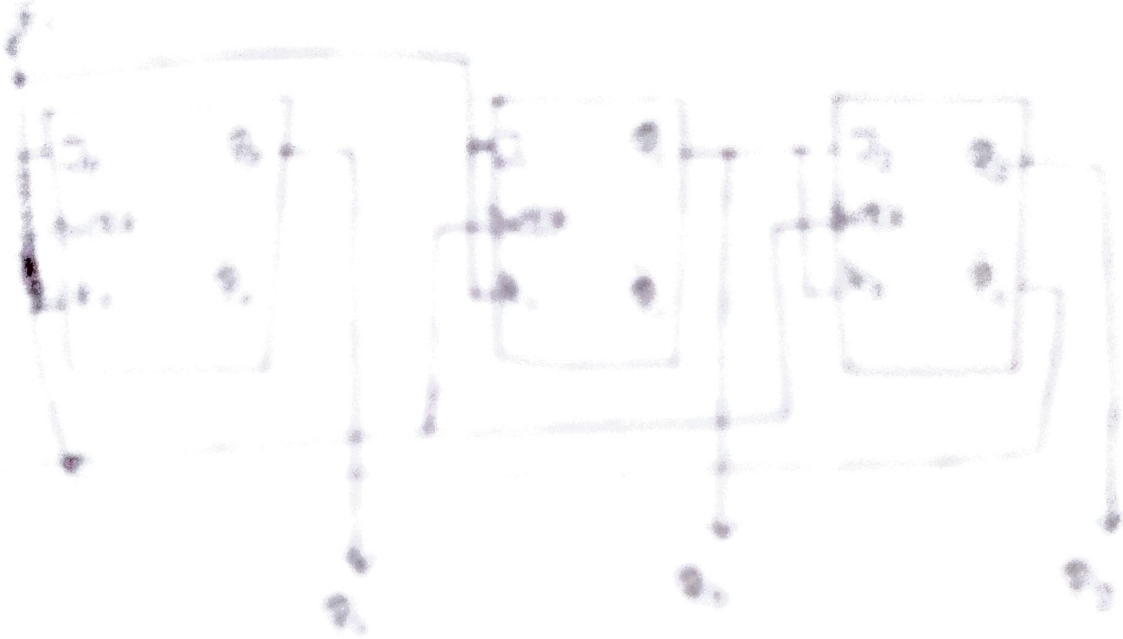


$J_2 = Q_1$

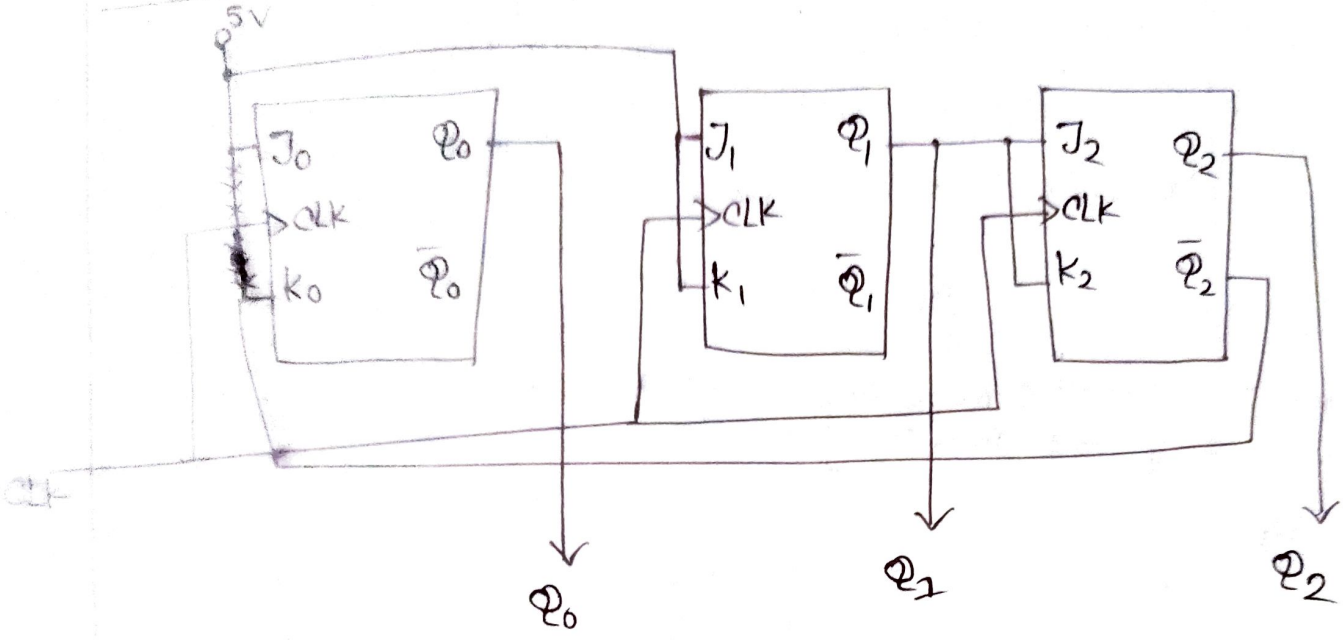
For K_2,



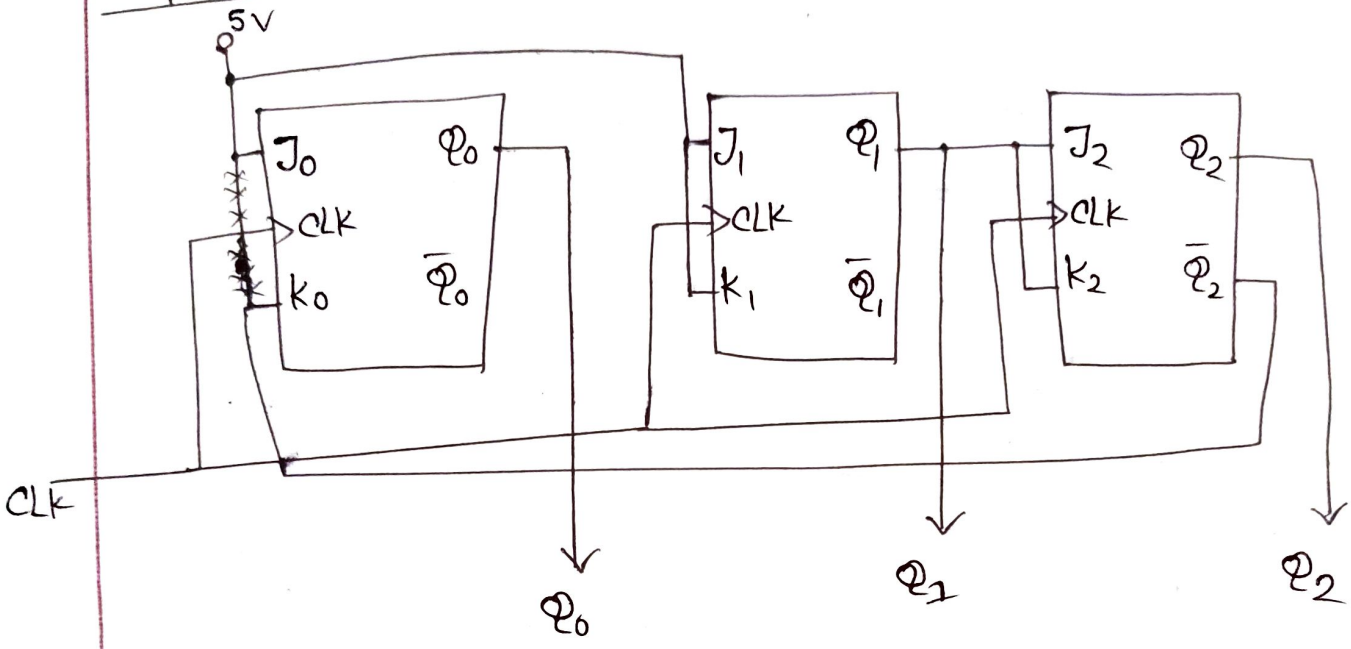
$K_2 = Q_1$



Step-5 : Circuit diagram



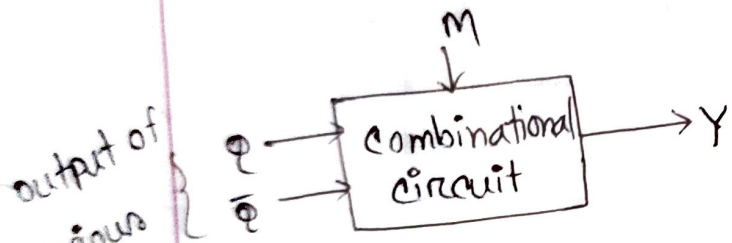
Step-5 : Circuit diagram



3-bit and 4-bit UP/DOWN Ripple Counter

⇒ A mode control input (M) is used to select either up or down mode.

⇒ A combinational circuit is required between each pair of flip flops.



M	Q	$\bar{Q}$	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

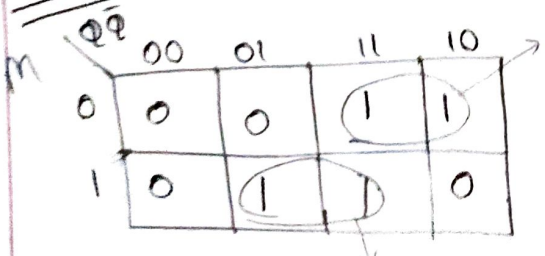
M = 0 , up counting

M = 1 , down counting.

→ Q connected to CLK of next flip-flop

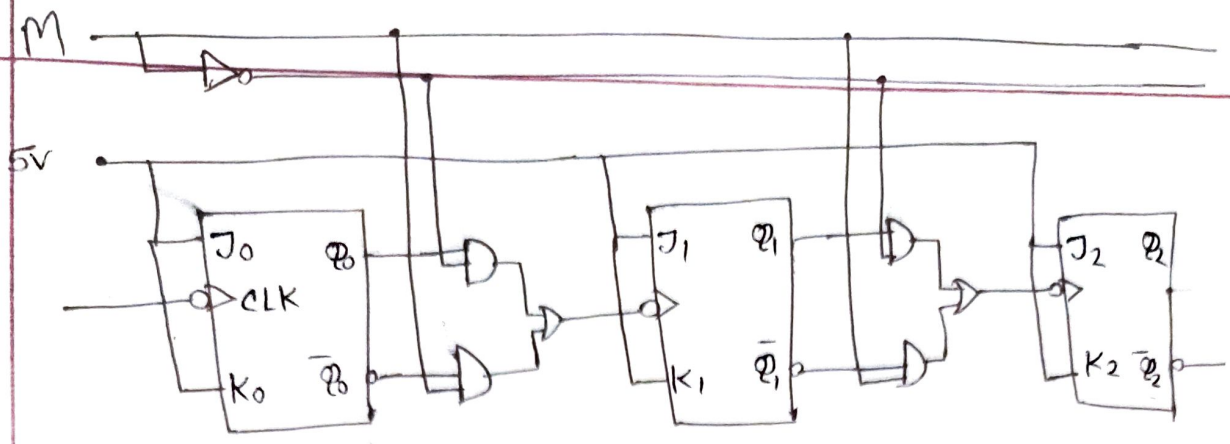
→ $\bar{Q}$ connected to CLK of next flip-flop

K Map



$y = \bar{M}Q + M\bar{Q}$

3-bit asynchronous up/down counter



4-bit asynchronous up/down counter

M = 0, up counting, Q connected to CLK next ff

M = 1, down counting, Q̄ ~ to CLK next ff

M	Q	Q̄	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
0			
0			
0			
0			
1			
1			
1			
1			
1			
1			
1			
1			

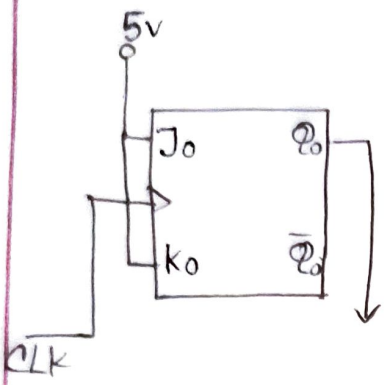
Problem

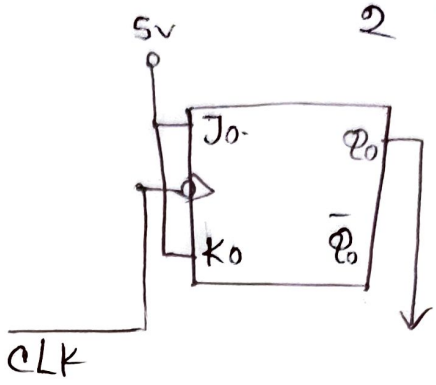
MOD-2 synchronous up counter

Flip-flop = 1

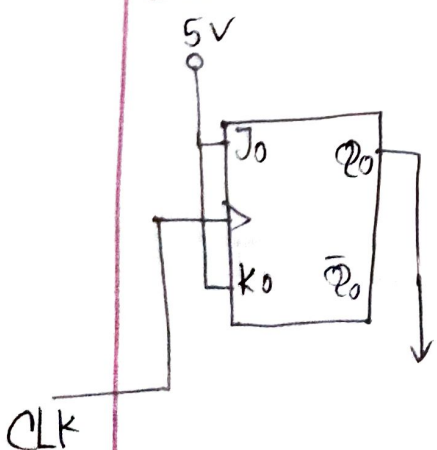
States = 2

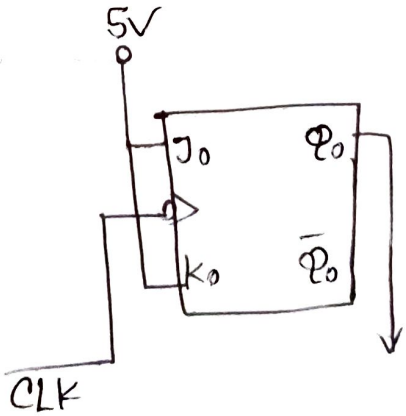
CLK	Q_0
Initially	0
1	1
2	0





MOD-2 synchronous down counter





MOD-3 Synchronous up counter

Flip-Flop = 2 ∴ States = 3

0 to 2

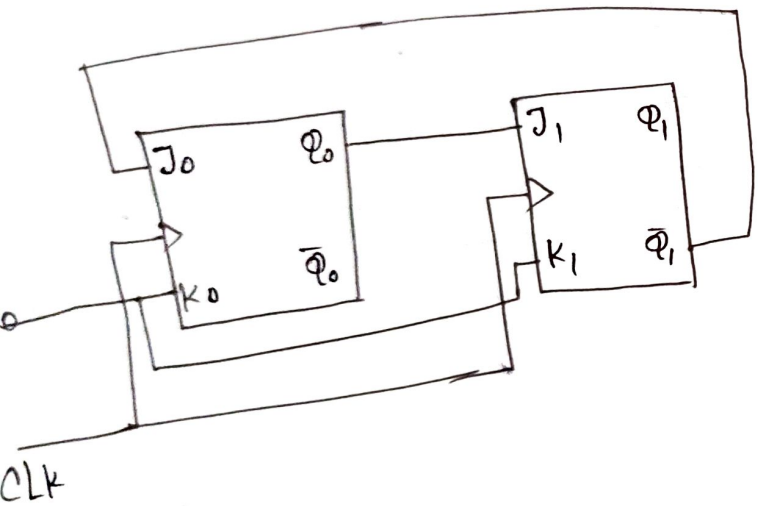
CLK	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0

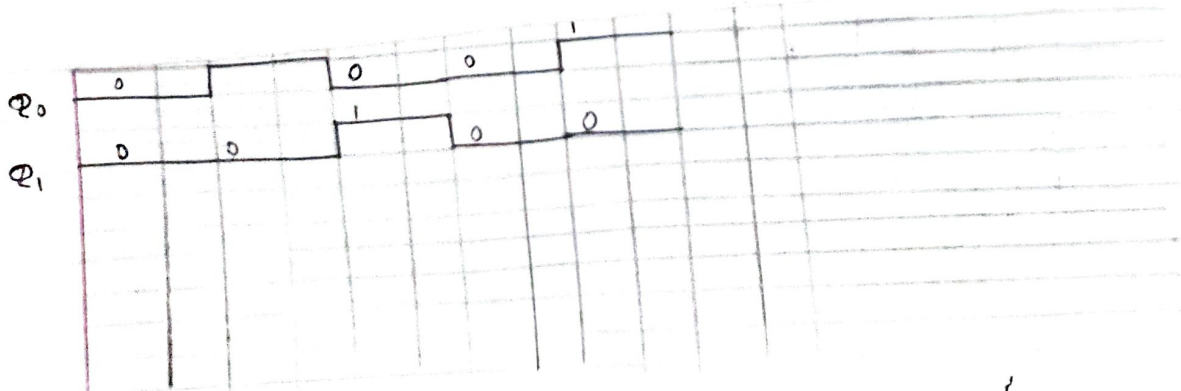
5V

$J_0 = \bar{Q}_1$, $K_0 = 1$

$J_1 = Q_0$, $K_1 = 1$

MOD-3 synchronous up counter



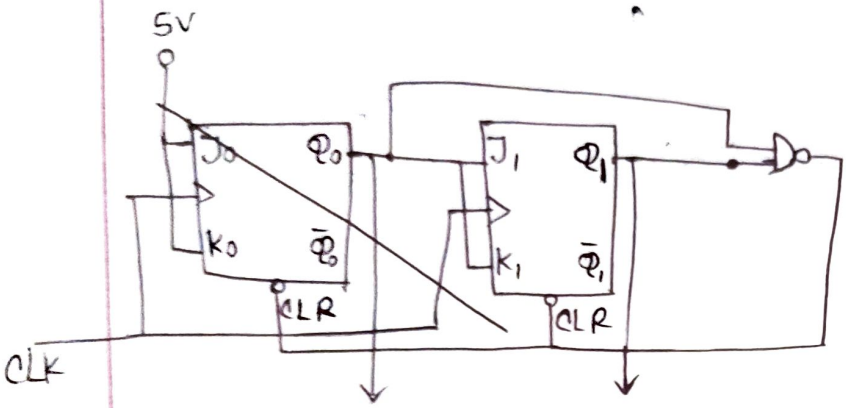


MOD-3 Synchronous up counter

Flip-Flop = 2 ∴ States = 3

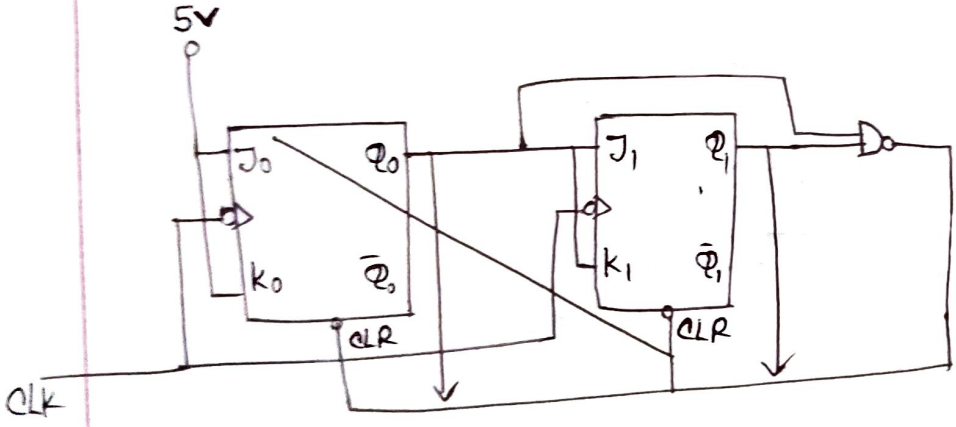
0 to 2

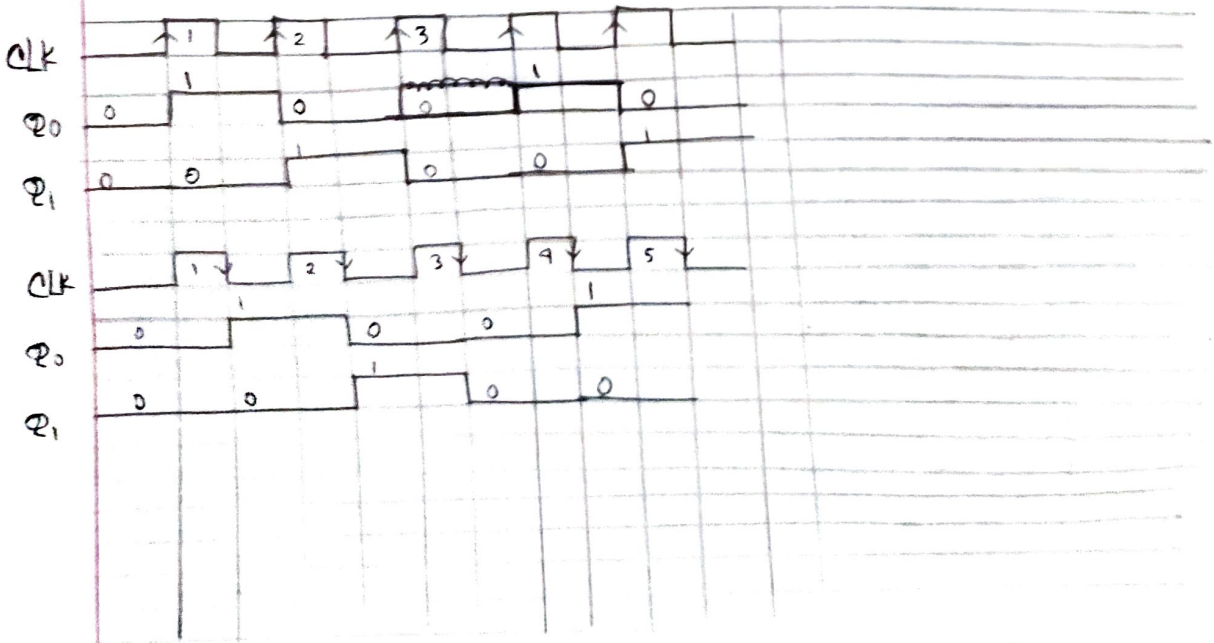
CLK	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	0	0



~~$J_0 = K_0 = 1$~~

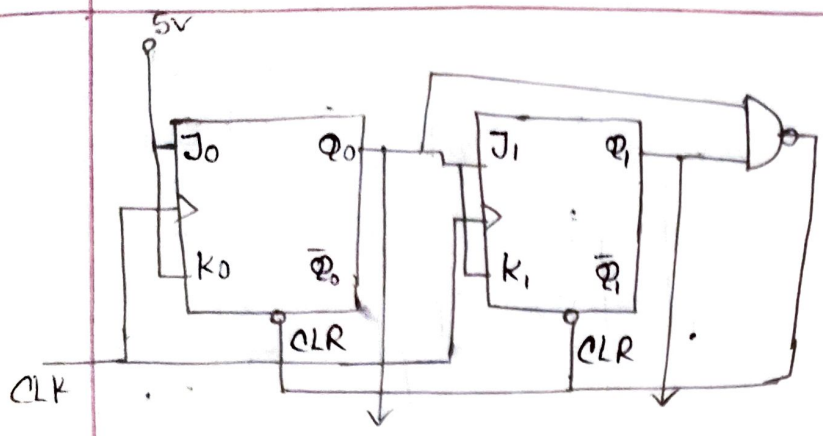
~~$J_1 = K_1 = Q_0$~~



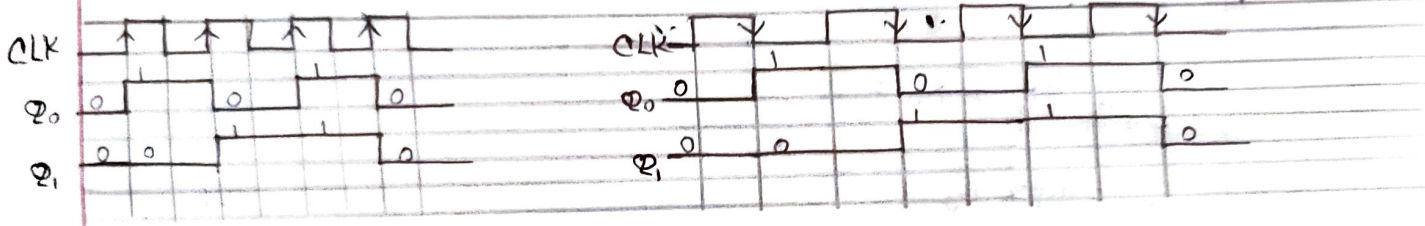


MOD-4 Synchronous up counter

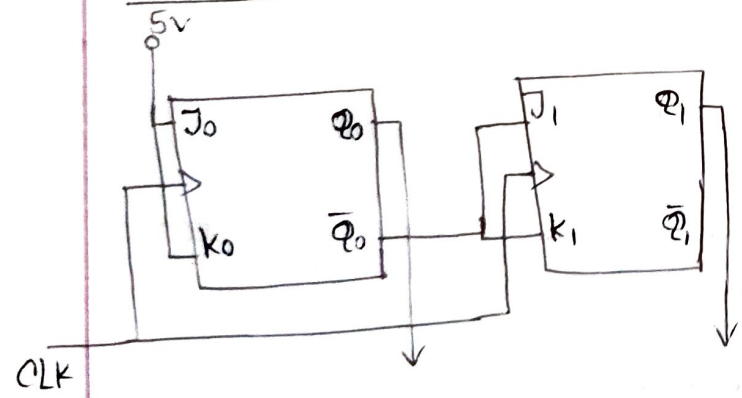
$J_0 = K_0 = 1$, $J_1 = K_1 = Q_0$



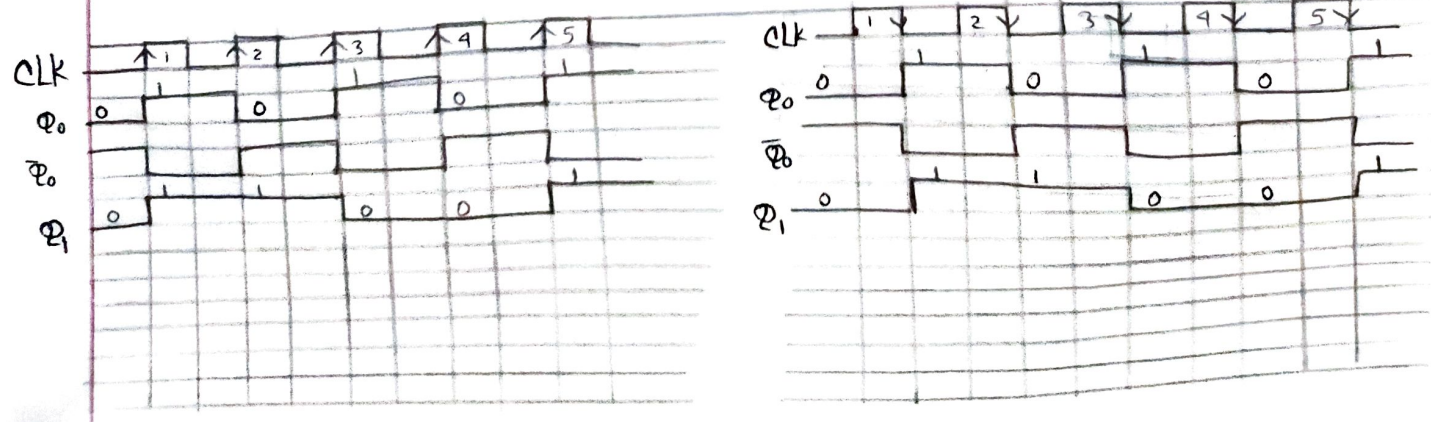
CLK	Q_1	Q_0
	0	0
1	0	1
2	1	0
3	1	1
4	0	0



MOD-4 Synchronous down counter

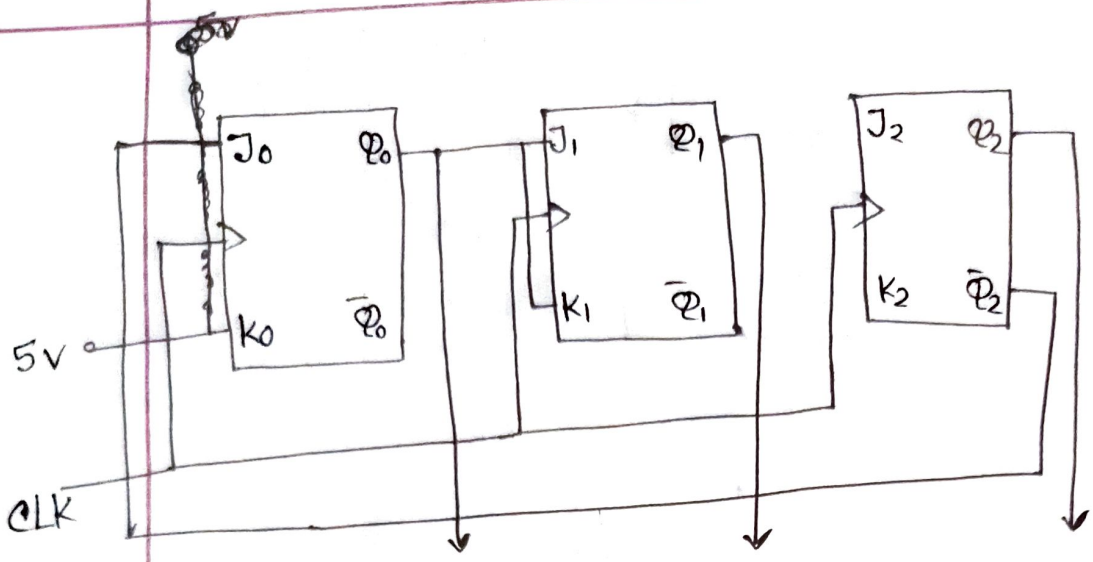


CLK	Q_1	Q_0
	0	0
1	1	1
2	1	0
3	0	1
4	0	0



MOD-5 synchronous up counter.

~~$J_0 = K_0 = 1$~~,

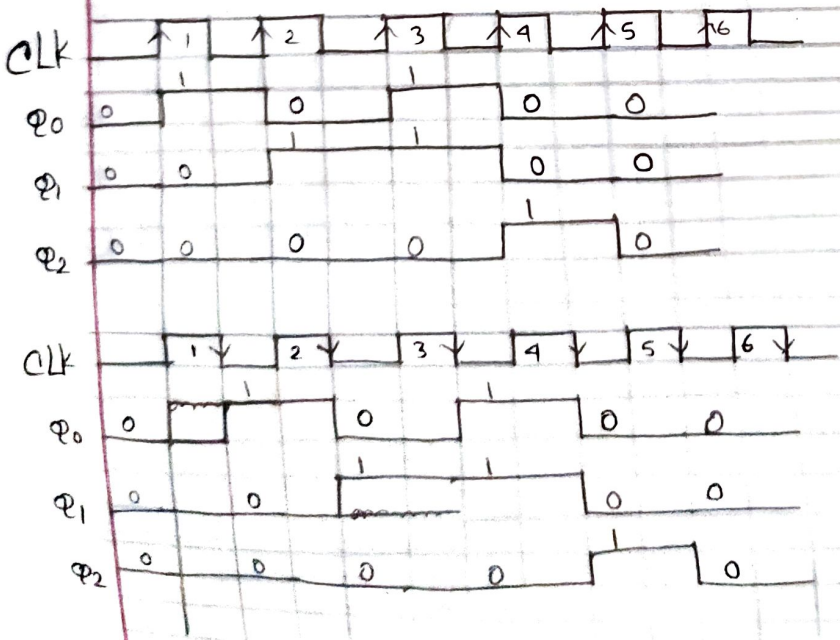


CLK	Q_2	Q_1	Q_0
	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	0	0	0

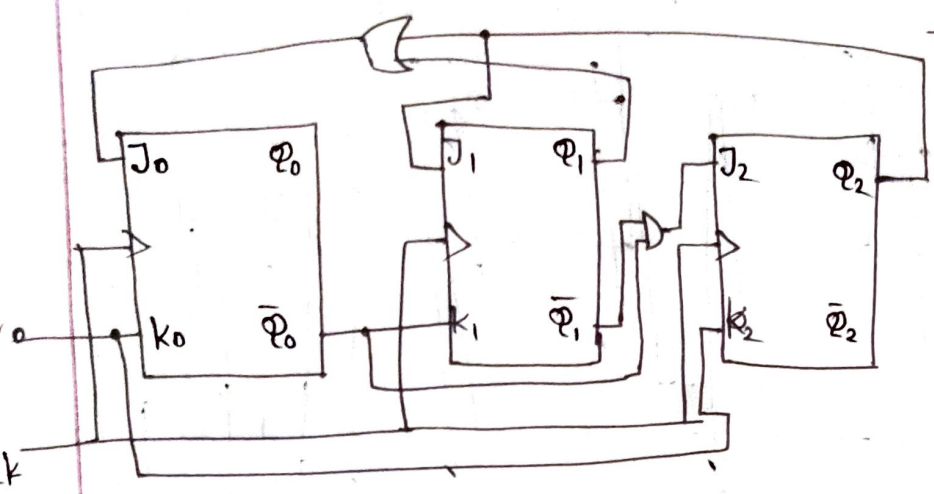
$J_0 = \bar{Q}_2$, $K_0 = 1$

$J_1 = K_1 = Q_0$

$J_2 = Q_1 Q_0$, $K_2 = 1$



Mod-5 synchronous down counter.

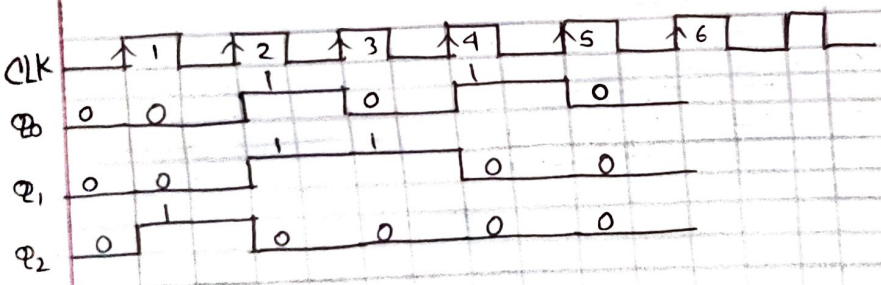


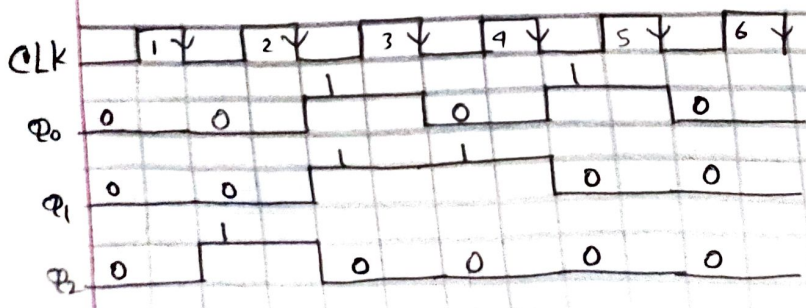
CLK	Q_2	Q_1	Q_0
	0	0	0
1	1	0	0
2	0	1	1
3	0	1	0
4	0	0	1
5	0	0	0

$J_0 = Q_2 + Q_1$, $K_0 = 1$

$J_1 = Q_2$, $K_1 = \bar{Q}_0$

$J_2 = \bar{Q}_1 \bar{Q}_0$, $K_2 = 1$.





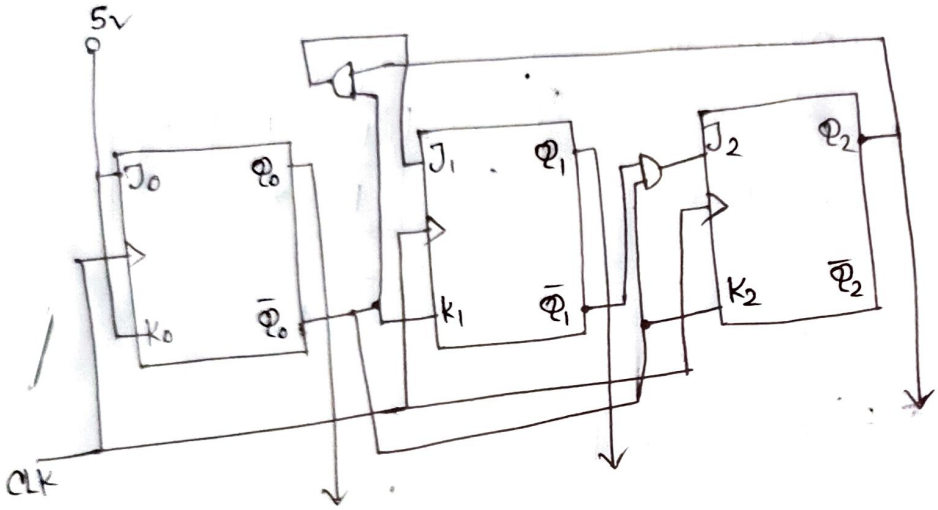
MOD-6 synchronous down counter

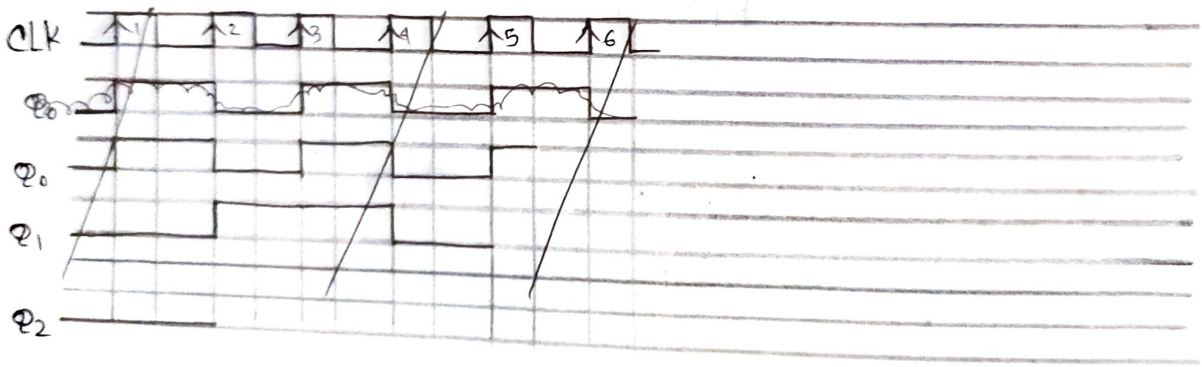
CLK	Q_2	Q_1	Q_0
	0	0	0
1	1	0	1
2	1	0	0
3	0	1	1
4	0	1	0
5	0	0	1
6	0	0	0

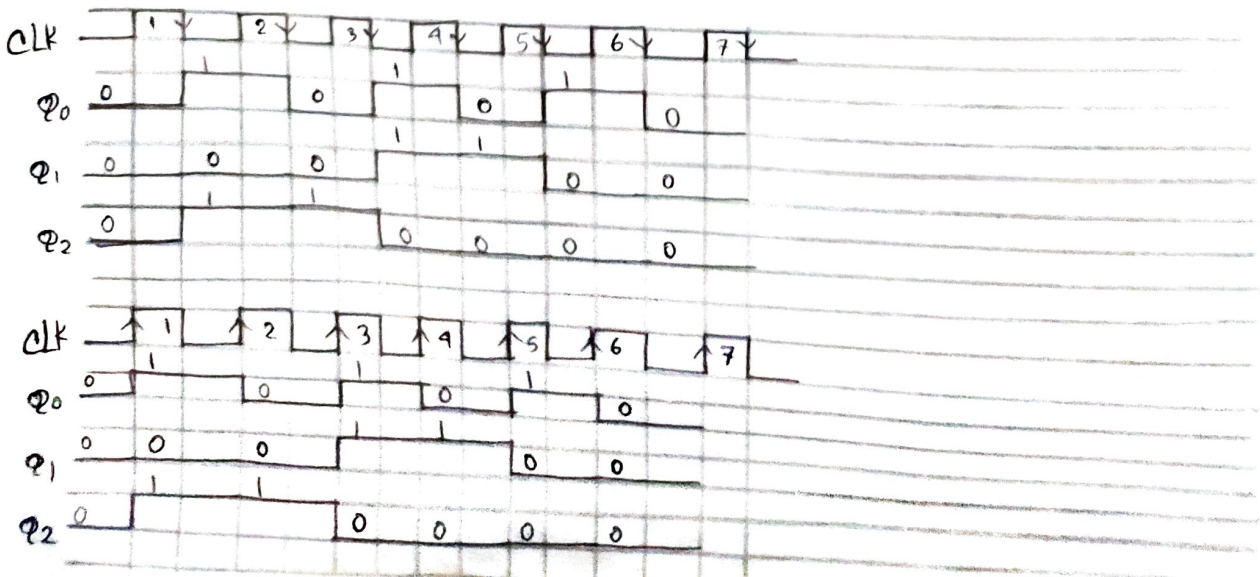
$J_0 = K_0 = 1$

$J_1 = Q_2 \bar{Q}_0$, $K_1 = \bar{Q}_0$

$J_2 = \bar{Q}_1 \bar{Q}_0$, $K_2 = \bar{Q}_0$







Present			Next		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	0	1	0	1
1	0	1	1	0	0
1	0	0	0	1	1
0	1	1	0	1	0
0	1	0	0	0	1
0	0	1	0	0	0

J_0,

Q_2 \ Q_1Q_0	00	01	11	10
0	1	*	*	1
1	1	*	*	*

$J_0 = 1$

K_0,

Q_2 \ Q_1Q_0	00	01	11	10
0	*	1	1	*
1	*	1	*	*

$K_0 = 1$

J_1,

Q_2 \ Q_1Q_0	00	01	11	10
0	0	0	*	*
1	1	0	*	*

$J_1 = \bar{Q}_0 Q_2$

K_1,

Q_2 \ Q_1Q_0	00	01	11	10
0	*	*	0	1
1	*	*	*	*

$K_1 = \bar{Q}_0$

J_2,

Q_2 \ Q_1Q_0	00	01	11	10
0	1	0	0	0
1	*	*	*	*

$J_2 = \bar{Q}_1 \bar{Q}_0$

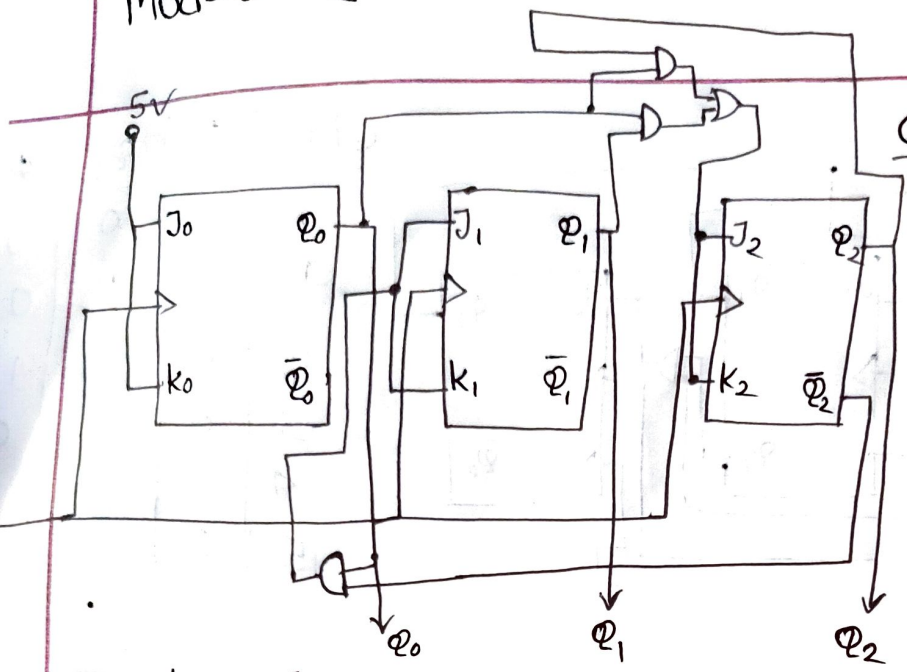
K_2,

Q_2 \ Q_1Q_0	00	01	11	10
0	*	*	*	*
1	1	0	*	*

$K_2 = \bar{Q}_0$

0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

Mod-6 synchronous up counter

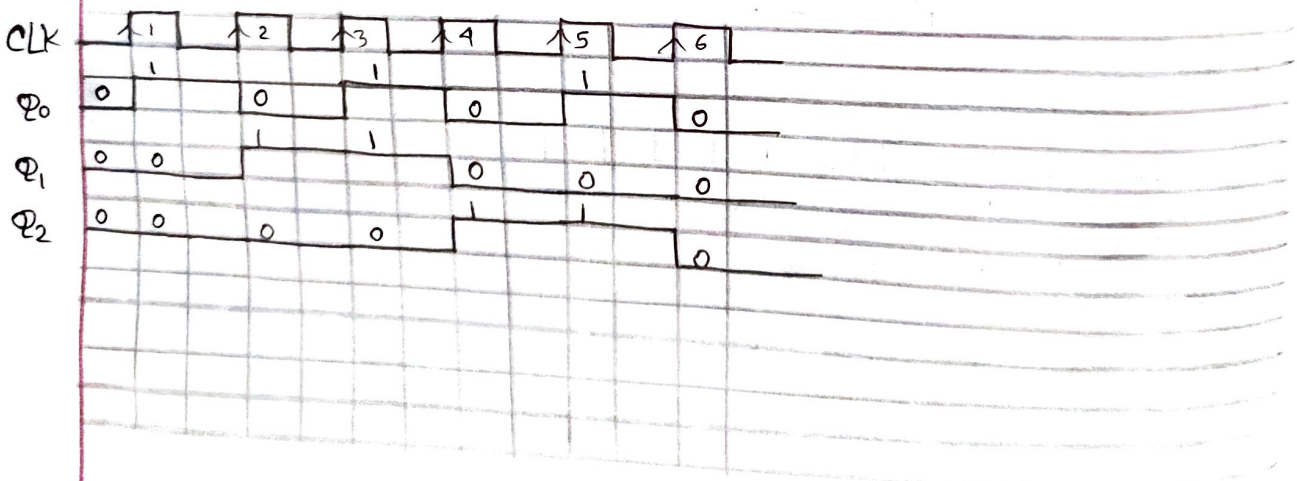


CLK	Q_2	Q_1	Q_0
	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	0	0	0

$J_0 = K_0 = 1$

$J_1 = K_1 = \bar{Q}_2 Q_0$

$J_2 = K_2 = Q_1 Q_0 + Q_2 Q_0$

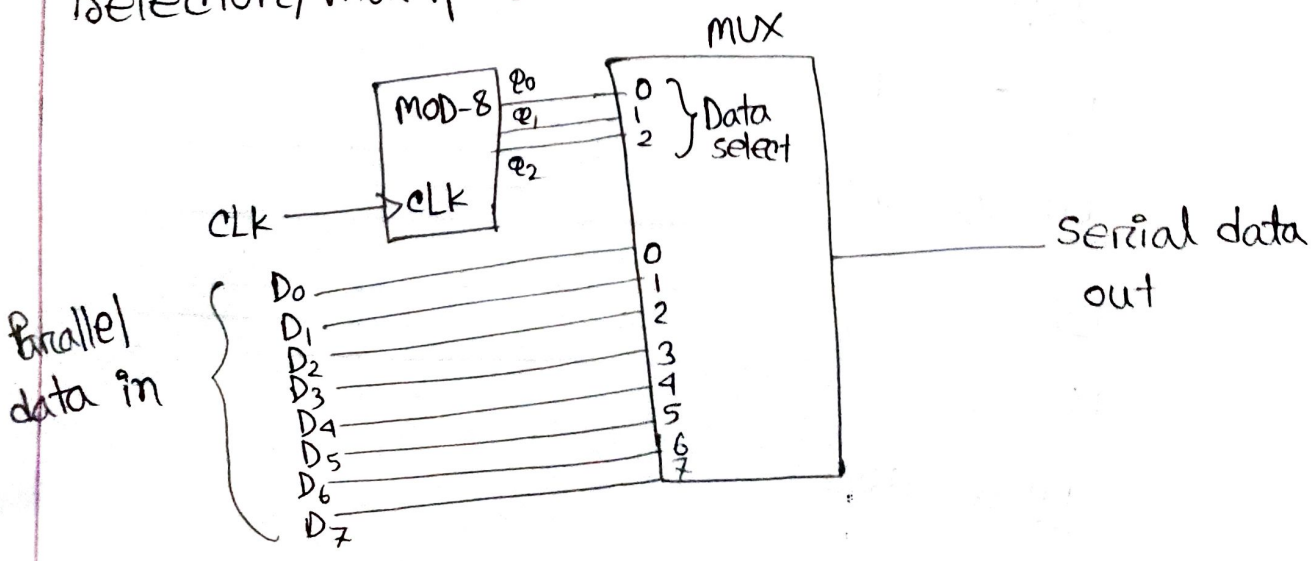


Counter Applications

I) A digital clock — A common example of a counter application is in timekeeping systems.

II) Automobile Parking Control —

III) Parallel-to-Serial data conversion (Multiplexing) — Parallel-to-serial conversion is normally accomplished by the use of a counter to provide a binary sequence for the data-select inputs of a data selector/multiplexer



Self-test

1] a) ripple counters

2] b) the method of clocking

3] b) the actual number of states in its sequence.

4] c) 8

5] a) 16

6] c) 4 flip-flops

7] b) Modulus 14

8] c) 48 ns

9] d) answers (b) and (c)

10] a) 1100

11] c) 1000

12]

13]

14] d) 1100

Chapter-9
Shift Registers

=> Shift Registers consist of an arrangement of flip-flops and are important in applications involving the storage and transfer of data in a digital system. A register, unlike a counter, has no specified sequence of states.

* A register, in general, is used solely for storing and shifting data entered into it from an external source and typically possesses no characteristic internal sequence of states.

* A register is a digital circuit with two basic functions: data storage and data movement. The storage capability of a register makes it an important type of

memory device. A register can consist of one or more flip-flops used to store and shift data.

* The storage capacity of a register is the total number of bits of digital data it can retain. Each stage in a shift register represents one bit of storage capacity. Therefore, the number of stages in a register determines its storage capacity.

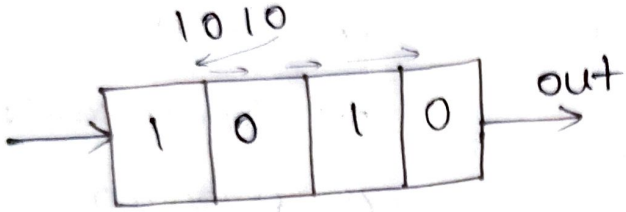
* The shifting capability of a register permits the movement of data from stage to stage within the register or into or out of the register upon application of clock pulses.

* Difference between register and counter:

Register	Counter
1] Register can hold data to use as a temporary memory storage.	1] Counter can only be loaded, stored, or incremented as a program counter.
2] It does not follow specific sequence of states.	2] Follow specific sequence of states.
3] It has the same clock.	3] It does not need a same clock.
4] Its path of stage is not predefined.	4] Its path of stage is predefined.
5] All register is not counter.	5] All counter is register.

* The types of data movement in shift registers:

1) Serial-in-Serial-out shift register.

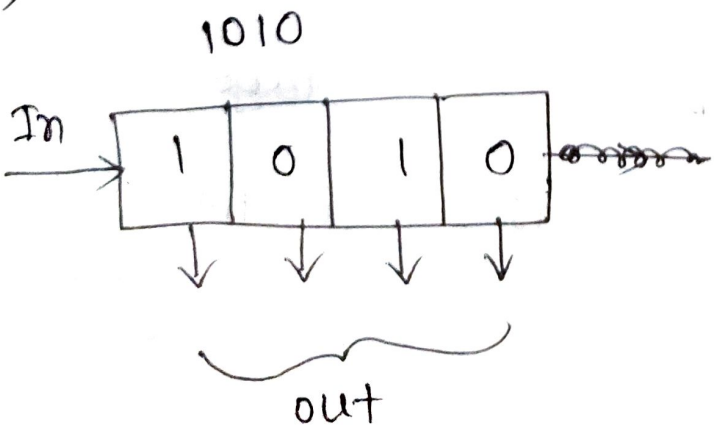


4 bit number,
4 clock for in
3 clock for out
total 7 clock needed

* Microprocessor ↓

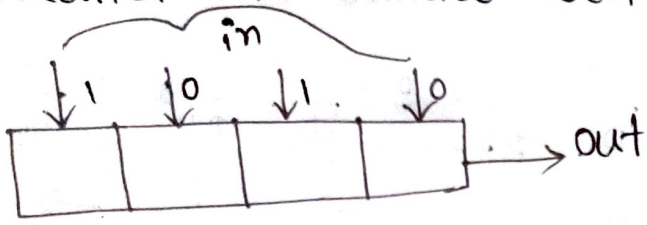
1) Big Endian format
11) Little Endian format.

11) Serial-in-Parallel-out shift register



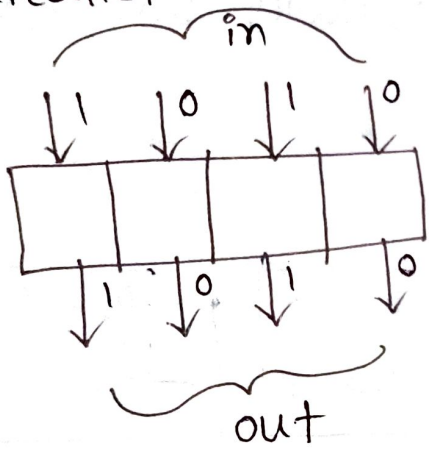
4 clock needed for in & out

iii) Parallel-in-Serial-out shift register.



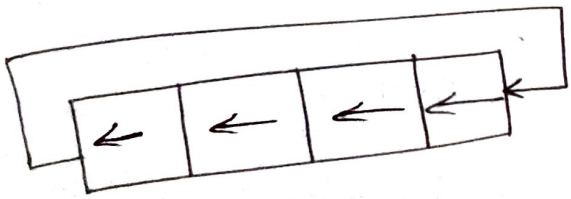
1 clock for in
3 clock for shift
total 4 clock needed.

iv) Parallel-in-Parallel-out shift register.

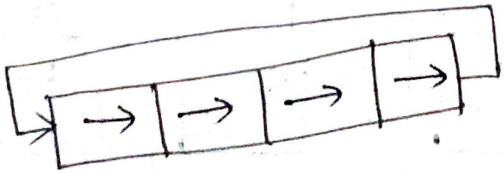


1 clock for in & out.

v) Rotate left shift register.

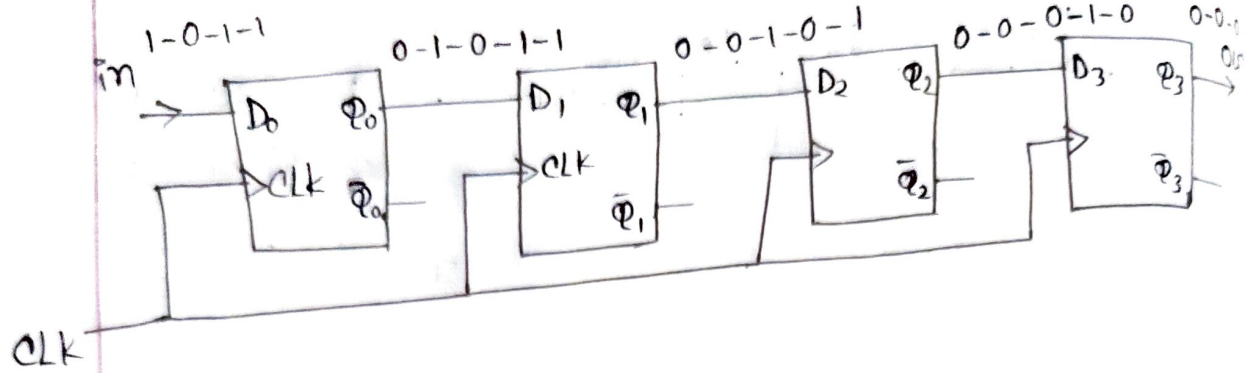


vi) Rotate right shift register.

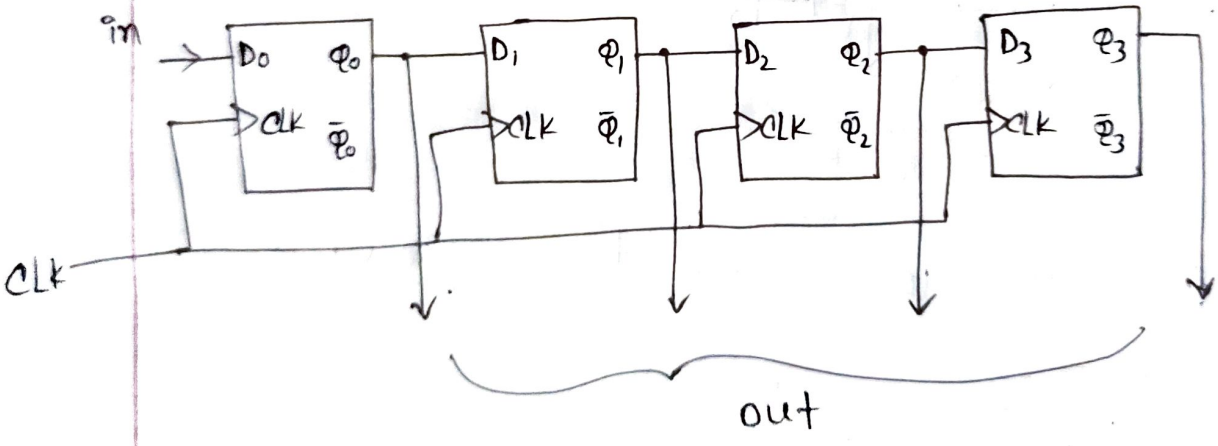


1101

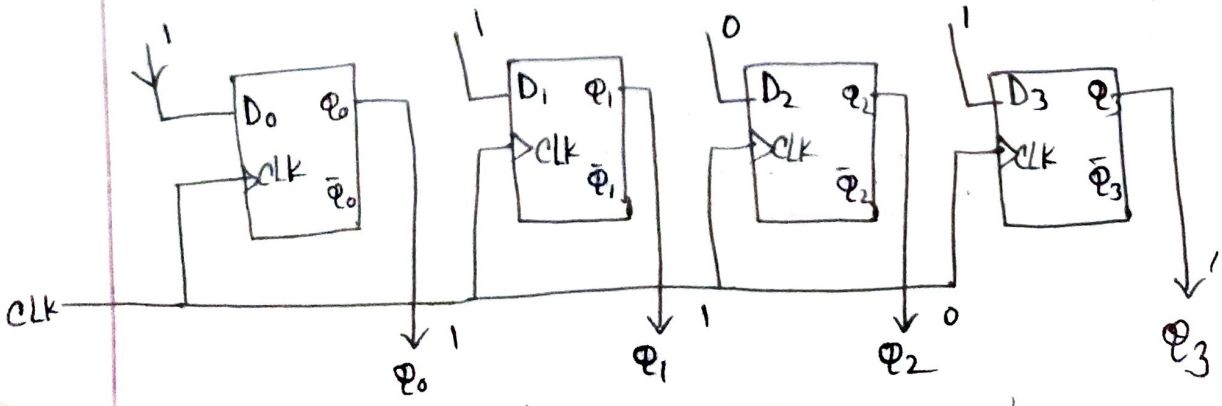
i) 4 bit Serial-in- Serial-out shift register



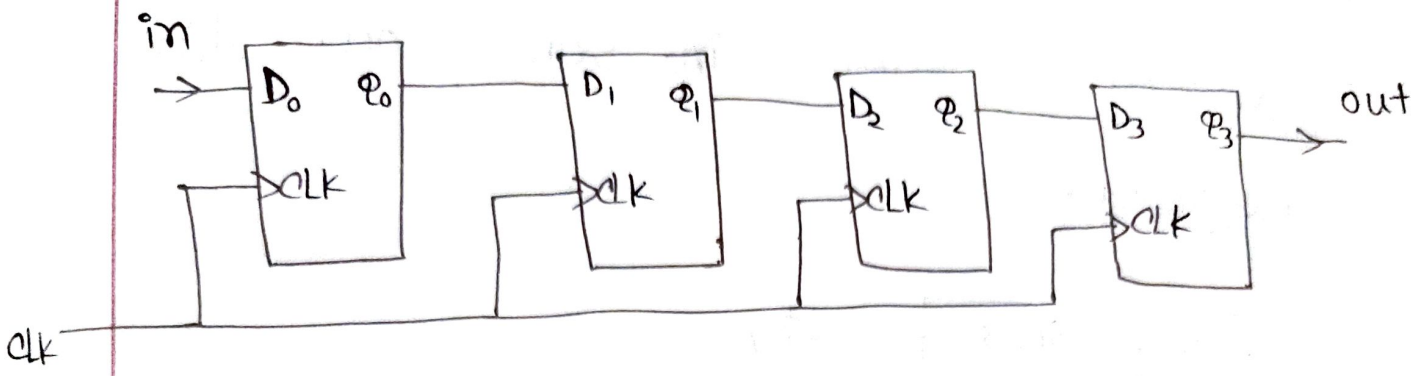
ii) 4 bit Serial-in- Parallel out shift register.



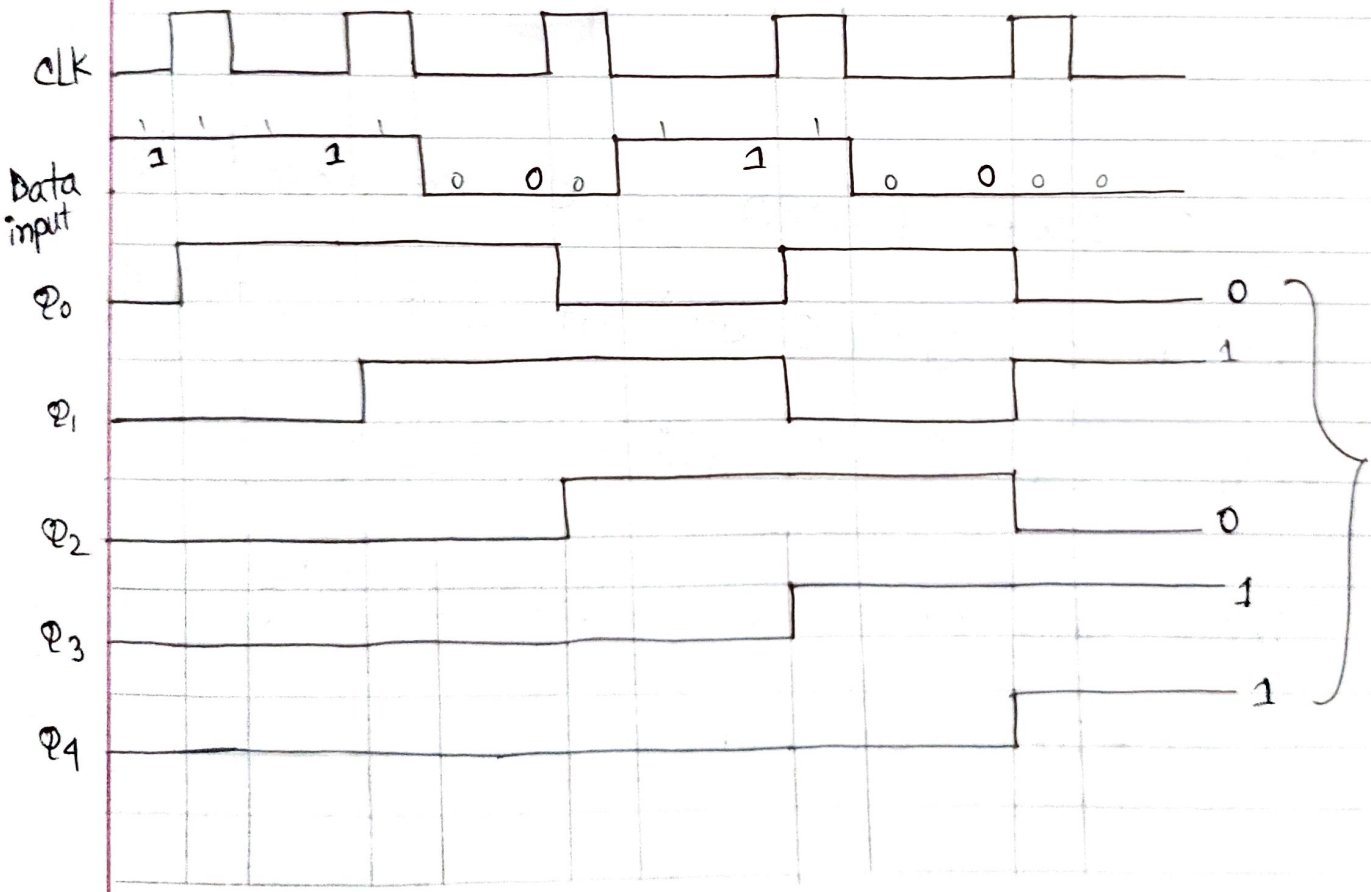
iv) Parallel-in - Parallel-out shift register



Example



Serial-in-Serial-out shift register.



* A traditional logic block symbol for an 8-bit serial-in-serial-out shift register.

Review

2) 8 clock pulses are required to enter a byte (8 bit) of data serially into an 8-bit shift register.

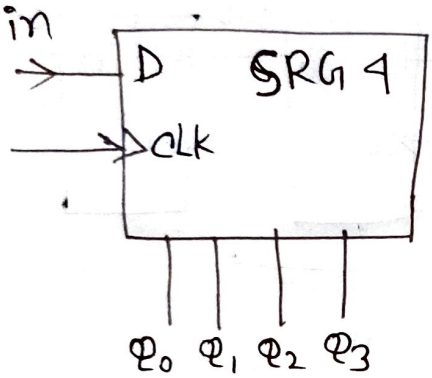
* Data bits are entered serially right-most bit first.

Data in
CLK
Q_0
Q_1
Q_2
Q_3

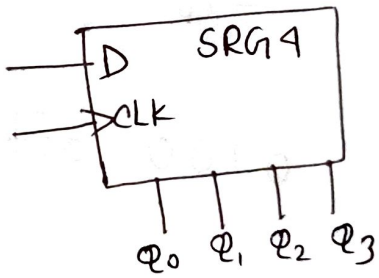
⇒ Serial-in- Parallel-out shift register.

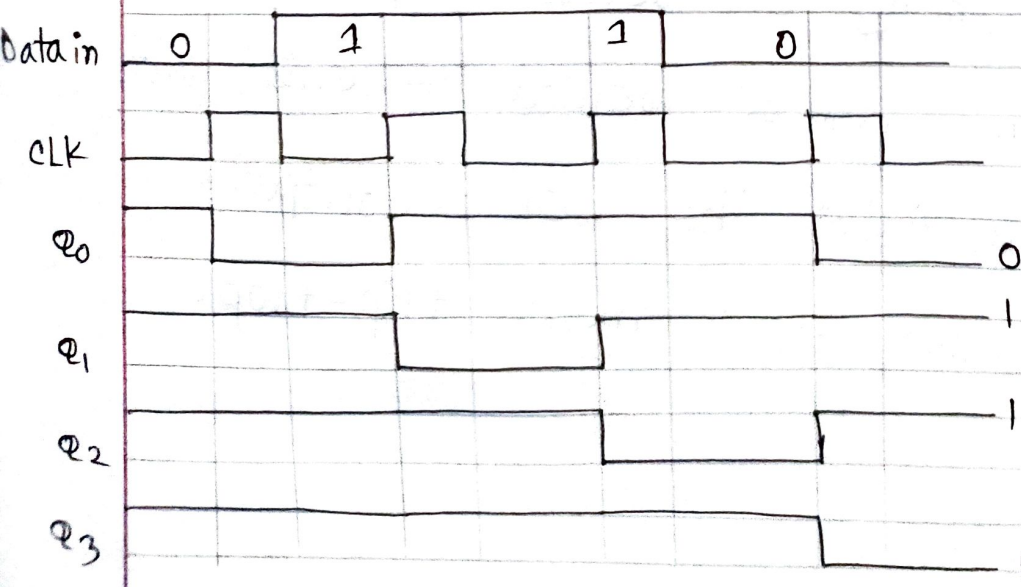
4 bit,

Block diagram



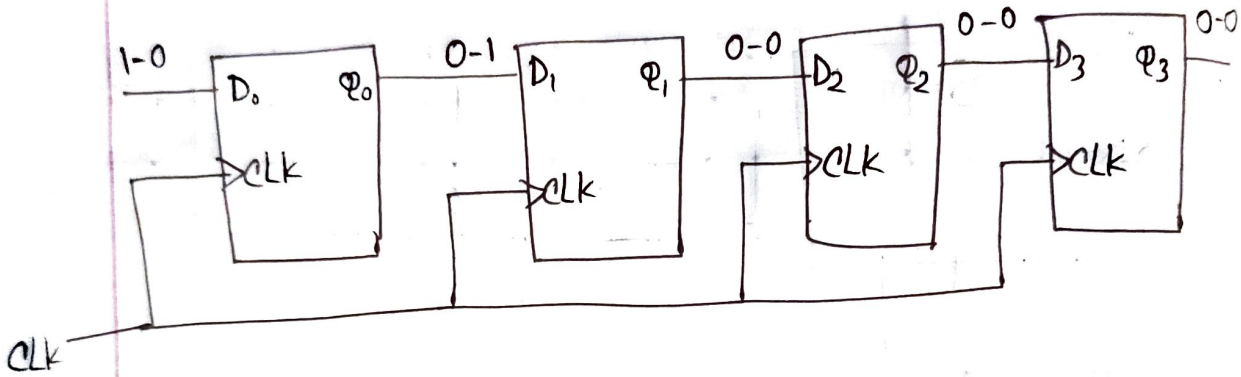
Example 9-2





Review ques

1] Bit sequence 1101 serial-in-parallel-out shift register.

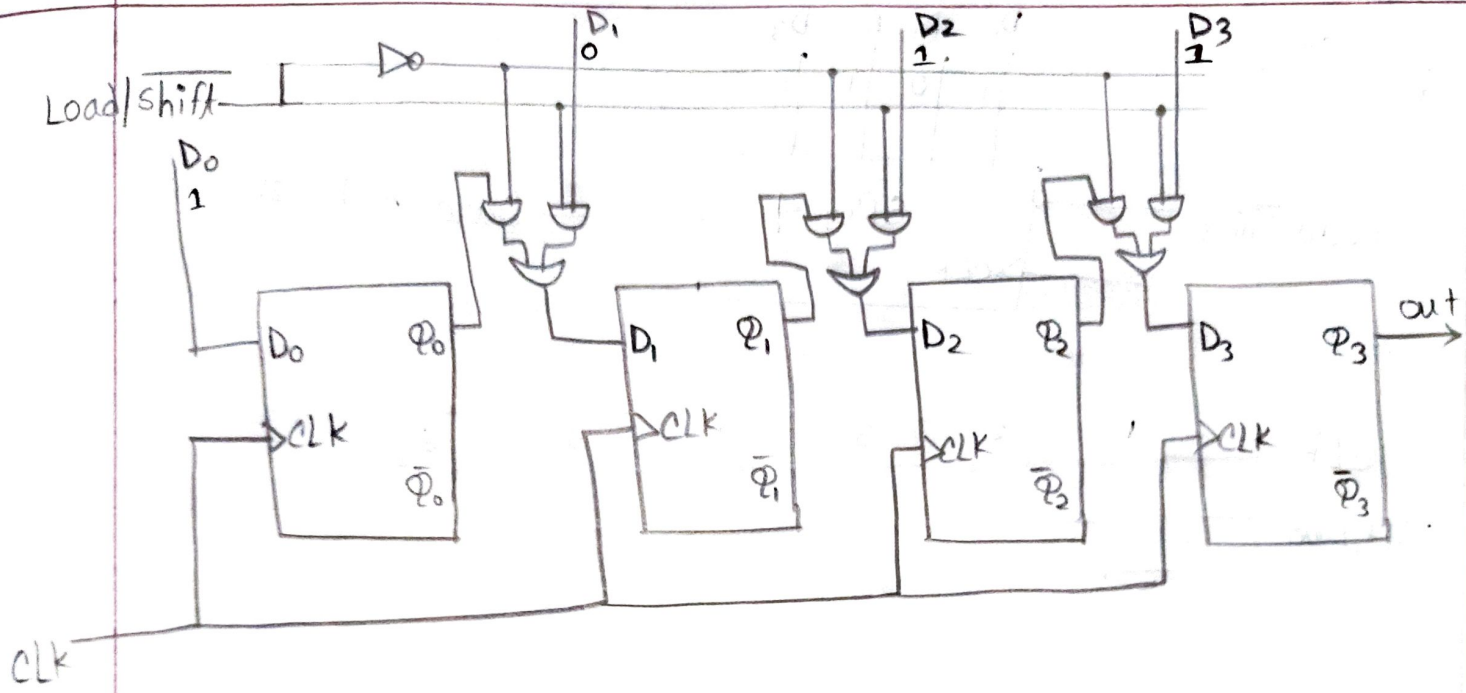


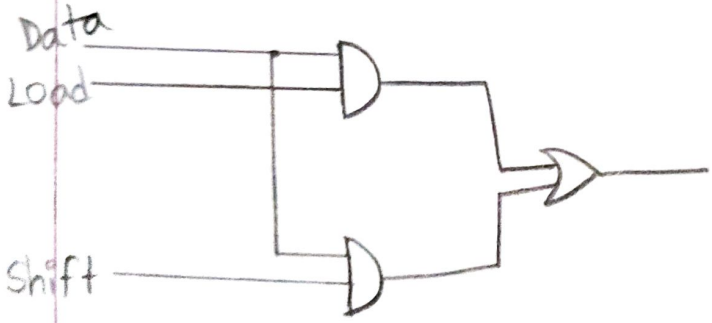
1-0 ——Q0 0-1 ——Q1 0-0 ——Q2 0-0 ——Q3

0100 after 2 clock pulses

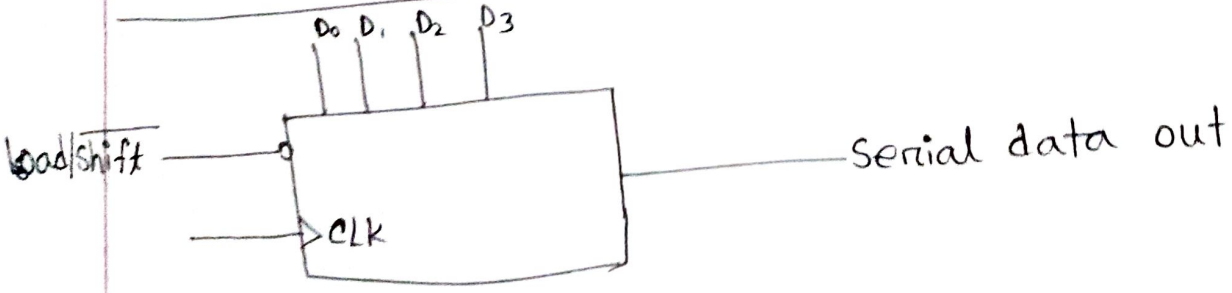
2] For using a serial-in-parallel-out register as a serial-in-serial-out register, take the serial output from the right most flip-flop.

Parallel-in-Serial-out shift registers

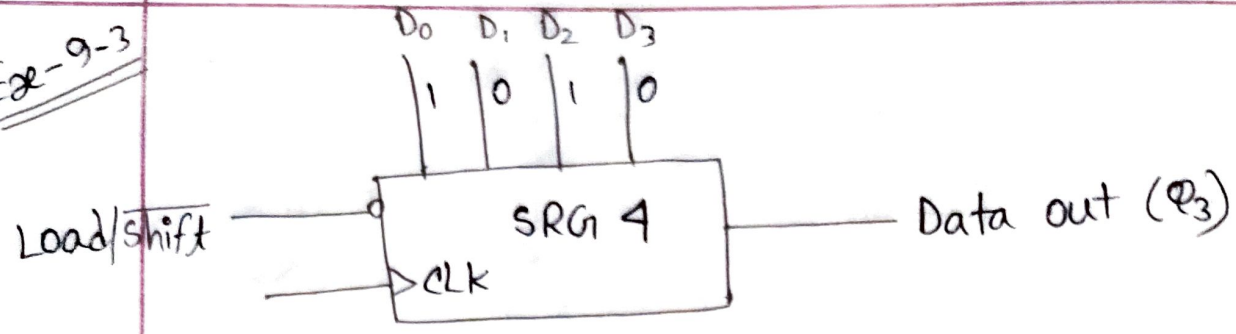


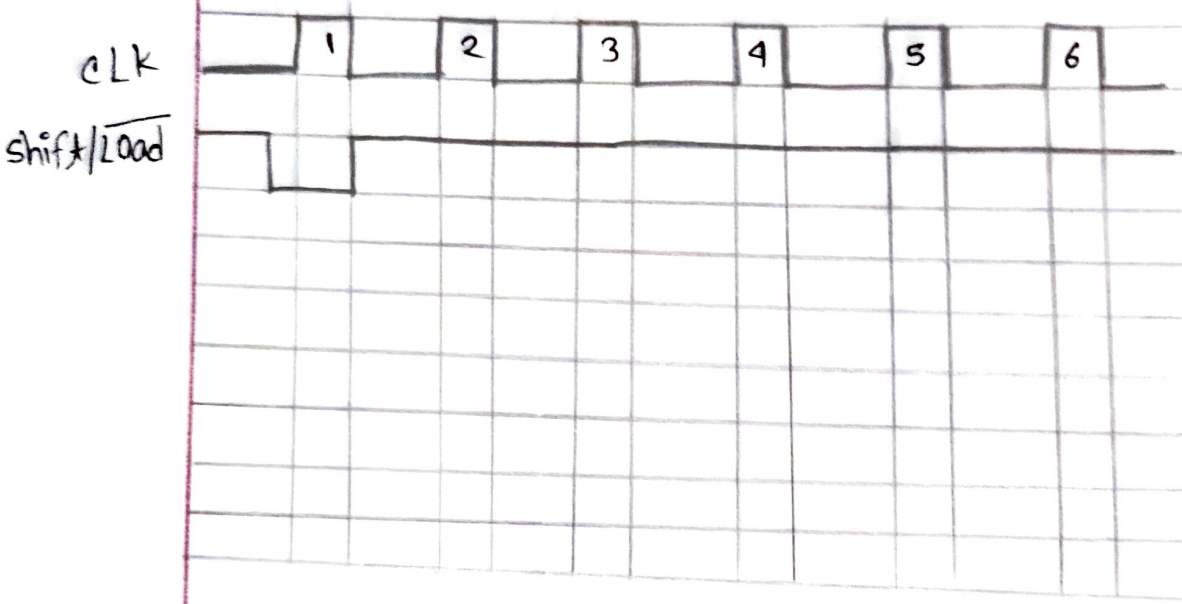


Logic symbol / Block diagram



Ex-9-3


D0 D1 D2 D3
1 0 1 0
Load/Shift
SRG 4
CLK
Data out (Q3)


CLK
1
2
3
4
5
6
Shift/Load

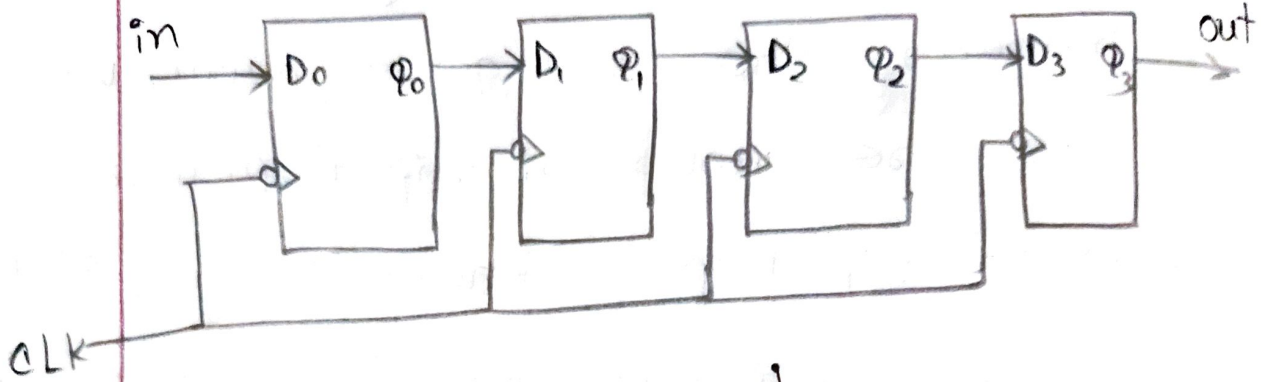
⇒ Flip-flop is 1-bit memory cell.
To increase the storage capacity, we have to use group of flip-flop. This group of flip-flop is known as Register.
The n-bit register consist of n number of flip-flops and is capable of storing n-bit word.

Classification of Register :-

i) Depending on input and output
   a) SISO b) SIPO c) PISO d) PIPO

ii) Depending on application
   a) Shift register b) Storage register.

Serial-in- Serial- out shift register



CLK	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
↓				